SECTION 1: BASIC CONCEPTS (Questions 1-25)

1. What is a Database?
A) A collection of related files
B) A collection of related data with inherent meaning
C) A software system
D) A programming language

Answer: B
Explanation: A database is a collection of related data that represents some aspect of the real world and has inherent meaning.

---

2. Which of the following is NOT a characteristic of the database approach?
A) Self-describing nature
B) Program-data dependence
C) Multiple views support
D) Data abstraction

Answer: B
Explanation: Program-data independence is a key feature of database approach, not dependence.

---

3. The overall description of a database is called:
A) Database schema
B) Database instance
C) Database state
D) Database model

Answer: A
Explanation: Schema is the overall description/blueprint of the database structure.

---

4. Which component of DBMS is responsible for storing metadata?
A) Query processor
B) Storage manager
C) Catalog
D) Transaction manager

Answer: C
Explanation: The catalog/dictionary stores metadata about the database structure.

---

5. The actual data in a database at a particular moment is called:
A) Schema
B) Instance
C) Subschema
D) DDL

Answer: B
Explanation: Instance or database state refers to the actual data at a specific point in time.

---

6. DBMS stands for:
A) Database Management System
B) Data Backup Management System
C) Database Modeling System
D) Data Base Management Software

Answer: A

---

7. Which language is used to define the database structure?
A) DML
B) DCL
C) DDL
D) TCL

Answer: C
Explanation: Data Definition Language (DDL) is used to define database schema.

---

8. Which of the following is an example of a commercial DBMS?
A) MySQL
B) Oracle
C) PostgreSQL
D) MongoDB

Answer: B
Explanation: Oracle is commercial, while MySQL, PostgreSQL are open-source, and MongoDB is NoSQL.

---

9. The property that allows multiple users to access database concurrently without problems is:

A) Atomicity
B) Isolation
C) Durability
D) Consistency

Answer: B
Explanation: Isolation ensures concurrent transactions don't interfere.

---

10. Which level of abstraction describes what data is stored?
A) Physical level
B) Logical level
C) View level
D) External level

Answer: B
Explanation: Logical level describes what data is stored and relationships.

---

11. The person who has central control over the database system is:
A) Database Administrator
B) End User
C) Application Programmer
D) System Analyst

Answer: A

---

12. Which of the following is NOT a function of DBA?
A) Schema definition
B) Writing application programs
C) Security management
D) Performance monitoring

Answer: B
Explanation: Application programmers write application programs, not DBA.

---

13. A database system is self-describing because:
A) It contains data
B) It contains metadata
C) It has indexes
D) It uses SQL

Answer: B
Explanation: The database contains both data and metadata about itself.

---

14. Which type of user interacts with database through forms without writing queries?
A) Naive users
B) Sophisticated users
C) Application programmers
D) DBA

Answer: A
Explanation: Naive users use pre-written applications/forms.

---

15. The three levels of database architecture are:
A) Internal, External, Conceptual
B) Physical, Logical, View
C) Schema, Subschema, Instance
D) User, Program, Data

Answer: A
Explanation: ANSI-SPARC architecture has Internal, Conceptual, and External levels.

SECTION 2: DATA MODELS (Questions 26-45)

16. Which data model organizes data in parent-child relationships?
A) Relational model
B) Network model
C) Hierarchical model
D) Object-oriented model

Answer: C
Explanation: Hierarchical model uses tree-like structure with parent-child relationships.

---

17. The relational model was proposed by:
A) Charles Bachman
B) Edgar F. Codd
C) Peter Chen
D) Ted Codd

Answer: B
Explanation: E.F. Codd proposed relational model in 1970.

---

18. In the relational model, data is organized in:
A) Trees
B) Tables
C) Graphs
D) Networks

Answer: B

---

19. A row in a relational table is called:
A) Attribute
B) Domain
C) Tuple
D) Degree

Answer: C

---

20. The number of attributes in a relation is called its:
A) Cardinality
B) Degree
C) Dimension
D) Size

Answer: B

---

21. The number of tuples in a relation is called its:
A) Degree
B) Cardinality
C) Arity
D) Scope

Answer: B

---

22. Which data model uses object-oriented concepts like encapsulation?
A) Relational
B) Network
C) Object-oriented
D) Hierarchical

Answer: C

---

23. The Entity-Relationship model was proposed by:
A) E.F. Codd
B) Peter Chen
C) James Martin
D) C.J. Date

Answer: B

---

24. In ER model, an entity is:
A) A table
B) A real-world object
C) A relationship
D) An attribute

Answer: B

---

25. Which model represents data as collection of records and relationships as sets?
A) Hierarchical
B) Relational
C) Network
D) Object-oriented

Answer: C
Explanation: Network model uses sets to represent relationships (CODASYL model).

---

26. A relationship type in ER model has:
A) Degree
B) Cardinality
C) Participation
D) All of the above

Answer: D

---

27. An attribute that can be further divided is called:
A) Simple attribute
B) Composite attribute

C) Derived attribute
D) Single-valued attribute

Answer: B

---

28. Which of the following is a valid relationship degree?
A) Unary
B) Binary
C) Ternary
D) All of the above

Answer: D

---

29. The network model allows:
A) One-to-one relationships only
B) Many-to-many relationships
C) Only hierarchical relationships
D) No relationships

Answer: B

---

30. In the hierarchical model, records are organized as:
A) Directed graphs
B) Trees
C) Tables
D) Linked lists

Answer: B

SECTION 3: SCHEMA & INSTANCE (Questions 46-60)

31. A database schema is also called:
A) Intension
B) Extension
C) Instance
D) State

Answer: A
Explanation: Schema (intension) remains constant, instance (extension) changes.

---

32. Which of the following changes most frequently?
A) Schema
B) Instance
C) Subschema
D) Metadata

Answer: B

---

33. The physical schema describes:
A) How data is stored
B) What data is stored
C) User views
D) Business rules

Answer: A

---

34. External schema corresponds to:
A) Physical storage
B) User views
C) Conceptual model
D) Internal model

Answer: B

---

35. The mapping between conceptual and internal schema provides:
A) Logical data independence
B) Physical data independence
C) Data security
D) Data integrity

Answer: B

---

36. Logical data independence is achieved through mapping between:
A) External and conceptual
B) Conceptual and internal
C) External and internal
D) User and physical

Answer: A

---

37. Which schema defines the stored data structures?
A) External schema
B) Conceptual schema
C) Internal schema
D) User schema

Answer: C

---

38. The three-schema architecture was proposed by:
A) IBM
B) ANSI/SPARC
C) Microsoft
D) Oracle

Answer: B

---

39. Which of the following is NOT a benefit of three-schema architecture?
A) Data independence
B) Data redundancy
C) Multiple user views
D) Centralized management

Answer: B
Explanation: Data redundancy is not a benefit; it's a problem to avoid.

---

40. In three-schema architecture, the middle level is:
A) Physical level
B) Conceptual level
C) External level
D) View level

Answer: B

SECTION 4: DBMS ARCHITECTURE (Questions 61-80)

41. Which architecture separates the DBMS into three tiers?
A) 1-tier
B) 2-tier

C) 3-tier
D) N-tier

Answer: C

---

42. In client-server architecture, the client handles:
A) Data storage
B) User interface
C) Query processing
D) Transaction management

Answer: B

---

43. Which component parses and analyzes queries?
A) Storage manager
B) Query processor
C) Transaction manager
D) File manager

Answer: B

---

44. The buffer manager is responsible for:
A) Managing main memory
B) Managing disk space
C) Managing users
D) Managing transactions

Answer: A

---

45. Which of the following is a storage manager component?
A) Authorization manager
B) Transaction manager
C) File manager
D) All of the above

Answer: D

---

46. The transaction manager ensures:
A) ACID properties

B) Fast queries
C) Data compression
D) Indexing

Answer: A

---

47. Which architecture is most suitable for web applications?
A) 1-tier
B) 2-tier
C) 3-tier
D) Centralized

Answer: C

---

48. In centralized DBMS architecture:
A) Data and DBMS on same machine
B) Data distributed across network
C) Multiple DBMS instances
D) Peer-to-peer processing

Answer: A

---

49. A distributed database has data:
A) At one location
B) At multiple locations
C) Only in memory
D) Only on cloud

Answer: B

---

50. The main advantage of client-server architecture over centralized is:
A) Lower cost
B) Better scalability
C) Simpler design
D) Less security needed

Answer: B

SECTION 5: DATA INDEPENDENCE (Questions 81-100)

51. Data independence means:
A) Data is independent of users
B) Applications independent of storage
C) Data independent of itself
D) Programs independent of data

Answer: B/D

---

52. Physical data independence protects from changes in:
A) User views
B) Storage structures
C) Table structures
D) Relationships

Answer: B

---

53. If we change the storage device from HDD to SSD, which independence applies?
A) Logical independence
B) Physical independence
C) Both
D) Neither

Answer: B

---

54. Adding a new attribute to a table should not affect existing programs due to:
A) Physical independence
B) Logical independence
C) Program independence
D) Data abstraction

Answer: B

---

55. Which is easier to achieve in DBMS?
A) Logical data independence
B) Physical data independence
C) Both equally
D) Neither

Answer: B
Explanation: Physical data independence is easier than logical data independence.

SECTION 6: DATABASE LANGUAGES (Questions 101-115)

56. DDL statements are used to:
A) Query data
B) Define structure
C) Manipulate data
D) Control access

Answer: B

---

57. Which of the following is a DDL command?
A) SELECT
B) INSERT
C) CREATE
D) UPDATE

Answer: C

---

58. DML commands are used for:
A) Defining schema
B) Data manipulation
C) Access control
D) Transaction control

Answer: B

---

59. Which is NOT a DML command?
A) INSERT
B) DELETE
C) UPDATE
D) ALTER

Answer: D

---

60. DCL commands include:
A) GRANT and REVOKE
B) COMMIT and ROLLBACK

C) SELECT and INSERT
D) CREATE and DROP

Answer: A

---

61. TCL commands handle:
A) Transactions
B) Tables
C) Users
D) Views

Answer: A

---

62. Which language is used to define views?
A) DDL
B) DML
C) DCL
D) TCL

Answer: A

---

63. SQL stands for:
A) Structured Query Language
B) Simple Query Language
C) Sequential Query Language
D) System Query Language

Answer: A

SECTION 7: DATABASE USERS (Questions 116-125)

64. Which user writes database application programs?
A) Naive users
B) Application programmers
C) Sophisticated users
D) Parametric users

Answer: B

---

65. Parametric users interact with database using:
A) SQL queries
B) Pre-written transactions
C) Programming languages
D) DDL commands

Answer: B

---

66. Sophisticated users interact using:
A) Forms
B) Query languages
C) Menus
D) GUI

Answer: B

---

67. The DBA is responsible for:
A) Writing programs
B) Designing forms
C) Overall administration
D) Data entry

Answer: C

SECTION 8: ADVANCED CONCEPTS (Questions 126-140)

68. A data dictionary contains:
A) User data
B) Metadata
C) Indexes
D) Queries

Answer: B

---

69. Which property ensures transactions are processed reliably?
A) ACID
B) BASE
C) CAP
D) CRUD

Answer: A

---

70. The "A" in ACID stands for:
A) Availability
B) Atomicity
C) Accessibility
D) Accuracy

Answer: B

---

71. The "C" in ACID stands for:
A) Consistency
B) Concurrency
C) Correctness
D) Completion

Answer: A

---

72. The "I" in ACID stands for:
A) Integrity
B) Isolation
C) Independence
D) Interaction

Answer: B

---

73. The "D" in ACID stands for:
A) Durability
B) Data
C) Database
D) Dependency

Answer: A

---

74. A file processing system lacks:
A) Data redundancy
B) Data inconsistency

C) Data integration
D) Data isolation

Answer: C

---

75. Which is an advantage of DBMS over file systems?
A) Lower cost
B) Less complexity
C) Better data sharing
D) Simpler design

Answer: C

---

76. Data abstraction in DBMS means:
A) Hiding complexity
B) Data encryption
C) Data compression
D) Data duplication

Answer: A

---

77. The level of abstraction closest to users is:
A) Physical
B) Logical
C) View
D) Internal

Answer: C

---

78. Which component handles concurrent access?
A) Query processor
B) Storage manager
C) Transaction manager
D) File manager

Answer: C

---

79. The recovery manager handles:
A) System crashes

B) User errors
C) Disk failures
D) All of the above

Answer: D

---

80. Integrity constraints ensure:
A) Data accuracy
B) Fast access
C) Less storage
D) More users

Answer: A

---

81. A candidate key is:
A) A primary key
B) A unique identifier
C) A foreign key
D) An alternate key

Answer: B

---

82. Which constraint ensures referential integrity?
A) Primary key
B) Foreign key
C) Unique key
D) Check constraint

Answer: B

---

83. The degree of a relationship type is:
A) Number of entities
B) Number of attributes
C) Number of relationships
D) Number of constraints

Answer: A

---

84. In ER model, a weak entity:
A) Has no attributes
B) Depends on another entity
C) Has no key
D) Both B and C

Answer: D

---

85. Cardinality ratio represents:
A) Number of attributes
B) Number of relationships
C) Maximum participation
D) Minimum participation

Answer: C

SECTION 9: MISCELLANEOUS (Questions 141-150)

86. Which is NOT a type of database?
A) Centralized
B) Distributed
C) Homogeneous
D) Sequential

Answer: D

---

87. A homogeneous distributed database means:
A) Same hardware
B) Same software
C) Same data
D) Same users

Answer: B

---

88. Which is a NoSQL database?
A) MongoDB
B) Oracle
C) MySQL
D) PostgreSQL

Answer: A

---

89. The main characteristic of NoSQL databases is:
A) ACID compliance
B) Fixed schema
C) Flexible schema
D) SQL support

Answer: C

---

90. Big data is characterized by:
A) Volume only
B) Velocity only
C) Variety only
D) All of the above

Answer: D

---

91. Which is a cloud database service?
A) Amazon RDS
B) MySQL
C) Oracle
D) SQL Server

Answer: A

---

92. In-memory databases store data in:
A) Hard disk
B) RAM
C) Cache
D) SSD

Answer: B

---

93. Which is an example of an open-source DBMS?
A) SQL Server
B) Oracle

C) MySQL
D) DB2

Answer: C

---

94. The first popular DBMS was:
A) IMS
B) System R
C) Oracle
D) DB2

Answer: A
Explanation: IBM's IMS (Information Management System) was the first popular DBMS.

---

95. The relational model was first implemented in:
A) IMS
B) System R
C) Oracle
D) INGRES

Answer: B

---

96. Which company developed System R?
A) IBM
B) Oracle
C) Microsoft
D) DEC

Answer: A

---

97. Oracle was the first to implement:
A) Hierarchical model
B) Network model
C) Relational model commercially
D) Object-oriented model

Answer: C

---

98. Which is NOT a function of DBMS?
A) Data storage
B) Data retrieval
C) Data compilation
D) Data security

Answer: C

---

99. Database redologs are used for:
A) Recovery
B) Indexing
C) Query optimization
D) Security

Answer: A

---

100. A database checkpoint is used for:
A) Faster recovery
B) Better indexing
C) Query optimization
D) Security

Answer: A

---

101. Which operation is NOT part of transaction?
A) Begin
B) Read
C) Write
D) Index

Answer: D

---

102. The transaction state diagram includes:
A) Active state
B) Committed state
C) Failed state
D) All of the above

Answer: D

---

103. A deadlock in database occurs when:
A) Two transactions wait for each other
B) System crashes
C) Disk is full
D) User logs off

Answer: A

---

104. The two-phase locking protocol ensures:
A) Serializability
B) Recovery
C) Security
D) Integrity

Answer: A

---

105. Which is a concurrency control protocol?
A) Timestamp ordering
B) Lock-based
C) Optimistic
D) All of the above

Answer: D

---

106. The log-based recovery uses:
A) Undo and Redo
B) Backup only
C) Mirroring only
D) RAID only

Answer: A

---

107. RAID in databases stands for:
A) Redundant Array of Inexpensive Disks
B) Rapid Access of Indexed Data
C) Random Array of Independent Disks
D) Reliable Access of Integrated Data

Answer: A

SECTION 1: THREE-TIER ARCHITECTURE (Questions 1-25)

1. The three levels of database architecture (ANSI/SPARC) are:
A) Network, Hierarchical, Relational
B) Internal, Conceptual, External
C) Physical, Logical, View
D) User, DBA, Developer

Answer: B
Explanation: ANSI/SPARC architecture defines three levels: Internal (physical storage), Conceptual (global view), and External (user views).

---

2. Which level describes how data is actually stored?
A) External level
B) Conceptual level
C) Internal level
D) View level

Answer: C
Explanation: The internal level deals with storage structures, file organization, and access methods.

---

3. The conceptual level represents:
A) Individual user views
B) Physical storage details
C) Entire database content
D) Index structures

Answer: C
Explanation: Conceptual level provides a community view of the entire database, hiding storage details.

---

4. Multiple user views are supported at which level?
A) Internal level

B) Conceptual level
C) External level
D) Storage level

Answer: C
Explanation: External level provides different views for different users, hiding irrelevant data.

---

5. The mapping between conceptual and internal levels provides:
A) Logical data independence
B) Physical data independence
C) View independence
D) Schema independence

Answer: B
Explanation: Changes in storage structures don't affect conceptual schema due to conceptual-internal mapping.

---

6. External-conceptual mapping provides:
A) Physical data independence
B) Logical data independence
C) Storage independence
D) Access independence

Answer: B
Explanation: Changes to conceptual schema don't affect external views due to this mapping.

---

7. In three-schema architecture, the schema at the lowest level is:
A) External schema
B) Conceptual schema
C) Internal schema
D) Subschema

Answer: C

---

8. Which schema defines stored record types and access methods?
A) External schema
B) Conceptual schema
C) Internal schema
D) User schema

Answer: C

---

9. The number of external schemas in a database system can be:
A) Only one
B) Multiple
C) Zero
D) Exactly two

Answer: B
Explanation: Multiple external schemas can exist for different user groups.

---

10. Which level is also known as the logical level?
A) External level
B) Conceptual level
C) Internal level
D) Storage level

Answer: B

SECTION 2: CLIENT-SERVER ARCHITECTURE (Questions 26-45)

11. In two-tier client-server architecture, the client directly communicates with:
A) Another client
B) Database server
C) Application server
D) Web server

Answer: B

---

12. Which tier in three-tier architecture handles business logic?
A) Client tier
B) Middle tier
C) Database tier
D) Presentation tier

Answer: B

---

13. The main advantage of three-tier over two-tier architecture is:
A) Simplicity
B) Scalability
C) Lower cost
D) Faster queries

Answer: B
Explanation: Three-tier architecture provides better scalability, security, and load balancing.

---

14. In client-server architecture, the server provides:
A) User interface
B) Data services
C) Application logic
D) Display management

Answer: B

---

15. A fat client contains:
A) Only display logic
B) Application logic and display logic
C) Only database logic
D) Only network logic

Answer: B

---

16. A thin client typically contains:
A) Full application logic
B) Only presentation logic

B) Database management logic
D) Business rules

Answer: B

---

17. Which architecture is most suitable for web-based applications?
A) 1-tier
B) 2-tier
C) 3-tier
D) Centralized

Answer: C

---

18. In three-tier architecture, the database server is in which tier?
A) Tier 1
B) Tier 2
C) Tier 3
D) Distributed across all tiers

Answer: C

---

19. The application server resides in which tier of three-tier architecture?
A) Client tier
B) Middle tier
C) Data tier
D) Presentation tier

Answer: B

---

20. Which architecture uses ODBC/JDBC for database connectivity?
A) 1-tier
B) 2-tier
C) 3-tier
D) Centralized

Answer: B
Explanation: Two-tier architecture typically uses ODBC/JDBC for direct client-server communication.

---

21. The presentation tier in web applications is implemented using:
A) SQL
B) HTML/CSS/JavaScript
C) Java/Python
D) Database

Answer: B

---

22. In n-tier architecture, the number of tiers is:
A) Exactly 3
B) Exactly 2
C) More than 3
D) Exactly 1

Answer: C

---

23. Which component in client-server architecture manages user authentication?
A) Client only
B) Server only
C) Both client and server
D) Network

Answer: B

---

24. The main disadvantage of two-tier architecture is:
A) Too many tiers
B) Scalability issues
C) Complex programming
D) No security

Answer: B

---

25. In distributed client-server architecture, the database can be:
A) Only centralized
B) Only replicated
C) Partitioned or replicated
D) Only partitioned

Answer: C

SECTION 3: DBMS COMPONENTS & MODULES (Questions 46-70)

26. Which component is responsible for parsing and validating queries?
A) Storage manager
B) Query processor
C) Transaction manager
D) File manager

Answer: B

---

27. The query optimizer's main function is to:
A) Execute queries
B) Choose most efficient execution plan
C) Store data
D) Manage transactions

Answer: B

---

28. Which of the following is NOT a component of storage manager?
A) File manager
B) Buffer manager
C) Authorization manager
D) Query parser

Answer: D
Explanation: Query parser is part of query processor, not storage manager.

---

29. The buffer manager manages:
A) Disk space
B) Main memory
C) CPU cache
D) Network buffers

Answer: B

---

30. Which component ensures ACID properties?
A) Query processor
B) Storage manager
C) Transaction manager
D) File manager

Answer: C

---

31. The transaction manager consists of:
A) Concurrency control manager
B) Recovery manager
C) Both A and B
D) Query optimizer

Answer: C

---

32. The concurrency control manager handles:
A) Multiple simultaneous transactions
B) Query optimization
C) Index creation
D) View management

Answer: A

---

33. Which component is responsible for crash recovery?
A) Concurrency manager
B) Recovery manager
C) Buffer manager
D) File manager

Answer: B

---

34. The file manager is responsible for:
A) Managing disk space allocation
B) Parsing SQL
C) Optimizing queries
D) Managing users

Answer: A

---

35. The authorization manager handles:
A) Query execution
B) Access permissions
C) Data storage
D) Buffer management

Answer: B

---

36. Which component maintains the data dictionary?
A) DDL compiler
B) DML compiler
C) Query optimizer
D) Transaction manager

Answer: A

---

37. The DML compiler translates DML statements into:
A) Machine code
B) Query evaluation plans
C) Assembly language
D) Byte code

Answer: B

---

38. The query execution engine:
A) Parses queries
B) Executes query plans
C) Optimizes queries
D) Compiles DDL

Answer: B

---

39. Which component interacts with the operating system file manager?
A) Query processor
B) Storage manager
C) Transaction manager
D) DDL compiler

Answer: B

---

40. The preprocessor in DBMS:
A) Processes DDL statements
B) Removes embedded DML commands
C) Optimizes queries
D) Manages transactions

Answer: B
Explanation: The preprocessor strips DML commands from host language code.

---

41. The database catalog stores:
A) User data
B) Metadata
C) Indexes
D) Query results

Answer: B

---

42. Which information is NOT typically stored in the catalog?
A) Schema definitions
B) Integrity constraints

C) User data
D) Authorization information

Answer: C

---

43. The runtime database processor handles:
A) Query compilation
B) Privileged commands
C) Query execution at runtime
D) Catalog updates

Answer: C

---

44. Stored procedures are managed by:
A) Query processor
B) Runtime manager
C) Transaction manager
D) File manager

Answer: B

---

45. Which component manages triggers?
A) Query processor
B) Storage manager
C) Transaction manager
D) Runtime database processor

Answer: D

SECTION 4: DATA INDEPENDENCE (Questions 71-85)

46. Physical data independence allows changes to:
A) External schemas
B) Storage structures
C) Conceptual schema
D) User views

Answer: B

---

47. If we add a new index on a table, which independence is demonstrated?
A) Logical independence
B) Physical independence
C) Both
D) Neither

Answer: B

---

48. Logical data independence is harder to achieve because:
A) Storage is complex
B) Multiple views depend on conceptual schema
C) Hardware changes frequently
D) Users don't cooperate

Answer: B

---

49. Which mapping is modified when we change storage organization?
A) External-conceptual
B) Conceptual-internal
C) External-internal
D) View-storage

Answer: B

---

50. When we split a table into two tables, which independence helps existing applications?
A) Physical independence
B) Logical independence
C) Storage independence
D) Network independence

Answer: B

---

51. The ability to modify conceptual schema without changing external schemas is:
A) Physical data independence
B) Logical data independence
C) Schema transparency
D) View transparency

Answer: B

---

52. Data independence is achieved through:
A) Multiple schemas
B) Mappings between schema levels
C) Data abstraction
D) All of the above

Answer: D

---

53. Which of the following changes affects physical data independence?
A) Adding a new attribute
B) Changing data type
C) Changing file organization from heap to sequential
D) Creating a new view

Answer: C

---

54. The highest level of data abstraction is:
A) Physical level
B) Logical level
C) View level
D) Internal level

Answer: C

---

55. Which statement about data independence is FALSE?
A) It reduces application maintenance
B) It makes systems more adaptable
C) It increases data redundancy
D) It allows changes without rewriting programs

Answer: C

SECTION 5: CENTRALIZED & DISTRIBUTED ARCHITECTURES (Questions 86-100)

56. In a centralized DBMS architecture:
A) Data distributed across sites
B) All components on single machine
C) Multiple DBMS instances
D) Peer-to-peer processing

Answer: B

---

57. Which is NOT an advantage of centralized architecture?
A) Simple management
B) High availability
C) Easy security control
D) Single point of failure

Answer: B
Explanation: Centralized systems have a single point of failure, making them less available.

---

58. A homogeneous distributed database means:
A) Same hardware at all sites
B) Same DBMS software at all sites
C) Same data at all sites
D) Same users at all sites

Answer: B

---

59. In heterogeneous distributed databases:
A) Different hardware
B) Different DBMS software

C) Different data models
D) All of the above

Answer: D

---

60. Data fragmentation in distributed databases includes:
A) Horizontal fragmentation
B) Vertical fragmentation
C) Mixed fragmentation
D) All of the above

Answer: D

---

61. Horizontal fragmentation divides:
A) Attributes
B) Tuples/rows
C) Tables
D) Databases

Answer: B

---

62. Vertical fragmentation divides:
A) Rows
B) Columns/attributes
C) Records
D) Files

Answer: B

---

63. Data replication in distributed databases improves:
A) Availability
B) Performance
C) Reliability
D) All of the above

Answer: D

---

64. The main challenge in data replication is:
A) Storage cost
B) Maintaining consistency
C) Network delay
D) Security

Answer: B

---

65. Which transparency hides the location of data in distributed databases?
A) Fragmentation transparency
B) Location transparency
C) Replication transparency
D) Network transparency

Answer: B

---

66. Fragmentation transparency means:
A) User doesn't know data is fragmented
B) Data is not fragmented
C) Fragments are hidden from DBA
D) Fragments are replicated

Answer: A

---

67. The client in a distributed database can access data:
A) Only from local site
B) Only from remote sites
C) From multiple sites transparently
D) From one site only

Answer: C

---

68. Which architecture uses middleware to connect different databases?
A) Centralized
B) Federated

C) Standalone
D) Single-tier

Answer: B

---

69. A federated database system:
A) Integrates multiple autonomous databases
B) Uses single DBMS
C) Has no autonomy
D) Is always homogeneous

Answer: A

---

70. The degree of autonomy in federated databases refers to:
A) Data size
B) Local control
C) Query speed
D) Storage capacity

Answer: B

SECTION 6: PARALLEL DATABASE ARCHITECTURE (Questions 101-115)

71. Shared-memory parallel architecture:
A) Each processor has private memory
B) All processors share main memory
C) No memory sharing
D) Shared disk only

Answer: B

---

72. In shared-disk architecture:
A) Each node has private disk
B) All nodes share disks

C) No disk sharing
D) Memory and disk both shared

Answer: B

---

73. Shared-nothing architecture is characterized by:
A) All resources shared
B) Each node has private memory and disk
C) Shared memory only
D) Shared disk only

Answer: B

---

74. Which architecture offers best scalability?
A) Shared-memory
B) Shared-disk
C) Shared-nothing
D) Shared-everything

Answer: C
Explanation: Shared-nothing scales best as nodes can be added independently.

---

75. The main disadvantage of shared-memory architecture is:
A) High cost
B) Limited scalability
C) Complex programming
D) Poor performance

Answer: B

---

76. In parallel databases, inter-query parallelism means:
A) Multiple queries execute in parallel
B) Single query uses multiple processors
C) Queries execute sequentially
D) Only updates are parallel

Answer: A

---

77. Intra-query parallelism refers to:
A) Multiple queries running together
B) One query using multiple processors
C) Queries waiting in queue
D) Parallel backups

Answer: B

---

78. Which type of parallelism is best for decision support queries?
A) Inter-query
B) Intra-query
C) Transaction
D) Serial

Answer: B
Explanation: Complex analytical queries benefit from intra-query parallelism.

---

79. In parallel databases, speedup refers to:
A) Handling more transactions
B) Completing tasks faster with more resources
C) Storing more data
D) Better security

Answer: B

---

80. Scaleup in parallel databases means:
A) Faster queries
B) Handling larger workloads with more resources
C) Reducing cost
D) Improving security

Answer: B

SECTION 7: BUFFER MANAGEMENT & STORAGE (Questions 116-130)

81. The buffer manager uses which replacement strategy?
A) LRU
B) MRU
C) FIFO
D) Any of these

Answer: D

---

82. The pin/unpin commands in buffer management:
A) Allocate disk space
B) Lock buffer pages in memory
C) Create indexes
D) Drop tables

Answer: B

---

83. A dirty page in buffer is:
A) Empty page
B) Page with errors
C) Modified page not written to disk
D) Deleted page

Answer: C

---

84. The forced output of buffer pages is called:
A) Pinning
B) Flushing
C) Loading
D) Caching

Answer: B

---

85. Buffer manager interacts with:
A) Only file manager
B) Only transaction manager
C) Both file and transaction managers
D) Only query processor

Answer: C

---

86. The storage manager maps:
A) Logical to physical addresses
B) Queries to results
C) Users to permissions
D) Tables to indexes

Answer: A

---

87. Which storage component manages free space?
A) Buffer manager
B) File manager
C) Recovery manager
D) Transaction manager

Answer: B

---

88. Record management functions include:
A) Inserting records
B) Deleting records
C) Modifying records
D) All of the above

Answer: D

SECTION 8: TRANSACTION MANAGEMENT COMPONENTS (Questions 131-140)

89. The transaction manager maintains:
A) Transaction table
B) Lock table
C) Log buffer
D) All of the above

Answer: D

---

90. The lock table is managed by:
A) Recovery manager
B) Concurrency control manager
C) Buffer manager
D) File manager

Answer: B

---

91. The log manager writes records to:
A) Buffer pool
B) Log file
C) Data file
D) Index file

Answer: B

---

92. Checkpoint information is maintained by:
A) Query processor
B) Recovery manager
C) DDL compiler
D) DML preprocessor

Answer: B

---

93. The deadlock detector is part of:
A) Recovery manager
B) Concurrency control manager
C) Buffer manager
D) File manager

Answer: B

---

94. Transaction rollback is handled by:
A) Concurrency manager
B) Recovery manager
C) Buffer manager
D) Query processor

Answer: B

---

95. The transaction commit process involves:
A) Writing log records
B) Releasing locks
C) Flushing buffers
D) All of the above

Answer: D

SECTION 9: QUERY PROCESSING COMPONENTS (Questions 141-150)

96. The query processing pipeline order is:
A) Parsing → Optimization → Execution
B) Optimization → Parsing → Execution
C) Execution → Parsing → Optimization
D) Parsing → Execution → Optimization

Answer: A

---

97. The query parser checks for:
A) Syntax errors
B) Semantic errors
C) Both A and B
D) Performance issues

Answer: C

---

98. Query optimization uses:
A) System catalogs
B) Statistical information
C) Cost formulas
D) All of the above

Answer: D

---

99. An execution plan includes:
A) Access methods
B) Join methods
C) Operator order
D) All of the above

Answer: D

---

100. The query result cache stores:
A) Frequently accessed query results
B) All query results
C) Table definitions
D) Index definitions

Answer: A

SECTION 10: DATA DICTIONARY & SYSTEM CATALOG (Questions 151-165)

101. The system catalog is automatically updated by:
A) Users
B) DBA
C) DBMS
D) Application programs

Answer: C

---

102. Which information is stored in the catalog?
A) Table names
B) Column names
C) Index definitions
D) All of the above

Answer: D

---

103. The catalog itself is stored as:
A) Regular tables
B) Special files
C) Memory structures
D) Indexes only

Answer: A
Explanation: System catalog is typically stored as regular tables that DBMS manages.

---

104. Statistical information in catalog helps:
A) Query optimization
B) Data security
C) Transaction management
D) Recovery

Answer: A

---

105. The catalog can be queried using:
A) SQL
B) Only special commands
C) Only DBA tools
D) Assembly language

Answer: A

---

106. Which system view shows all tables in Oracle?
A) USER_TABLES
B) ALL_TABLES
C) DBA_TABLES
D) All of the above

Answer: D

---

107. In MySQL, the catalog information is in:
A) INFORMATION_SCHEMA
B) SYSTEM_CATALOG
C) MASTER_TABLES
D) DATA_DICTIONARY

Answer: A

---

108. Authorization information is stored in:
A) Catalog
B) Data files
C) Index files
D) Log files

Answer: A

SECTION 11: DATABASE MIDDLEWARE (Questions 166-175)

109. Database middleware provides:
A) Connectivity between applications and DBMS
B) User interface
C) Data storage
D) Backup services

Answer: A

---

110. ODBC stands for:
A) Open Database Connectivity

B) Object Database Connection
C) Oracle Database Connectivity
D) Open Data Bridge Connection

Answer: A

---

111. JDBC is used for connecting:
A) Java applications to databases
B) JavaScript to databases
C) Python to databases
D) C++ to databases

Answer: A

---

112. The ODBC architecture includes:
A) Application
B) Driver Manager
C) Driver
D) All of the above

Answer: D

---

113. A JDBC driver that converts JDBC calls to DBMS protocol is:
A) Type 1 driver
B) Type 2 driver
C) Type 3 driver
D) Type 4 driver

Answer: D

---

114. Connection pooling is managed by:
A) DBMS
B) Middleware
C) Application
D) Operating System

Answer: B

115. Which is NOT a function of database middleware?
A) Connection management
B) Query translation
C) Data storage
D) Result formatting

Answer: C

SECTION 12: ARCHITECTURE TRENDS (Questions 176-185)

116. Cloud database architecture provides:
A) Scalability on demand
B) Pay-per-use model
C) Managed services
D) All of the above

Answer: D

---

117. Database-as-a-Service (DBaaS) is:
A) On-premise database
B) Cloud database service
C) Desktop database
D) Embedded database

Answer: B

---

118. In-memory database architecture stores data primarily in:
A) Hard disk
B) RAM
C) SSD
D) Tape

Answer: B

---

119. NewSQL databases combine:
A) NoSQL scalability with ACID
B) Relational with hierarchical
C) Network with object-oriented
D) Centralized with distributed

Answer: A

---

120. Microservices architecture typically uses:
A) Single shared database
B) Database per service
C) No database
D) Centralized database only

Answer: B

SECTION 1: INTRODUCTION TO DATA MODELS (Questions 1-20)

1. A data model is:
A) A collection of data
B) A set of concepts to describe data structure, relationships, and constraints
C) A database management system
D) A programming language

Answer: B
Explanation: Data models provide a way to describe the structure, operations, and constraints of data.

---

2. Which of the following is NOT a category of data models?
A) High-level/conceptual models
B) Representational/models
C) Low-level/physical models
D) Execution models

Answer: D
Explanation: The three main categories are conceptual, representational, and physical models.

---

3. Conceptual data models provide:
A) Implementation details
B) Storage specifications
C) Concepts close to user perception
D) Index structures

Answer: C

---

4. Representational data models hide:
A) All data
B) Some storage details
C) User views
D) Relationships

Answer: B

---

5. Physical data models describe:
A) How data is stored
B) User requirements
C) Business rules
D) Entity relationships

Answer: A

---

6. The Entity-Relationship model is an example of:
A) Physical model
B) Conceptual model
C) Representational model
D) Implementation model

Answer: B

---

7. The relational model is an example of:
A) Conceptual model
B) Physical model
C) Representational model
D) Object model

Answer: C

---

8. Which model is closest to how users perceive data?
A) Physical model
B) Conceptual model
C) Internal model
D) Storage model

Answer: B

---

9. The hierarchical model organizes data as:
A) Tables
B) Tree structures
C) Graphs
D) Objects

Answer: B

---

10. The network model organizes data as:
A) Trees
B) Tables
C) Graphs/lattices
D) Objects

Answer: C

SECTION 2: HIERARCHICAL MODEL (Questions 21-35)

11. In hierarchical model, the relationship between parent and child is:
A) Many-to-many

B) One-to-many
C) One-to-one
D) Many-to-one

Answer: B

---

12. Which company developed IMS (Information Management System)?
A) Oracle
B) IBM
C) Microsoft
D) DEC

Answer: B

---

13. In hierarchical model, a child segment can have:
A) Multiple parents
B) Only one parent
C) No parent
D) Any number of parents

Answer: B

---

14. The hierarchical model uses which data structure?
A) Linked list
B) Tree
C) Graph
D) Array

Answer: B

---

15. The root segment in hierarchical model:
A) Has no parent
B) Has multiple parents
C) Has no children
D) Is always a leaf

Answer: A

---

16. A leaf segment in hierarchical model:
A) Has no parent
B) Has no children
C) Has multiple parents
D) Is the root

Answer: B

---

17. The hierarchical model supports which type of relationship?
A) M:N
B) 1:N with strict parent-child
C) Only 1:1
D) Any relationship

Answer: B

---

18. The main disadvantage of hierarchical model is:
A) Fast access
B) Redundancy due to M:N limitations
C) Simple structure
D) Data independence

Answer: B
Explanation: M:N relationships require duplication leading to redundancy.

---

19. In hierarchical model, access paths are:
A) Flexible
B) Predetermined
C) Dynamic
D) Optional

Answer: B

---

20. Which operation is used to traverse in hierarchical model?
A) JOIN

B) GET NEXT
C) SELECT
D) PROJECT

Answer: B

---

21. The hierarchical model was primarily used in:
A) Web applications
B) Mainframe systems
C) Mobile apps
D) Desktop databases

Answer: B

---

22. Virtual records in hierarchical model are used for:
A) Storing data
B) Representing relationships without duplication
C) Indexing
D) Security

Answer: B

---

23. The hierarchical model's data manipulation language includes:
A) GET, GET NEXT, GET UNIQUE
B) SELECT, INSERT, UPDATE
C) FIND, FETCH, RETRIEVE
D) QUERY, SEARCH, SCAN

Answer: A

---

24. In IMS, a database record is called:
A) Segment
B) Tuple
C) Row
D) Object

Answer: A

---

25. The hierarchical model lacks:
A) Data independence
B) Parent-child relationships
C) Tree structure
D) Fast access

Answer: A

SECTION 3: NETWORK MODEL (Questions 36-55)

26. The network model was standardized by:
A) ANSI
B) CODASYL
C) ISO
D) IEEE

Answer: B

---

27. In network model, the basic data structure is:
A) Table
B) Set
C) Tree
D) Object

Answer: B

---

28. A set in network model consists of:
A) Owner record and member record
B) Two tables
C) Index and data
D) Parent and children only

Answer: A

---

29. Which type of relationships does network model support?
A) Only 1:1
B) Only 1:N
C) M:N and others
D) Only hierarchical

Answer: C

---

30. In network model, a member record can have:
A) Only one owner
B) Multiple owners
C) No owner
D) Fixed owner

Answer: B
Explanation: Network model allows multiple owners (sets), enabling M:N relationships.

---

31. The network model was also known as:
A) Relational model
B) CODASYL model
C) Hierarchical model
D) Object model

Answer: B

---

32. DBTG stands for:
A) Database Task Group
B) Data Base Transaction Group
C) Distributed Base Task Group
D) Data Block Transfer Group

Answer: A

---

33. The network model's schema is defined using:
A) SQL
B) DDL (Data Definition Language)

C) Java
D) XML

Answer: B

---

34. Which of the following is NOT a feature of network model?
A) Sets with owner and members
B) Many-to-many relationships
C) Single parent constraint
D) Network of records

Answer: C
Explanation: Multiple parents are allowed in network model.

---

35. In network model, the access path must be:
A) Hidden from programmers
B) Known to programmers
C) Automatically chosen
D) Dynamic

Answer: B

---

36. The network model's main advantage over hierarchical is:
A) Simplicity
B) Better M:N relationship handling
C) Less complexity
D) Faster queries

Answer: B

---

37. A CODASYL set represents:
A) A 1:N relationship
B) A table
C) An index
D) A view

Answer: A

---

38. The currency indicators in network model track:
A) Current record position
B) Money values
C) Transaction amounts
D) Index values

Answer: A

---

39. Which of the following is a DML command in network model?
A) FIND
B) SELECT
C) JOIN
D) GROUP BY

Answer: A

---

40. The network model's record-at-a-time processing means:
A) Multiple records processed together
B) One record processed at a time
C) All records processed
D) No processing

Answer: B

---

41. The schema in network model describes:
A) Storage details
B) Logical structure
C) User views
D) Indexes

Answer: B

---

42. The subschema in network model corresponds to:
A) Physical storage
B) User view

C) System catalog
D) Index structure

Answer: B

---

43. Which model influenced the development of modern OODBMS?
A) Hierarchical
B) Network
C) Relational
D) Flat file

Answer: B

---

44. The network model's complexity arises from:
A) Simple structure
B) Pointer-based navigation
C) Table structure
D) Object orientation

Answer: B

---

45. In network model, deletion of an owner record requires:
A) Automatic deletion of members
B) Handling member records first
C) No special handling
D) System restart

Answer: B

SECTION 4: RELATIONAL MODEL (Questions 56-80)

46. The relational model was proposed by:
A) Charles Bachman
B) Edgar F. Codd
C) Peter Chen
D) James Martin

Answer: B

---

47. In which year was the relational model proposed?
A) 1960
B) 1970
C) 1980
D) 1990

Answer: B

---

48. A relation in relational model is represented as:
A) Tree
B) Table
C) Graph
D) Set

Answer: B

---

49. A row in a relational table is called:
A) Attribute
B) Tuple
C) Domain
D) Key

Answer: B

---

50. A column in a relational table is called:
A) Tuple
B) Attribute
C) Record
D) Field

Answer: B

---

51. The number of attributes in a relation is called:
A) Cardinality
B) Degree
C) Arity
D) Both B and C

Answer: D

---

52. The number of tuples in a relation is called:
A) Cardinality
B) Degree
C) Size
D) Dimension

Answer: A

---

53. A domain in relational model is:
A) A table
B) Set of allowable values for an attribute
C) A key
D) A relationship

Answer: B

---

54. Which property ensures that each row is unique?
A) Domain constraint
B) Key constraint
C) Referential constraint
D) Check constraint

Answer: B

---

55. A superkey is:
A) A set of attributes that uniquely identifies a tuple
B) The primary key
C) A foreign key
D) An index

Answer: A

---

56. A candidate key is:
A) A superkey with no proper subset that is a superkey
B) Any set of attributes
C) Always a single attribute
D) Never used as primary key

Answer: A

---

57. The primary key is:
A) Any candidate key
B) A candidate key chosen to identify tuples
C) Always the first attribute
D) A foreign key

Answer: B

---

58. A foreign key is:
A) A key from another table
B) A set of attributes referencing another relation's primary key
C) Not a key at all
D) Always null

Answer: B

---

59. Referential integrity ensures:
A) All foreign key values match a primary key value
B) All attributes have values
C) No duplicates
D) Fast access

Answer: A

---

60. Entity integrity constraint states that:
A) Foreign keys can be null
B) Primary key attributes cannot be null
C) All attributes can be null
D) No constraints on keys

Answer: B

---

61. Which of the following is a relational algebra operation?
A) SELECT
B) PROJECT
C) JOIN
D) All of the above

Answer: D

---

62. The SELECT operation in relational algebra:
A) Chooses columns
B) Chooses rows
C) Combines tables
D) Renames attributes

Answer: B

---

63. The PROJECT operation in relational algebra:
A) Chooses rows
B) Chooses columns
C) Joins tables
D) Divides relations

Answer: B

---

64. The JOIN operation combines:
A) Two relations based on a condition
B) Rows from same relation
C) Attributes from same relation
D) Keys only

Answer: A

---

65. The DIVISION operation is used for queries involving:
A) All conditions
B) Any condition
C) The phrase "for all"
D) Existence

Answer: C

---

66. Relational calculus is:
A) Procedural
B) Non-procedural
C) Implementation-oriented
D) Storage-oriented

Answer: B

---

67. Tuple relational calculus uses:
A) Variables that range over tuples
B) Variables that range over attributes
C) Domain variables
D) Table variables

Answer: A

---

68. Domain relational calculus uses:
A) Tuple variables
B) Domain variables
C) Relation variables
D) Attribute variables

Answer: B

---

69. Codd's 12 rules define:
A) Normalization rules
B) What constitutes a fully relational DBMS
C) SQL standards
D) Indexing methods

Answer: B

---

70. Which Codd's rule requires that data be represented only as table values?
A) Rule 1: Information rule
B) Rule 2: Guaranteed access rule
C) Rule 3: Systematic treatment of nulls
D) Rule 4: Active online catalog

Answer: A

SECTION 5: ENTITY-RELATIONSHIP MODEL (Questions 81-100)

71. The ER model was proposed by:
A) E.F. Codd
B) Peter Chen
C) Charles Bachman
D) Ted Codd

Answer: B

---

72. In ER model, an entity is:
A) A table
B) A real-world object distinguishable from others
C) A relationship
D) An attribute

Answer: B

---

73. An entity set is:
A) Collection of similar entities
B) A single entity
C) A relationship
D) A key

Answer: A

---

74. Attributes in ER model describe:
A) Relationships
B) Properties of entities
C) Keys only
D) Constraints

Answer: B

---

75. A simple attribute is:
A) Divisible
B) Atomic/not divisible
C) Multi-valued
D) Derived

Answer: B

---

76. A composite attribute:
A) Cannot be divided
B) Can be divided into smaller attributes
C) Has multiple values
D) Is calculated

Answer: B

---

77. A multi-valued attribute:
A) Has single value
B) Can have multiple values for an entity
C) Is always composite
D) Is always derived

Answer: B

---

78. A derived attribute:
A) Is stored in database
B) Is calculated from other attributes
C) Is always a key
D) Cannot be null

Answer: B

---

79. A relationship type in ER model has:
A) Degree
B) Cardinality
C) Participation
D) All of the above

Answer: D

---

80. The degree of a relationship type is:
A) Number of attributes
B) Number of participating entity types
C) Number of relationships
D) Number of keys

Answer: B

---

81. A unary relationship involves:
A) One entity type
B) Two entity types
C) Three entity types
D) Multiple entity types

Answer: A

---

82. A binary relationship involves:
A) One entity type
B) Two entity types
C) Three entity types
D) Four entity types

Answer: B

---

83. Cardinality ratio represents:
A) Minimum participation
B) Maximum participation
C) Average participation
D) Total participation

Answer: B

---

84. The 1:N cardinality means:
A) One entity in first set related to many in second
B) Many entities related to many
C) One-to-one relationship
D) No relationship

Answer: A

---

85. Participation constraint can be:
A) Total
B) Partial
C) Both A and B
D) Neither

Answer: C

---

86. Total participation means:
A) Every entity participates
B) Some entities participate
C) No participation
D) Optional participation

Answer: A

---

87. A weak entity:
A) Has its own key
B) Depends on another entity for identification
C) Has no attributes
D) Is always total

Answer: B

---

88. The identifying relationship connects:
A) Two strong entities
B) Weak entity with owner entity
C) Two weak entities
D) Attributes

Answer: B

---

89. A partial key in weak entity:
A) Uniquely identifies weak entities
B) Identifies weak entities with owner's key
C) Is always a foreign key
D) Is not needed

Answer: B

---

90. Which symbol represents total participation in ER diagram?
A) Single line
B) Double line
C) Dashed line
D) Arrow

Answer: B

SECTION 6: OBJECT-ORIENTED & OBJECT-RELATIONAL MODELS (Questions 101-120)

91. Object-oriented data model incorporates:
A) Encapsulation
B) Inheritance
C) Polymorphism
D) All of the above

Answer: D

---

92. An object has:
A) State and behavior
B) Only state
C) Only behavior
D) Neither

Answer: A

---

93. Encapsulation means:
A) Hiding implementation details
B) Showing all details
C) No methods
D) Public attributes

Answer: A

---

94. Inheritance allows:
A) Creating new classes from existing ones
B) Deleting classes
C) Modifying all classes
D) No relationships

Answer: A

---

95. Polymorphism means:
A) Same operation different implementations
B) Different operations same name
C) No operations
D) Fixed operations

Answer: A

---

96. ODL stands for:
A) Object Definition Language
B) Object Data Language
C) Object Description Language
D) Object Development Language

Answer: A

---

97. OQL stands for:
A) Object Query Language
B) Object Quality Language
C) Object Quick Language
D) Object Quantitative Language

Answer: A

---

98. Object-relational model extends:
A) Hierarchical model
B) Relational model
C) Network model
D) Flat files

Answer: B

---

99. User-defined types in object-relational model allow:
A) Custom data types
B) Only built-in types
C) No new types
D) Only simple types

Answer: A

---

100. Methods in object-relational databases:
A) Cannot be stored
B) Can be stored with data
C) Are not allowed
D) Only in applications

Answer: B

SECTION 7: SEMISTRUCTURED DATA MODEL (Questions 121-135)

101. Semistructured data has:
A) Rigid schema
B) Flexible schema
C) No schema
D) Fixed structure

Answer: B

---

102. XML is an example of:
A) Structured data model
B) Semistructured data model
C) Network model
D) Hierarchical model

Answer: B

---

103. JSON is primarily used for:
A) Data exchange
B) Only storage
C) Only queries
D) Only indexing

Answer: A

---

104. In semistructured data, schema is:
A) Fixed
B) Self-describing
C) External
D) Not needed

Answer: B

---

105. Which is a characteristic of semistructured data?
A) Irregular structure
B) No fixed schema
C) Nested elements
D) All of the above

Answer: D

---

106. XPath is used to:
A) Navigate XML documents
B) Store XML
C) Validate XML
D) Transform XML

Answer: A

---

107. XQuery is used for:
A) Querying XML data
B) Storing XML
C) Validating XML
D) Displaying XML

Answer: A

---

108. DTD stands for:
A) Document Type Definition
B) Data Type Definition
C) Document Transfer Definition
D) Data Transfer Definition

Answer: A

---

109. XML Schema is used for:
A) Defining structure of XML documents
B) Storing data
C) Querying data
D) Transforming data

Answer: A

---

110. JSON Schema is used for:
A) Validating JSON structure
B) Storing JSON
C) Querying JSON
D) Formatting JSON

Answer: A

SECTION 8: COMPARISON OF DATA MODELS (Questions 136-150)

111. Which model provides the highest data independence?
A) Hierarchical
B) Network
C) Relational
D) Flat file

Answer: C

---

112. Which model is most user-friendly?
A) Network
B) Hierarchical
C) Relational
D) CODASYL

Answer: C

---

113. Which model requires navigation through pointers?
A) Relational
B) Network and Hierarchical
C) Object-relational
D) Semistructured

Answer: B

---

114. Which model supports set-at-a-time processing?
A) Network
B) Hierarchical
C) Relational
D) All of the above

Answer: C

---

115. Which model has the simplest conceptual representation?
A) Network
B) Hierarchical
C) Relational
D) Object-oriented

Answer: C

---

116. Which model is best suited for complex data with inheritance?
A) Relational
B) Object-oriented
C) Network
D) Hierarchical

Answer: B

---

117. Which model is most widely used today?
A) Hierarchical
B) Network
C) Relational
D) Object-oriented

Answer: C

---

118. Which model handles M:N relationships most naturally?
A) Hierarchical
B) Network
C) Relational (through tables)
D) Flat file

Answer: C

---

119. Which model requires predefined access paths?
A) Relational
B) Hierarchical and Network
C) Object-relational
D) Semistructured

Answer: B

---

120. Which model provides the best data abstraction?
A) Hierarchical
B) Network
C) Relational
D) Physical

Answer: C

SECTION 9: ADVANCED & MODERN DATA MODELS (Questions 151-165)

121. NoSQL data models include:
A) Key-value stores
B) Document stores
C) Column-family stores
D) All of the above

Answer: D

---

122. Key-value stores are optimized for:
A) Complex queries
B) Simple lookups
C) Joins
D) Transactions

Answer: B

---

123. Document stores store data as:
A) Tables
B) Documents (JSON, BSON, etc.)
C) Graphs
D) Trees

Answer: B

---

124. MongoDB is an example of:
A) Key-value store
B) Document store
C) Column-family store
D) Graph database

Answer: B

---

125. Column-family stores are inspired by:
A) Google BigTable
B) IBM IMS
C) Oracle
D) SQL Server

Answer: A

---

126. Cassandra is an example of:
A) Document store
B) Column-family store
C) Graph database
D) Key-value store

Answer: B

---

127. Graph databases use:
A) Tables
B) Nodes and edges
C) Documents
D) Columns

Answer: B

---

128. Neo4j is an example of:
A) Document store
B) Graph database
C) Key-value store
D) Column-family store

Answer: B

---

129. The property graph model includes:
A) Nodes with properties
B) Relationships with properties
C) Labels
D) All of the above

Answer: D

---

130. RDF stands for:
A) Resource Description Framework
B) Rapid Data Format
C) Relational Data Framework
D) Record Definition Format

Answer: A

---

131. Graph databases excel at:
A) Simple lookups
B) Complex relationship queries
C) Aggregations
D) Full-text search

Answer: B

---

132. Time-series data models are optimized for:
A) Text data
B) Time-stamped data
C) Binary data
D) Spatial data

Answer: B

---

133. InfluxDB is an example of:
A) Graph database
B) Time-series database
C) Document store
D) Key-value store

Answer: B

---

134. Spatial data models handle:
A) Geographic data
B) Text data
C) Time data
D) Binary data

Answer: A

---

135. PostGIS extends PostgreSQL for:
A) Time-series data
B) Spatial data
C) Graph data
D) Document data

Answer: B

SECTION 10: DATA MODEL CONSTRAINTS (Questions 166-175)

136. Inherent constraints are:
A) Defined by users
B) Implicit in data model
C) Added by DBA
D) Application-based

Answer: B

---

137. Schema-based constraints are defined:
A) In the schema using DDL
B) Implicitly
C) At runtime
D) By applications

Answer: A

---

138. Domain constraints specify:
A) Allowable values for attributes
B) Table names
C) Index structures
D) Storage locations

Answer: A

---

139. Key constraints ensure:
A) Uniqueness of tuples
B) Referential integrity
C) Domain validity
D) Fast access

Answer: A

---

140. NOT NULL is an example of:
A) Key constraint
B) Domain constraint
C) Referential constraint
D) Assertion

Answer: B

SECTION 1: INTRODUCTION TO RELATIONAL MODEL (Questions 1-25)

1. The relational model was proposed by:
A) Charles Bachman
B) Edgar F. Codd
C) Peter Chen
D) James Martin

Answer: B
Explanation: E.F. Codd proposed the relational model in 1970 while working at IBM.

---

2. In which year was the relational model first proposed?
A) 1960
B) 1970
C) 1980
D) 1990

Answer: B

---

3. A relation in the relational model is represented as:
A) Tree
B) Table
C) Graph
D) Linked list

Answer: B

---

4. The term "relation" in mathematics refers to:
A) A function
B) A subset of Cartesian product
C) A matrix
D) A vector

Answer: B
Explanation: In mathematics, a relation is a subset of the Cartesian product of domains.

---

5. Which of the following is NOT a characteristic of a relation?
A) Each row is unique
B) Order of rows matters
C) Each column has a unique name
D) All values in a column are from same domain

Answer: B
Explanation: In relational model, order of rows is immaterial.

SECTION 2: BASIC TERMINOLOGY (Questions 26-45)

6. A row in a relational table is formally called:
A) Attribute
B) Tuple
C) Domain
D) Key

Answer: B

---

7. A column in a relational table is formally called:
A) Tuple
B) Attribute
C) Record
D) Field

Answer: B

---

8. The number of attributes in a relation is called its:
A) Cardinality
B) Degree
C) Arity
D) Both B and C

Answer: D

---

9. The number of tuples in a relation is called its:
A) Cardinality
B) Degree
C) Size
D) Dimension

Answer: A

---

10. A relation schema R(A1, A2, ..., An) defines:
A) The structure of the relation
B) The actual data
C) The indexes
D) The storage location

Answer: A

---

11. A relation instance refers to:
A) The schema definition
B) The set of tuples at a particular time
C) The database itself
D) The query

Answer: B

---

12. The Cartesian product of two sets A and B is:
A) Set of ordered pairs (a,b) where a∈A, b∈B
B) Set of all elements in A and B
C) Intersection of A and B
D) Union of A and B

Answer: A

---

13. A domain in relational model is:
A) A table
B) Set of allowable values for an attribute
C) A key
D) A relationship

Answer: B

---

14. Which of the following is true about domains?
A) Domains must be atomic
B) Domains can be composite
C) Domains are always integer
D) Domains cannot be specified

Answer: A
Explanation: Relational model requires atomic (simple) domains.

---

15. A relation is in first normal form (1NF) if:
A) It has a primary key
B) All attributes are atomic
C) It has no duplicate rows
D) All columns are from same domain

Answer: B

SECTION 3: KEYS AND CONSTRAINTS (Questions 46-70)

16. A superkey is:
A) A set of attributes that uniquely identifies a tuple
B) The primary key
C) A foreign key
D) An index

Answer: A

---

17. A candidate key is:
A) A superkey with no proper subset that is a superkey
B) Any set of attributes
C) Always a single attribute
D) Never used as primary key

Answer: A

---

18. A candidate key must satisfy which properties?
A) Uniqueness and minimality
B) Only uniqueness
C) Only minimality
D) Referential integrity

Answer: A

---

19. The primary key is:
A) Any candidate key

B) A candidate key chosen to identify tuples
C) Always the first attribute
D) A foreign key

Answer: B

---

20. A foreign key is:
A) A key from another table
B) A set of attributes referencing another relation's primary key
C) Not a key at all
D) Always null

Answer: B

---

21. Referential integrity ensures:
A) All foreign key values match a primary key value or are null
B) All attributes have values
C) No duplicates exist
D) Fast access to data

Answer: A

---

22. Entity integrity constraint states that:
A) Foreign keys can be null
B) Primary key attributes cannot be null
C) All attributes can be null
D) No constraints on keys

Answer: B

---

23. A composite key is:
A) A key with multiple attributes
B) A foreign key
C) An alternate key
D) A superkey

Answer: A

---

24. A simple key consists of:
A) Multiple attributes
B) Single attribute
C) Foreign key only
D) Composite attributes

Answer: B

---

25. An alternate key is:
A) A candidate key not chosen as primary key
B) The primary key
C) A foreign key
D) Any superkey

Answer: A

---

26. A surrogate key is:
A) A natural key
B) An artificially generated unique identifier
C) A composite key
D) A foreign key

Answer: B
Explanation: Surrogate keys are system-generated (e.g., auto-increment columns).

---

27. Which constraint ensures that attribute values come from a specific set?
A) Key constraint
B) Domain constraint
C) Referential constraint
D) Check constraint

Answer: B

---

28. CHECK constraint is used for:
A) Defining primary keys

B) Specifying condition on attribute values
C) Creating foreign keys
D) Creating indexes

Answer: B

---

29. The UNIQUE constraint ensures:
A) No null values
B) All values are distinct
C) Referential integrity
D) Domain integrity

Answer: B

---

30. NOT NULL constraint ensures:
A) Attribute must have a value
B) Attribute can be null
C) Values are unique
D) Values are from a domain

Answer: A

SECTION 4: RELATIONAL ALGEBRA OPERATIONS (Questions 71-95)

31. Relational algebra is:
A) A procedural query language
B) A non-procedural query language
C) A programming language
D) A data definition language

Answer: A

---

32. The SELECT operation in relational algebra:
A) Chooses columns (vertical subset)

B) Chooses rows (horizontal subset)
C) Combines tables
D) Renames attributes

Answer: B

---

33. The PROJECT operation in relational algebra:
A) Chooses rows
B) Chooses columns
C) Joins tables
D) Divides relations

Answer: B

---

34. Which symbol is used for SELECT operation?
A) π (pi)
B) σ (sigma)
C) ⋈ (bowtie)
D) ρ (rho)

Answer: B

---

35. Which symbol is used for PROJECT operation?
A) σ (sigma)
B) π (pi)
C) ⋈ (bowtie)
D) ρ (rho)

Answer: B

---

36. The RENAME operation uses which symbol?
A) σ
B) π
C) ρ
D) ⋈

Answer: C

---

37. The UNION operation requires:
A) Compatible relations (same degree and domain compatibility)
B) Any two relations
C) Relations with primary keys
D) Relations with indexes

Answer: A

---

38. Which operation returns all tuples that are in both relations?
A) UNION
B) INTERSECTION
C) DIFFERENCE
D) CARTESIAN PRODUCT

Answer: B

---

39. The CARTESIAN PRODUCT combines:
A) Tuples from two relations
B) Attributes from same relation
C) Keys only
D) Indexes

Answer: A

---

40. The JOIN operation is a combination of:
A) SELECT and PROJECT
B) CARTESIAN PRODUCT and SELECT
C) UNION and INTERSECTION
D) DIFFERENCE and UNION

Answer: B

---

41. NATURAL JOIN eliminates:
A) Duplicate tuples
B) Duplicate attributes

C) Null values
D) Foreign keys

Answer: B
Explanation: Natural join removes duplicate join attributes.

---

42. THETA JOIN uses:
A) Equality condition only
B) Any comparison condition
C) No condition
D) Only less than condition

Answer: B

---

43. EQUIJOIN is a theta join with:
A) Any condition
B) Equality condition
C) No condition
D) Greater than condition

Answer: B

---

44. OUTER JOIN preserves:
A) Only matching tuples
B) Non-matching tuples from one or both relations
C) No tuples
D) Only null values

Answer: B

---

45. LEFT OUTER JOIN preserves:
A) All tuples from left relation
B) All tuples from right relation
C) All tuples from both relations
D) Only matching tuples

Answer: A

---

46. RIGHT OUTER JOIN preserves:
A) All tuples from left relation
B) All tuples from right relation
C) All tuples from both relations
D) Only matching tuples

Answer: B

---

47. FULL OUTER JOIN preserves:
A) All tuples from left relation only
B) All tuples from right relation only
C) All tuples from both relations
D) Only matching tuples

Answer: C

---

48. The DIVISION operation is used for queries involving:
A) "All" conditions
B) "Any" conditions
C) "Exists" conditions
D) "Not" conditions

Answer: A
Explanation: Division handles "for all" type queries.

---

49. Which operation is not primitive (can be expressed using others)?
A) SELECT
B) PROJECT
C) JOIN
D) UNION

Answer: C
Explanation: JOIN can be expressed as Cartesian product + SELECT.

---

50. The AGGREGATE operations include:
A) SUM, COUNT, AVG
B) SELECT, PROJECT
C) JOIN, UNION
D) DIVIDE, RENAME

Answer: A

SECTION 5: RELATIONAL CALCULUS (Questions 96-110)

51. Relational calculus is:
A) Procedural
B) Non-procedural/declarative
C) Implementation-oriented
D) Storage-oriented

Answer: B

---

52. Tuple relational calculus uses:
A) Variables that range over tuples
B) Variables that range over attributes
C) Domain variables
D) Table variables

Answer: A

---

53. Domain relational calculus uses:
A) Tuple variables
B) Domain variables (attribute values)
C) Relation variables
D) Attribute variables

Answer: B

---

54. Which quantifier is used in tuple relational calculus?
A) FOR ALL (∀)

B) EXISTS (∃)
C) Both A and B
D) Neither

Answer: C

---

55. A tuple relational calculus expression has the form:
A) { t | Condition(t) }
B) { a1, a2 | Condition }
C) SELECT * FROM table
D) π attributes (table)

Answer: A

---

56. A domain relational calculus expression has the form:
A) { t | Condition(t) }
B) { <a1, a2, ..., an> | Condition(a1, a2, ..., an) }
C) { * | Condition }
D) { relation | Condition }

Answer: B

---

57. Relational calculus is equivalent in expressive power to:
A) Relational algebra
B) Hierarchical model
C) Network model
D) File systems

Answer: A
Explanation: Relational algebra and calculus are equivalent (Codd's theorem).

---

58. Safety of expressions in relational calculus means:
A) Expression returns finite results
B) Expression is secure
C) Expression is fast
D) Expression has no errors

Answer: A
Explanation: Safe expressions guarantee finite results.

SECTION 6: CODD'S RULES (Questions 111-125)

59. How many rules define a fully relational DBMS according to Codd?
A) 10
B) 12
C) 15
D) 20

Answer: B

---

60. Rule 1 (Information rule) states:
A) Data represented only as table values
B) Access via combination of table name, key, and column
C) Systematic treatment of nulls
D) Active online catalog

Answer: A

---

61. Rule 2 (Guaranteed access rule) requires:
A) Access to each data via table, key, and column
B) Data compression
C) Encryption
D) Indexing

Answer: A

---

62. Rule 3 (Systematic treatment of nulls) requires:
A) No nulls allowed
B) Consistent handling of missing information
C) Nulls as zero
D) Nulls as spaces

Answer: B

---

63. Rule 4 (Active online catalog) requires:
A) Catalog accessible via same query language
B) Catalog on paper
C) Catalog only for DBA
D) No catalog needed

Answer: A

---

64. Rule 5 (Comprehensive data sublanguage) requires:
A) Multiple languages
B) One language supporting all operations
C) No query language
D) Only programming languages

Answer: B

---

65. Rule 6 (View updating rule) requires:
A) All views updatable
B) All theoretically updatable views should be updatable
C) No views updatable
D) Only base tables updatable

Answer: B

---

66. Rule 8 (Physical data independence) requires:
A) Changes to storage don't affect applications
B) Changes to tables don't affect storage
C) No changes allowed
D) Applications must change with storage

Answer: A

---

67. Rule 9 (Logical data independence) requires:
A) Changes to tables don't affect applications
B) Changes to storage don't affect applications

C) No logical changes allowed
D) Applications must change

Answer: A

---

68. Rule 11 (Distribution independence) requires:
A) Applications unaffected by distribution
B) Data must be distributed
C) No distribution allowed
D) Only centralized databases

Answer: A

---

69. Rule 12 (Nonsubversion rule) requires:
A) Cannot bypass integrity rules
B) Can bypass integrity rules
C) No integrity rules
D) Only application-level integrity

Answer: A

SECTION 7: INTEGRITY CONSTRAINTS (Questions 126-140)

70. Domain integrity ensures:
A) Values are from defined domain
B) Primary keys are unique
C) Foreign keys reference valid values
D) Business rules are followed

Answer: A

---

71. Entity integrity ensures:
A) Each row is unique entity
B) Each table has a primary key
C) No duplicate rows
D) All of the above

Answer: D

---

72. Referential integrity ensures:
A) Foreign key values reference existing primary keys
B) All foreign keys are non-null
C) Foreign keys are unique
D) Foreign keys are indexed

Answer: A

---

73. A referential integrity constraint involves:
A) One table only
B) Two or more tables
C) No tables
D) Only indexes

Answer: B

---

74. The action "CASCADE" for foreign key means:
A) Delete/update propagates to referencing rows
B) Prevent deletion
C) Set foreign key to null
D) Do nothing

Answer: A

---

75. The action "SET NULL" for foreign key means:
A) Delete parent, set child FK to null
B) Delete child only
C) Prevent deletion
D) Cascade deletion

Answer: A

---

76. The action "RESTRICT" for foreign key means:
A) Prevent deletion if child exists
B) Delete anyway
C) Set to default
D) Cascade

Answer: A

---

77. "NO ACTION" in SQL foreign key means:
A) Similar to RESTRICT
B) No constraint
C) Allow any action
D) Delete all

Answer: A

---

78. Assertions are:
A) Complex constraints involving multiple tables
B) Simple column constraints
C) Index definitions
D) View definitions

Answer: A

---

79. Triggers are used for:
A) Enforcing complex business rules
B) Creating indexes
C) Defining views
D) Querying data

Answer: A

SECTION 8: RELATIONAL DATABASE DESIGN (Questions 141-155)

80. A relation schema R is in 1NF if:
A) All attributes are atomic
B) It has no transitive dependencies
C) It has a candidate key
D) All non-key attributes depend on the key

Answer: A

---

81. Functional dependency X → Y means:
A) X determines Y uniquely
B) Y determines X
C) X and Y are related
D) X equals Y

Answer: A

---

82. A functional dependency is trivial if:
A) Y is subset of X
B) X is subset of Y
C) X = Y
D) X and Y are unrelated

Answer: A

---

83. Armstrong's axioms are used for:
A) Computing closure of functional dependencies
B) Creating indexes
C) Query optimization
D) Data encryption

Answer: A

---

84. The closure of a set of attributes F+ is:
A) All FDs that can be derived from F
B) All attributes in the relation
C) The primary key
D) All foreign keys

Answer: A

---

85. A candidate key can be found by computing:
A) Attribute closure
B) FD closure
C) Relation cardinality
D) Degree of relation

Answer: A

---

86. Normalization is the process of:
A) Removing redundancy and anomalies
B) Adding redundancy
C) Creating indexes
D) Encrypting data

Answer: A

---

87. Anomalies in database include:
A) Insertion anomaly
B) Deletion anomaly
C) Update anomaly
D) All of the above

Answer: D

---

88. 2NF is violated if there is:
A) Partial dependency
B) Transitive dependency
C) Multivalued dependency
D) Join dependency

Answer: A

---

89. 3NF is violated if there is:
A) Partial dependency
B) Transitive dependency
C) Multivalued dependency
D) Join dependency

Answer: B

---

90. BCNF is stronger than 3NF because:
A) Every determinant must be a candidate key
B) No transitive dependencies
C) No partial dependencies
D) All attributes atomic

Answer: A

SECTION 9: VIEWS IN RELATIONAL DATABASES (Questions 156-165)

91. A view is:
A) A virtual table based on query result
B) A physical table
C) An index
D) A constraint

Answer: A

---

92. Views provide:
A) Security by hiding data
B) Complexity reduction
C) Logical data independence
D) All of the above

Answer: D

---

93. Materialized views store:
A) Query results physically
B) Only query definition
C) Indexes only
D) Constraints only

Answer: A

---

94. Updatable views must be:
A) Based on single table without aggregates
B) Based on multiple tables
C) With GROUP BY clause
D) With DISTINCT clause

Answer: A

---

95. WITH CHECK OPTION in views:
A) Ensures modifications meet view condition
B) Checks syntax
C) Verifies permissions
D) Optimizes queries

Answer: A

SECTION 10: ADVANCED CONCEPTS (Questions 166-180)

96. A tuple is also known as:
A) Record
B) Row
C) Both A and B
D) Column

Answer: C

---

97. The degree of a relation can change when we:
A) Add/remove columns

B) Add/remove rows
C) Update values
D) Create index

Answer: A

---

98. The cardinality of a relation changes when we:
A) Add/remove rows
B) Add/remove columns
C) Change column names
D) Create views

Answer: A

---

99. A null value represents:
A) Missing or unknown information
B) Zero
C) Empty string
D) Space

Answer: A

---

100. Three-valued logic in SQL includes:
A) TRUE, FALSE, UNKNOWN
B) TRUE, FALSE only
C) TRUE, FALSE, NULL
D) YES, NO, MAYBE

Answer: A

---

101. The information principle states that:
A) All information is represented as data in tables
B) All information is in indexes
C) Information is in programs
D) Information is in constraints

Answer: A

---

102. A relation is in 5NF if:
A) It has no join dependency
B) It has no multivalued dependency
C) It has no transitive dependency
D) All attributes are atomic

Answer: A

---

103. Multivalued dependency leads to which normal form?
A) 2NF
B) 3NF
C) 4NF
D) BCNF

Answer: C

---

104. Join dependency leads to which normal form?
A) 4NF
B) 5NF
C) 3NF
D) BCNF

Answer: B

---

105. The lossless join property ensures:
A) No spurious tuples after join
B) All data is preserved
C) No redundancy
D) Fast queries

Answer: A

---

106. Dependency preservation ensures:
A) All FDs can be checked on individual relations
B) All data is preserved

C) No redundancy
D) Fast queries

Answer: A

---

107. The chase algorithm is used to test:
A) Lossless join property
B) Query optimization
C) Index selection
D) View updatability

Answer: A

---

108. Inclusion dependencies generalize:
A) Foreign keys
B) Primary keys
C) Candidate keys
D) Superkeys

Answer: A

---

109. Template dependencies are used for:
A) Defining semantic constraints
B) Creating indexes
C) Optimizing queries
D) Partitioning data

Answer: A

---

110. The universal relation assumption assumes:
A) All attributes come from one universal table
B) Multiple tables
C) No tables
D) Only views

Answer: A

SECTION 1: INTRODUCTION TO RELATIONAL ALGEBRA (Questions 1-20)

1. Relational algebra is:
A) A procedural query language
B) A non-procedural query language
C) A programming language
D) A data definition language

Answer: A
Explanation: Relational algebra specifies how to retrieve data step by step.

---

2. Relational algebra operations are performed on:
A) Single relation only
B) One or more relations to produce a new relation
C) Files
D) Indexes only

Answer: B
Explanation: Operations take relations as input and produce a new relation as output (closure property).

---

3. The closure property of relational algebra means:
A) Output of operation is also a relation
B) Operations are closed to users
C) Relations are closed files
D) Operations are hidden

Answer: A

---

4. Which of the following is NOT a fundamental operation of relational algebra?
A) SELECT (σ)
B) PROJECT (π)

C) JOIN (⋈)
D) RENAME (ρ)

Answer: C
Explanation: JOIN can be derived from Cartesian product and SELECT.

---

5. The five fundamental operations of relational algebra are:
A) SELECT, PROJECT, UNION, SET DIFFERENCE, CARTESIAN PRODUCT
B) SELECT, PROJECT, JOIN, INTERSECTION, DIVISION
C) UNION, INTERSECTION, JOIN, DIVIDE, RENAME
D) SELECT, PROJECT, JOIN, UNION, INTERSECTION

Answer: A

SECTION 2: SELECT OPERATION (Questions 21-35)

6. The SELECT operation in relational algebra is denoted by:
A) π (pi)
B) σ (sigma)
C) ⋈ (bowtie)
D) ρ (rho)

Answer: B

---

7. SELECT operation in relational algebra corresponds to which SQL clause?
A) SELECT
B) WHERE
C) FROM
D) GROUP BY

Answer: B

---

8. The SELECT operation returns:
A) Vertical subset (columns)
B) Horizontal subset (rows)

C) Combined tables
D) Renamed attributes

Answer: B

---

9. Which of the following is a valid SELECT operation?
A) σ salary > 50000 (Employee)
B) π salary > 50000 (Employee)
C) ⋈ salary > 50000 (Employee)
D) ρ salary > 50000 (Employee)

Answer: A

---

10. The degree of the result after SELECT operation:
A) Decreases
B) Increases
C) Remains same
D) Becomes zero

Answer: C
Explanation: SELECT filters rows, number of attributes remains same.

---

11. SELECT operation with multiple conditions uses:
A) AND (∧), OR (∨), NOT (¬)
B) +, -, *
C) >, <, =
D) Only AND

Answer: A

---

12. σ condition1 ∧ condition2 (R) is equivalent to:
A) σ condition1 (σ condition2 (R))
B) σ condition1 (R) ∪ σ condition2 (R)
C) σ condition1 (R) ∩ σ condition2 (R)
D) π condition1 (σ condition2 (R))

Answer: A

---

13. Which comparison operators can be used in SELECT?
A) =, ≠, >, <, ≥, ≤
B) Only =
C) Only > and <
D) Only ≠

Answer: A

---

14. The number of tuples after SELECT operation:
A) Always decreases or stays same
B) Always increases
C) May increase
D) Becomes infinite

Answer: A
Explanation: SELECT filters rows, so count never increases.

---

15. σ true (R) returns:
A) Empty relation
B) All tuples of R
C) No tuples
D) Error

Answer: B

SECTION 3: PROJECT OPERATION (Questions 36-50)

16. The PROJECT operation in relational algebra is denoted by:
A) σ (sigma)
B) π (pi)
C) ⋈ (bowtie)
D) ρ (rho)

Answer: B

---

17. PROJECT operation corresponds to which SQL clause?
A) SELECT (column list)
B) WHERE
C) FROM
D) ORDER BY

Answer: A

---

18. The PROJECT operation returns:
A) Horizontal subset (rows)
B) Vertical subset (columns)
C) Combined tables
D) Filtered rows

Answer: B

---

19. Which of the following is a valid PROJECT operation?
A) π name, salary (Employee)
B) σ name, salary (Employee)
C) ⋈ name, salary (Employee)
D) ρ name, salary (Employee)

Answer: A

---

20. PROJECT operation eliminates:
A) Duplicate rows automatically
B) Duplicate columns
C) Null values
D) Foreign keys

Answer: A
Explanation: In relational algebra, PROJECT eliminates duplicate tuples.

---

21. The degree of the result after PROJECT operation:
A) May decrease (fewer attributes)

B) Always increases
C) Remains same
D) Becomes zero

Answer: A

---

22. π A,B (R) where A and B are attributes, the result contains:
A) Only attributes A and B
B) All attributes of R
C) Only attribute A
D) Only attribute B

Answer: A

---

23. π A (π A,B (R)) is equivalent to:
A) π A (R)
B) π A,B (R)
C) σ A (R)
D) R

Answer: A

---

24. If R has attributes A, B, C and 10 tuples, π A,B (R) will have maximum:
A) 10 tuples (if A,B combinations are unique)
B) 5 tuples
C) 20 tuples
D) 100 tuples

Answer: A

---

25. PROJECT operation can reduce cardinality because:
A) Duplicate tuples are eliminated
B) Rows are filtered
C) Columns are removed
D) Keys are lost

Answer: A

SECTION 4: RENAME OPERATION (Questions 51-60)

26. The RENAME operation in relational algebra is denoted by:
A) σ
B) π
C) ρ
D) ⋈

Answer: C

---

27. Which of the following correctly renames relation R to S?
A) ρ(S, R)
B) σ(S, R)
C) π(S, R)
D) ⋈(S, R)

Answer: A

---

28. To rename attributes of relation R from A,B to X,Y:
A) ρ (X,Y), (A,B) (R)
B) σ (X,Y) (R)
C) π (X,Y) (R)
D) ⋈ (X,Y) (R)

Answer: A

---

29. RENAME operation is needed for:
A) Self-joins
B) Avoiding attribute name conflicts
C) Both A and B
D) Deleting relations

Answer: C

---

30. ρ Employee2(Employee) creates:
A) A copy of Employee relation with new name
B) A view
C) A new table with no data
D) An index

Answer: A

SECTION 5: UNION, INTERSECTION, DIFFERENCE (Questions 61-80)

31. UNION operation in relational algebra requires:
A) Compatible relations (same degree and domain compatibility)
B) Any two relations
C) Relations with same name
D) Relations with primary keys

Answer: A

---

32. UNION compatibility means:
A) Same number of attributes and compatible domains
B) Same number of tuples
C) Same primary key
D) Same relation name

Answer: A

---

33. R ∪ S returns:
A) Tuples in both R and S
B) Tuples in R or S or both
C) Tuples in R but not S
D) All combinations of tuples

Answer: B

---

34. INTERSECTION operation (R ∩ S) returns:
A) Tuples in R or S
B) Tuples in both R and S
C) Tuples in R but not S
D) All combinations

Answer: B

---

35. SET DIFFERENCE operation (R - S) returns:
A) Tuples in R but not in S
B) Tuples in S but not in R
C) Tuples in both
D) All tuples

Answer: A

---

36. Which operation is not commutative?
A) UNION
B) INTERSECTION
C) SET DIFFERENCE
D) Both A and B

Answer: C
Explanation: R - S ≠ S - R generally.

---

37. If R and S are union compatible, R ∪ S = S ∪ R demonstrates:
A) Commutativity of UNION
B) Associativity of UNION
C) Distributivity
D) Idempotence

Answer: A

---

38. R ∪ (S ∪ T) = (R ∪ S) ∪ T demonstrates:
A) Commutativity
B) Associativity

C) Distributivity
D) Idempotence

Answer: B

---

39. $R \cap R = R$ demonstrates:
A) Commutativity
B) Associativity
C) Idempotence
D) Distributivity

Answer: C

---

40. The identity element for UNION is:
A) Empty relation
B) Universal relation
C) R itself
D) None

Answer: A
Explanation: $R \cup \emptyset = R$

---

41. Which of the following is true?
A) $R \cap (S \cup T) = (R \cap S) \cup (R \cap T)$
B) $R \cap (S \cup T) = (R \cup S) \cap (R \cup T)$
C) $R \cup (S \cap T) = (R \cup S) \cap (R \cup T)$
D) All of the above

Answer: D
Explanation: These are distributive laws.

---

42. The cardinality of $R \cup S$ is at most:
A) $|R| + |S|$
B) $|R| \times |S|$
C) $|R| - |S|$
D) $\min(|R|, |S|)$

Answer: A

---

43. The cardinality of R ∩ S is at most:
A) min(|R|, |S|)
B) max(|R|, |S|)
C) |R| + |S|
D) |R| × |S|

Answer: A

---

44. If R and S are disjoint, then |R ∪ S| =:
A) |R| + |S|
B) |R| - |S|
C) min(|R|, |S|)
D) max(|R|, |S|)

Answer: A

---

45. Which operation can be expressed using other operations?
A) UNION
B) INTERSECTION
C) SET DIFFERENCE
D) CARTESIAN PRODUCT

Answer: B
Explanation: R ∩ S = R - (R - S)

SECTION 6: CARTESIAN PRODUCT (Questions 81-95)

46. CARTESIAN PRODUCT (R × S) combines:
A) Tuples from R with tuples from S
B) Attributes from R with attributes from S
C) Both tuples and attributes
D) Only matching tuples

Answer: C
Explanation: Each tuple of R pairs with each tuple of S, producing |R| × |S| tuples with degree = degree(R) + degree(S)

---

47. If R has m tuples and n attributes, S has p tuples and q attributes, then R × S has:
A) m × p tuples, n + q attributes
B) m + p tuples, n + q attributes
C) m × p tuples, n × q attributes
D) m + p tuples, n × q attributes

Answer: A

---

48. CARTESIAN PRODUCT without any selection usually produces:
A) Meaningless results
B) Useful combined data
C) Empty relation
D) Index

Answer: A
Explanation: Pure Cartesian product without conditions produces many meaningless combinations.

---

49. R × S = S × R demonstrates:
A) Commutativity
B) Associativity
C) Distributivity
D) Idempotence

Answer: A

---

50. (R × S) × T = R × (S × T) demonstrates:
A) Commutativity
B) Associativity
C) Distributivity
D) Idempotence

Answer: B

SECTION 7: JOIN OPERATIONS (Questions 96-125)

51. The JOIN operation is derived from:
A) CARTESIAN PRODUCT followed by SELECT
B) UNION followed by PROJECT
C) INTERSECTION followed by RENAME
D) DIFFERENCE followed by UNION

Answer: A

---

52. THETA JOIN ($R \bowtie_\theta S$) uses:
A) Any comparison condition θ
B) Only equality condition
C) Only less than condition
D) No condition

Answer: A

---

53. EQUIJOIN is a THETA JOIN with:
A) Equality condition only
B) Any condition
C) Greater than condition
D) Less than condition

Answer: A

---

54. NATURAL JOIN ($R \bowtie S$) eliminates:
A) Duplicate tuples
B) Duplicate attributes (common columns)
C) Null values
D) Foreign keys

Answer: B

---

55. Natural join automatically joins on:
A) Attributes with same name in both relations
B) Primary keys
C) Foreign keys
D) First attribute

Answer: A

---

56. If R and S have no common attributes, then R ⋈ S =:
A) R × S
B) R ∪ S
C) R ∩ S
D) R - S

Answer: A

---

57. OUTER JOIN preserves:
A) Only matching tuples
B) Non-matching tuples from one or both relations
C) No tuples
D) Only null values

Answer: B

---

58. LEFT OUTER JOIN (R ⟕ S) preserves:
A) All tuples from left relation (R)
B) All tuples from right relation (S)
C) All tuples from both
D) Only matching tuples

Answer: A

---

59. RIGHT OUTER JOIN (R ⟖ S) preserves:
A) All tuples from left relation
B) All tuples from right relation
C) All tuples from both
D) Only matching tuples

Answer: B

---

60. FULL OUTER JOIN (R ⟗ S) preserves:
A) All tuples from left relation only
B) All tuples from right relation only
C) All tuples from both relations
D) Only matching tuples

Answer: C

---

61. In LEFT OUTER JOIN, non-matching tuples from left relation have:
A) NULL values for right relation attributes
B) Zero values
C) Empty strings
D) Deleted

Answer: A

---

62. SEMI JOIN (R ⋉ S) returns:
A) Tuples in R that match with some tuple in S
B) All tuples from both
C) Only matching attributes
D) Cartesian product

Answer: A
Explanation: Semi join returns only attributes from left relation.

---

63. ANTI JOIN returns:
A) Tuples in R with no match in S
B) Tuples in both
C) All tuples
D) Cartesian product

Answer: A

---

64. The number of tuples in natural join is at most:
A) |R| × |S|
B) |R| + |S|
C) min(|R|, |S|)
D) max(|R|, |S|)

Answer: A

---

65. Which join is most commonly used in practice?
A) Natural join
B) Equijoin
C) Outer join
D) Theta join

Answer: B

SECTION 8: DIVISION OPERATION (Questions 126-140)

66. The DIVISION operation (R ÷ S) is used for queries involving:
A) "All" conditions
B) "Any" conditions
C) "Exists" conditions
D) "Not" conditions

Answer: A
Explanation: Division handles "for all" type queries (e.g., find suppliers who supply all parts).

---

67. For R ÷ S to be valid:
A) Attributes of S must be subset of attributes of R
B) R and S must have same attributes
C) S must have more attributes than R
D) No condition

Answer: A

---

68. If R has attributes (A, B) and S has attribute (B), then R ÷ S returns:
A) A values that are associated with every B in S
B) All A values
C) All B values
D) Cartesian product

Answer: A

---

69. The division operation can be expressed as:
A) $R \div S = \pi A(R) - \pi A((\pi A(R) \times S) - R)$
B) $R \div S = R \times S$
C) $R \div S = R - S$
D) $R \div S = R \cup S$

Answer: A

---

70. If S is empty, then R ÷ S =:
A) π attributes of R that are not in S (R)
B) Empty relation
C) R itself
D) Error

Answer: A
Explanation: Division by empty set returns all tuples of R.

SECTION 9: AGGREGATE OPERATIONS (Questions 141-150)

71. Aggregate operations in relational algebra include:
A) SUM, COUNT, AVG, MAX, MIN
B) SELECT, PROJECT
C) JOIN, UNION
D) DIVIDE, RENAME

Answer: A

---

72. GROUP BY in relational algebra is represented using:
A) Aggregation with grouping attributes
B) Simple SELECT
C) PROJECT only
D) JOIN only

Answer: A

---

73. $\mathcal{G}$ function, attributes (R) represents:
A) Grouping and aggregation
B) Cartesian product
C) Union
D) Difference

Answer: A

---

74. COUNT(*) returns:
A) Number of tuples
B) Sum of values
C) Average
D) Maximum

Answer: A

---

75. Aggregation operations are:
A) Not part of pure relational algebra
B) Part of basic algebra
C) Fundamental operations
D) Unnecessary

Answer: A

SECTION 10: RELATIONAL CALCULUS (Questions 151-175)

76. Relational calculus is:
A) Procedural

B) Non-procedural/declarative
C) Implementation-oriented
D) Storage-oriented

Answer: B

---

77. Tuple relational calculus uses:
A) Variables that range over tuples
B) Variables that range over attributes
C) Domain variables
D) Table variables

Answer: A

---

78. A tuple relational calculus expression has the form:
A) { t | Condition(t) }
B) { a1, a2 | Condition }
C) SELECT * FROM table
D) π attributes (table)

Answer: A

---

79. Domain relational calculus uses:
A) Tuple variables
B) Domain variables (attribute values)
C) Relation variables
D) Attribute variables

Answer: B

---

80. A domain relational calculus expression has the form:
A) { t | Condition(t) }
B) { <a1, a2, ..., an> | Condition(a1, a2, ..., an) }
C) { * | Condition }
D) { relation | Condition }

Answer: B

---

81. Which quantifier is used in relational calculus?
A) FOR ALL (∀)
B) EXISTS (∃)
C) Both A and B
D) Neither

Answer: C

---

82. ∃ t ∈ R (P(t)) means:
A) There exists a tuple in R satisfying P
B) All tuples in R satisfy P
C) No tuples satisfy P
D) R is empty

Answer: A

---

83. ∀ t ∈ R (P(t)) means:
A) There exists a tuple satisfying P
B) All tuples in R satisfy P
C) Some tuples satisfy P
D) P is false

Answer: B

---

84. Relational calculus is equivalent in expressive power to:
A) Relational algebra
B) Hierarchical model
C) Network model
D) File systems

Answer: A
Explanation: Codd's theorem proves relational algebra and calculus are equivalent.

---

85. Safety of expressions in relational calculus means:
A) Expression returns finite results

B) Expression is secure
C) Expression is fast
D) Expression has no errors

Answer: A

---

86. An unsafe expression in relational calculus might:
A) Return infinite results
B) Cause syntax error
C) Be slow
D) Use wrong syntax

Answer: A

---

87. The expression { t | ¬ (t ∈ R) } is:
A) Unsafe (infinite)
B) Safe
C) Valid
D) Syntax error

Answer: A
Explanation: This would return all tuples not in R, which is infinite.

SECTION 11: QUERY EXAMPLES (Questions 176-190)

88. Find names of employees working in department 'IT' (Relational Algebra):
A) π name (σ dept='IT' (Employee))
B) σ dept='IT' (π name (Employee))
C) π name ⋈ σ dept='IT' (Employee)
D) ρ name (σ dept='IT' (Employee))

Answer: A

---

89. Find names of employees working in 'IT' or earning >50000:
A) π name (σ dept='IT' ∨ salary>50000 (Employee))

B) π name (σ dept='IT' ∧ salary>50000 (Employee))
C) σ dept='IT' (π name (Employee)) ∪ σ salary>50000 (π name (Employee))
D) Both A and similar expressions

Answer: A

---

90. Find names of employees who work on project 'P1' (Employee ⋈ Works_on):
A) π name (Employee ⋈ (σ project='P1' (Works_on)))
B) σ project='P1' (Employee × Works_on)
C) π name (σ project='P1' (Employee × Works_on))
D) All of the above are equivalent

Answer: D

---

91. Find suppliers who supply all parts (Division query):
A) π supplier (Supplies ÷ π part (Parts))
B) π supplier (Supplies) × Parts
C) σ supplier (Supplies ÷ Parts)
D) π supplier (Supplies ⋈ Parts)

Answer: A

---

92. In tuple relational calculus, find names of employees earning >50000:
A) { t.name | Employee(t) ∧ t.salary > 50000 }
B) { t | Employee(t) ∧ t.salary > 50000 }
C) { name | ∃ e (Employee(e) ∧ e.salary > 50000 ∧ name = e.name) }
D) All of the above with appropriate syntax

Answer: D

---

93. Find names of employees who work on all projects:
A) { e.name | Employee(e) ∧ ∀ p (Project(p) → ∃ w (Works_on(w) ∧ w.emp_id = e.id ∧ w.proj_id = p.id)) }
B) { e.name | Employee(e) ∧ ∃ p (Project(p) ∧ Works_on(w)) }
C) { e.name | ∀ p (Project(p) ∧ Works_on(w)) }
D) { e.name | Works_on(e) }

Answer: A

---

94. The query { t | Employee(t) ∧ ∃ d (Department(d) ∧ d.dept_name = 'IT' ∧ t.dept_id = d.id) } finds:
A) Employees in IT department
B) All employees
C) IT department details
D) Department names

Answer: A

SECTION 12: COMPARISON AND PROPERTIES (Questions 191-200)

95. Which of the following is NOT a property of relational algebra?
A) Closure property
B) Procedural nature
C) Set-at-a-time processing
D) Record-at-a-time processing

Answer: D
Explanation: Relational algebra operates on sets of tuples, not individual records.

---

96. Relational algebra is considered:
A) Procedural
B) Declarative
C) Object-oriented
D) Functional

Answer: A

---

97. Relational calculus is considered:
A) Procedural
B) Declarative

C) Implementation-oriented
D) Storage-oriented

Answer: B

---

98. The expressive power of relational algebra and relational calculus is:
A) Equal
B) Algebra is more powerful
C) Calculus is more powerful
D) Cannot be compared

Answer: A

---

99. SQL is based on:
A) Relational algebra only
B) Relational calculus only
C) Both relational algebra and calculus
D) Neither

Answer: C

---

100. Which operation cannot be expressed in pure relational algebra?
A) Transitive closure
B) SELECT
C) PROJECT
D) UNION

Answer: A
Explanation: Transitive closure requires recursive queries or extensions to relational algebra.

---

101. Recursive queries require:
A) Extended relational algebra
B) Basic algebra
C) Only SELECT
D) Only PROJECT

Answer: A

---

102. The MINUS operation in SQL corresponds to which relational algebra operation?
A) SET DIFFERENCE
B) UNION
C) INTERSECTION
D) CARTESIAN PRODUCT

Answer: A

---

103. The INTERSECT operation in SQL corresponds to:
A) SET DIFFERENCE
B) UNION
C) INTERSECTION
D) CARTESIAN PRODUCT

Answer: C

---

104. Codd's theorem states:
A) Relational algebra and calculus are equivalent
B) SQL is complete
C) Normalization is necessary
D) Indexes improve performance

Answer: A

SECTION 1: INTRODUCTION TO SQL (Questions 1-20)

1. SQL stands for:
A) Structured Query Language
B) Simple Query Language
C) Sequential Query Language
D) System Query Language

Answer: A

---

2. SQL was originally developed at:
A) IBM
B) Oracle
C) Microsoft
D) Apple

Answer: A
Explanation: SQL was developed at IBM in the 1970s for System R.

---

3. SQL is a:
A) Procedural language
B) Non-procedural language
C) Object-oriented language
D) Functional language

Answer: B
Explanation: SQL is declarative - you specify WHAT, not HOW.

---

4. Which organization standardized SQL first?
A) ISO
B) ANSI
C) IEEE
D) W3C

Answer: B
Explanation: ANSI standardized SQL in 1986, followed by ISO in 1987.

---

5. The latest SQL standard is:
A) SQL-92
B) SQL:1999
C) SQL:2003
D) SQL:2016 (with ongoing updates)

Answer: D

---

6. Which of the following is NOT a category of SQL commands?
A) DDL

B) DML
C) DCL
D) DPL

Answer: D
Explanation: Main categories are DDL, DML, DCL, and TCL.

---

7. DDL stands for:
A) Data Definition Language
B) Data Manipulation Language
C) Data Control Language
D) Data Description Language

Answer: A

---

8. DML stands for:
A) Data Definition Language
B) Data Manipulation Language
C) Data Management Language
D) Data Modeling Language

Answer: B

---

9. DCL stands for:
A) Data Control Language
B) Data Command Language
C) Data Communication Language
D) Data Connection Language

Answer: A

---

10. TCL stands for:
A) Transaction Control Language
B) Table Control Language
C) Total Control Language
D) Transfer Control Language

Answer: A

SECTION 2: DATA DEFINITION LANGUAGE (DDL) - Questions 21-45

11. Which command is used to create a new table?
A) CREATE TABLE
B) NEW TABLE
C) ADD TABLE
D) MAKE TABLE

Answer: A

---

12. The correct syntax to create a table is:
A) CREATE TABLE table_name (column1 datatype, column2 datatype, ...)
B) CREATE table_name (column1 datatype, column2 datatype, ...)
C) NEW TABLE table_name (column1 datatype, column2 datatype, ...)
D) TABLE CREATE table_name (column1 datatype, column2 datatype, ...)

Answer: A

---

13. Which command is used to modify an existing table structure?
A) MODIFY TABLE
B) ALTER TABLE
C) CHANGE TABLE
D) UPDATE TABLE

Answer: B

---

14. To add a new column to an existing table:
A) ALTER TABLE table_name ADD column_name datatype
B) ALTER TABLE table_name ADD COLUMN column_name datatype
C) UPDATE TABLE table_name ADD column_name datatype
D) Both A and B are valid in different DBMS

Answer: D

---

15. To drop a column from a table:
A) ALTER TABLE table_name DROP COLUMN column_name
B) ALTER TABLE table_name REMOVE COLUMN column_name
C) UPDATE TABLE table_name DROP column_name
D) DELETE COLUMN column_name FROM table_name

Answer: A

---

16. Which command removes a table from the database?
A) DELETE TABLE
B) DROP TABLE
C) REMOVE TABLE
D) CLEAR TABLE

Answer: B

---

17. What is the difference between DROP and TRUNCATE?
A) DROP removes structure, TRUNCATE removes data only
B) TRUNCATE removes structure, DROP removes data only
C) Both are same
D) DROP is faster than TRUNCATE

Answer: A

---

18. TRUNCATE TABLE command:
A) Removes all rows but keeps table structure
B) Removes table structure
C) Removes specific rows
D) Updates table data

Answer: A

---

19. Which command removes all rows from a table without logging individual row deletions?
A) DELETE FROM table_name
B) TRUNCATE TABLE table_name
C) DROP TABLE table_name
D) REMOVE TABLE table_name

Answer: B

---

20. The PRIMARY KEY constraint:
A) Uniquely identifies each row
B) Allows NULL values
C) Allows duplicate values
D) Is optional

Answer: A

---

21. A PRIMARY KEY must be:
A) Unique and NOT NULL
B) Unique only
C) NOT NULL only
D) Either unique or NOT NULL

Answer: A

---

22. The FOREIGN KEY constraint:
A) References another table's primary key
B) Creates an index
C) Ensures data uniqueness
D) Prevents NULL values

Answer: A

---

23. Which constraint ensures that all values in a column are different?
A) PRIMARY KEY
B) UNIQUE

C) FOREIGN KEY
D) CHECK

Answer: B

---

24. The CHECK constraint is used to:
A) Validate data based on a condition
B) Create foreign keys
C) Create indexes
D) Define primary keys

Answer: A

---

25. The DEFAULT constraint:
A) Provides a default value for a column
B) Sets default table name
C) Creates default index
D) Sets default schema

Answer: A

---

26. Which statement is correct about NULL?
A) NULL = 0
B) NULL = ''
C) NULL is unknown/missing value
D) NULL = 'NULL'

Answer: C

---

27. The NOT NULL constraint ensures:
A) Column cannot have NULL values
B) Column must have unique values
C) Column must have default value
D) Column is primary key

Answer: A

---

28. Which command renames a table?
A) RENAME TABLE old_name TO new_name
B) ALTER TABLE old_name RENAME TO new_name
C) Both A and B (syntax varies by DBMS)
D) CHANGE TABLE old_name TO new_name

Answer: C

---

29. To rename a column in SQL Server:
A) EXEC sp_rename 'table.old_column', 'new_column', 'COLUMN'
B) ALTER TABLE table_name RENAME COLUMN old_name TO new_name
C) RENAME COLUMN old_name TO new_name
D) Both A and B (syntax varies)

Answer: D

---

30. The CREATE INDEX command is used to:
A) Improve query performance
B) Create primary key
C) Create foreign key
D) Create table

Answer: A

SECTION 3: DATA MANIPULATION LANGUAGE (DML) - Questions 46-75

31. Which command is used to insert new rows into a table?
A) INSERT INTO
B) ADD ROW
C) UPDATE
D) CREATE ROW

Answer: A

---

32. Correct syntax for INSERT:
A) INSERT INTO table_name (column1, column2) VALUES (value1, value2)
B) INSERT (column1, column2) INTO table_name VALUES (value1, value2)
C) INSERT table_name (column1, column2) VALUES (value1, value2)
D) INSERT VALUES (value1, value2) INTO table_name

Answer: A

---

33. To insert all columns in order, you can use:
A) INSERT INTO table_name VALUES (value1, value2, ...)
B) INSERT INTO table_name ALL VALUES (value1, value2, ...)
C) INSERT table_name VALUES (value1, value2, ...)
D) ADD INTO table_name VALUES (value1, value2, ...)

Answer: A

---

34. Which command retrieves data from the database?
A) SELECT
B) GET
C) FETCH
D) RETRIEVE

Answer: A

---

35. The basic SELECT syntax is:
A) SELECT column1, column2 FROM table_name
B) GET column1, column2 FROM table_name
C) RETRIEVE column1, column2 FROM table_name
D) DISPLAY column1, column2 FROM table_name

Answer: A

---

36. SELECT * FROM table_name returns:
A) All columns and all rows
B) All columns but no rows

C) First column only
D) Table structure only

Answer: A

---

37. The WHERE clause is used to:
A) Filter rows based on conditions
B) Group rows
C) Sort rows
D) Join tables

Answer: A

---

38. Which operator is used for pattern matching in SQL?
A) LIKE
B) MATCH
C) PATTERN
D) SIMILAR

Answer: A

---

39. The wildcard '%' in LIKE represents:
A) Any sequence of characters
B) Any single character
C) Any digit
D) Any letter

Answer: A

---

40. The wildcard '_' in LIKE represents:
A) Any sequence of characters
B) Any single character
C) Any digit
D) Any letter

Answer: B

---

41. Which query finds names starting with 'J'?
A) WHERE name LIKE 'J%'
B) WHERE name LIKE '%J'
C) WHERE name LIKE 'J_'
D) WHERE name = 'J%'

Answer: A

---

42. The IN operator is used to:
A) Check if value matches any in a list
B) Check if value is between two values
C) Check for NULL
D) Check pattern

Answer: A

---

43. The BETWEEN operator is used to:
A) Check if value is within a range
B) Check if value is in a list
C) Join tables
D) Group data

Answer: A

---

44. Which query finds employees with salary between 50000 and 70000?
A) WHERE salary BETWEEN 50000 AND 70000
B) WHERE salary > 50000 AND < 70000
C) WHERE salary IN (50000, 70000)
D) WHERE salary >= 50000 AND salary <= 70000

Answer: A and D are both correct

---

45. The IS NULL operator checks for:
A) Missing values
B) Zero values

C) Empty strings
D) Spaces

Answer: A

---

46. To find employees who do not have a manager:
A) WHERE manager_id IS NULL
B) WHERE manager_id = NULL
C) WHERE manager_id = 0
D) WHERE manager_id = ''

Answer: A
Explanation: NULL cannot be compared with = operator.

---

47. The UPDATE command is used to:
A) Modify existing rows
B) Insert new rows
C) Delete rows
D) Create tables

Answer: A

---

48. Correct UPDATE syntax:
A) UPDATE table_name SET column1 = value1 WHERE condition
B) UPDATE table_name column1 = value1 WHERE condition
C) SET table_name column1 = value1 WHERE condition
D) MODIFY table_name SET column1 = value1 WHERE condition

Answer: A

---

49. Without WHERE clause, UPDATE will:
A) Update all rows
B) Update no rows
C) Cause error
D) Update first row only

Answer: A

---

50. The DELETE command is used to:
A) Remove rows from a table
B) Remove table structure
C) Remove database
D) Remove columns

Answer: A

---

51. Correct DELETE syntax:
A) DELETE FROM table_name WHERE condition
B) DELETE table_name WHERE condition
C) REMOVE FROM table_name WHERE condition
D) DROP FROM table_name WHERE condition

Answer: A

---

52. Without WHERE clause, DELETE will:
A) Delete all rows
B) Delete no rows
C) Cause error
D) Delete first row only

Answer: A

---

53. What's the difference between DELETE and TRUNCATE?
A) DELETE can use WHERE, TRUNCATE cannot
B) DELETE logs row deletions, TRUNCATE doesn't
C) DELETE is DML, TRUNCATE is DDL
D) All of the above

Answer: D

---

54. The DISTINCT keyword is used to:
A) Eliminate duplicate rows from result
B) Select distinct columns

C) Create unique index
D) Count distinct values

Answer: A

---

55. SELECT DISTINCT department FROM employees returns:
A) Unique department names
B) All department names with duplicates
C) Department names and employee count
D) First department only

Answer: A

SECTION 4: DATA CONTROL LANGUAGE (DCL) - Questions 76-85

56. DCL commands include:
A) GRANT and REVOKE
B) COMMIT and ROLLBACK
C) SELECT and INSERT
D) CREATE and DROP

Answer: A

---

57. The GRANT command is used to:
A) Give privileges to users
B) Take away privileges
C) Create users
D) Delete users

Answer: A

---

58. The REVOKE command is used to:
A) Give privileges
B) Remove privileges

C) Modify privileges
D) View privileges

Answer: B

---

59. Which privilege allows reading data?
A) SELECT
B) INSERT
C) UPDATE
D) DELETE

Answer: A

---

60. GRANT SELECT ON employees TO user1:
A) Allows user1 to query employees table
B) Allows user1 to modify employees
C) Creates user1
D) Deletes employees

Answer: A

---

61. WITH GRANT OPTION allows:
A) User to grant privileges to others
B) User to revoke privileges
C) User to create tables
D) User to drop tables

Answer: A

---

62. To grant all privileges on a table:
A) GRANT ALL PRIVILEGES ON table TO user
B) GRANT ALL ON table TO user
C) Both A and B
D) GRANT * ON table TO user

Answer: C

SECTION 5: TRANSACTION CONTROL LANGUAGE (TCL) - Questions 86-100

63. TCL commands include:
A) COMMIT, ROLLBACK, SAVEPOINT
B) GRANT, REVOKE
C) SELECT, INSERT
D) CREATE, DROP

Answer: A

---

64. COMMIT command:
A) Makes all changes permanent
B) Undoes all changes
C) Saves intermediate point
D) Begins transaction

Answer: A

---

65. ROLLBACK command:
A) Undoes changes since last COMMIT
B) Makes changes permanent
C) Saves intermediate point
D) Begins transaction

Answer: A

---

66. SAVEPOINT is used to:
A) Create intermediate points for partial rollback
B) Save data permanently
C) Create checkpoints
D) Begin transaction

Answer: A

---

67. ROLLBACK TO SAVEPOINT sp1:
A) Undoes changes to savepoint sp1
B) Commits to sp1
C) Deletes sp1
D) Creates sp1

Answer: A

---

68. By default, SQL commands are:
A) Auto-committed
B) Not auto-committed
C) Rolled back
D) Saved to log

Answer: A
Explanation: Most SQL implementations auto-commit DML commands.

---

69. SET AUTOCOMMIT OFF:
A) Disables automatic commits
B) Enables automatic commits
C) Commits all transactions
D) Rolls back all transactions

Answer: A

SECTION 6: ADVANCED QUERIES - Questions 101-135

70. The ORDER BY clause is used to:
A) Sort results
B) Filter results
C) Group results
D) Join tables

Answer: A

---

71. ORDER BY column DESC sorts in:
A) Descending order
B) Ascending order
C) Random order
D) No order

Answer: A

---

72. Default sort order in ORDER BY is:
A) ASC (ascending)
B) DESC (descending)
C) Random
D) As inserted

Answer: A

---

73. The GROUP BY clause is used to:
A) Group rows with same values
B) Sort rows
C) Filter rows
D) Join tables

Answer: A

---

74. GROUP BY is typically used with:
A) Aggregate functions
B) WHERE clause only
C) ORDER BY only
D) JOIN only

Answer: A

---

75. Aggregate functions in SQL include:
A) COUNT, SUM, AVG, MAX, MIN
B) SELECT, INSERT, UPDATE
C) GROUP BY, ORDER BY
D) JOIN, UNION

Answer: A

---

76. COUNT(*) returns:
A) Number of rows including NULLs
B) Number of non-NULL values
C) Sum of values
D) Average of values

Answer: A

---

77. COUNT(column) returns:
A) Number of non-NULL values in column
B) Number of rows including NULLs
C) Sum of column values
D) Distinct values

Answer: A

---

78. The HAVING clause is used to:
A) Filter groups after GROUP BY
B) Filter rows before GROUP BY
C) Sort groups
D) Join tables

Answer: A

---

79. Difference between WHERE and HAVING:
A) WHERE filters rows, HAVING filters groups
B) WHERE filters groups, HAVING filters rows
C) Both are same
D) WHERE is for SELECT only

Answer: A

---

80. Which query finds departments with average salary > 60000?
A) SELECT dept_id, AVG(salary) FROM emp GROUP BY dept_id HAVING AVG(salary) > 60000
B) SELECT dept_id, AVG(salary) FROM emp WHERE AVG(salary) > 60000 GROUP BY dept_id
C) SELECT dept_id, AVG(salary) FROM emp HAVING AVG(salary) > 60000
D) SELECT dept_id, AVG(salary) FROM emp GROUP BY dept_id WHERE AVG(salary) > 60000

Answer: A

---

81. The JOIN clause is used to:
A) Combine rows from multiple tables
B) Sort results
C) Group results
D) Filter results

Answer: A

---

82. INNER JOIN returns:
A) Matching rows from both tables
B) All rows from left table
C) All rows from right table
D) All rows from both tables

Answer: A

---

83. LEFT JOIN returns:
A) All rows from left table, matching from right
B) All rows from right table, matching from left
C) Only matching rows
D) All rows from both

Answer: A

---

84. RIGHT JOIN returns:
A) All rows from right table, matching from left

B) All rows from left table, matching from right
C) Only matching rows
D) All rows from both

Answer: A

---

85. FULL OUTER JOIN returns:
A) All rows from both tables
B) Only matching rows
C) All rows from left only
D) All rows from right only

Answer: A

---

86. Correct syntax for INNER JOIN:
A) SELECT * FROM t1 INNER JOIN t2 ON t1.id = t2.id
B) SELECT * FROM t1, t2 WHERE t1.id = t2.id
C) Both A and B (old and new syntax)
D) SELECT * FROM t1 JOIN t2 WHERE t1.id = t2.id

Answer: C

---

87. Self-join is:
A) Joining a table with itself
B) Joining two different tables
C) Joining three tables
D) Joining with system tables

Answer: A

---

88. UNION operator combines:
A) Results of two or more SELECT statements
B) Two tables horizontally
C) Two tables with join condition
D) Two columns

Answer: A

---

89. UNION removes:
A) Duplicate rows
B) All rows
C) NULL values
D) Columns

Answer: A

---

90. UNION ALL:
A) Includes duplicates
B) Removes duplicates
C) Removes NULLs
D) Sorts results

Answer: A

---

91. For UNION to work:
A) Number of columns must match and data types compatible
B) Table names must match
C) Column names must match
D) Database names must match

Answer: A

---

92. INTERSECT returns:
A) Common rows from both queries
B) All rows from first query
C) All rows from second query
D) Combined rows

Answer: A

---

93. EXCEPT (or MINUS) returns:
A) Rows in first query not in second
B) Rows in second query not in first

C) Common rows
D) All rows

Answer: A

---

94. Subquery is:
A) A query inside another query
B) A small query
C) A fast query
D) A system query

Answer: A

---

95. A correlated subquery:
A) References outer query
B) Is independent of outer query
C) Runs only once
D) Cannot use outer table

Answer: A

---

96. EXISTS operator:
A) Checks if subquery returns any rows
B) Checks if value exists
C) Checks for NULL
D) Checks pattern

Answer: A

---

97. ANY operator:
A) Compares value to any value in subquery
B) Compares to all values
C) Checks existence
D) Checks pattern

Answer: A

---

98. ALL operator:
A) Compares value to all values in subquery
B) Compares to any value
C) Checks existence
D) Checks pattern

Answer: A

---

99. Which query finds employees earning more than all in dept 10?
A) SELECT * FROM emp WHERE salary > ALL (SELECT salary FROM emp WHERE dept_id = 10)
B) SELECT * FROM emp WHERE salary > ANY (SELECT salary FROM emp WHERE dept_id = 10)
C) SELECT * FROM emp WHERE salary IN (SELECT salary FROM emp WHERE dept_id = 10)
D) SELECT * FROM emp WHERE salary EXISTS (SELECT salary FROM emp WHERE dept_id = 10)

Answer: A

---

100. The WITH clause (CTE) is used for:
A) Defining temporary named result sets
B) Creating permanent tables
C) Creating views
D) Creating indexes

Answer: A

SECTION 7: VIEWS AND INDEXES - Questions 136-150

101. A view is:
A) Virtual table based on SELECT statement
B) Physical table
C) Index structure
D) Constraint

Answer: A

---

102. CREATE VIEW command:
A) Creates a virtual table
B) Creates physical table
C) Creates index
D) Creates constraint

Answer: A

---

103. Views can be used to:
A) Simplify complex queries
B) Provide security
C) Hide data complexity
D) All of the above

Answer: D

---

104. An updatable view must be based on:
A) Single table without aggregates
B) Multiple tables
C) Aggregates
D) GROUP BY

Answer: A

---

105. WITH CHECK OPTION in views:
A) Ensures modifications satisfy view condition
B) Checks syntax
C) Checks permissions
D) Optimizes queries

Answer: A

---

106. An index is used to:
A) Speed up data retrieval
B) Store data
C) Create tables
D) Define constraints

Answer: A

---

107. CREATE INDEX creates:
A) An index on table columns
B) A new table
C) A view
D) A constraint

Answer: A

---

108. A unique index ensures:
A) All values in indexed column(s) are unique
B) Fast access
C) Referential integrity
D) Data security

Answer: A

---

109. Composite index is:
A) Index on multiple columns
B) Index on single column
C) Index on view
D) Index on table

Answer: A

---

110. DROP INDEX command:
A) Removes an index
B) Removes a table
C) Removes a view
D) Removes a constraint

Answer: A

SECTION 8: SQL FUNCTIONS - Questions 151-165

111. String functions in SQL include:
A) CONCAT, SUBSTRING, LENGTH, UPPER
B) SUM, AVG, COUNT
C) MAX, MIN
D) JOIN, UNION

Answer: A

---

112. Date functions in SQL include:
A) NOW(), DATE(), YEAR(), MONTH()
B) SUM(), AVG()
C) COUNT()
D) CONCAT()

Answer: A

---

113. Numeric functions include:
A) ROUND, CEIL, FLOOR, ABS
B) UPPER, LOWER
C) CONCAT
D) SUBSTRING

Answer: A

---

114. The COALESCE function returns:
A) First non-NULL value in list
B) First value
C) Last non-NULL value
D) NULL if all NULL

Answer: A

---

115. NULLIF(a, b) returns:
A) NULL if a = b, else a
B) a if a = b, else NULL
C) b if a = b, else a
D) NULL always

Answer: A

---

116. CASE statement in SQL is used for:
A) Conditional logic
B) Looping
C) Error handling
D) Transaction control

Answer: A

---

117. Correct CASE syntax:
A) CASE WHEN condition THEN result ELSE result END
B) IF condition THEN result ELSE result
C) SWITCH condition CASE value THEN result
D) CHOOSE condition

Answer: A

SECTION 9: ADVANCED TOPICS - Questions 166-180

118. Window functions in SQL include:
A) ROW_NUMBER(), RANK(), DENSE_RANK()
B) SUM(), AVG() (as aggregate)
C) COUNT()
D) JOIN

Answer: A

---

119. ROW_NUMBER() assigns:
A) Unique sequential number to each row
B) Same rank for ties
C) No gaps for ties
D) Random numbers

Answer: A

---

120. RANK() vs DENSE_RANK():
A) RANK has gaps, DENSE_RANK has no gaps for ties
B) Both same
C) DENSE_RANK has gaps
D) Neither handles ties

Answer: A

---

121. The PARTITION BY clause in window functions:
A) Divides result set into partitions
B) Sorts results
C) Filters results
D) Groups results (like GROUP BY)

Answer: A

---

122. Stored procedures are:
A) Precompiled SQL code stored in database
B) Regular SQL queries
C) Views
D) Indexes

Answer: A

---

123. Functions vs Procedures:
A) Functions return value, procedures may not
B) Procedures return value, functions may not
C) Both are same
D) Functions cannot have parameters

Answer: A

---

124. Triggers are:
A) Automatically executed code on events
B) Manual queries
C) Views
D) Indexes

Answer: A

---

125. Triggers can be fired:
A) BEFORE or AFTER INSERT, UPDATE, DELETE
B) Only AFTER
C) Only BEFORE
D) Only on SELECT

Answer: A

SECTION 10: SQL OPTIMIZATION - Questions 181-190

126. Which is faster?
A) SELECT specific columns vs SELECT *
B) SELECT * is faster
C) Both same
D) Depends on data

Answer: A

---

127. Using indexes improves performance for:
A) SELECT queries with WHERE on indexed columns
B) INSERT operations
C) UPDATE of all rows
D) DELETE of all rows

Answer: A

---

128. EXPLAIN command is used to:
A) Show query execution plan
B) Execute query
C) Explain syntax
D) Document query

Answer: A

---

129. Which join is typically most efficient?
A) INNER JOIN
B) OUTER JOIN
C) CROSS JOIN
D) All same

Answer: A

---

130. EXISTS vs IN:
A) EXISTS often faster for large result sets
B) IN always faster
C) Both same
D) EXISTS cannot use indexes

Answer: A

SECTION 1: WINDOW FUNCTIONS (Questions 1-25)

1. Window functions in SQL operate on:
A) Entire table
B) A set of rows related to current row
C) Single row only
D) Multiple tables

Answer: B
Explanation: Window functions perform calculations across a set of rows related to the current row.

---

2. Which clause defines the window for window functions?
A) OVER()
B) WINDOW()
C) PARTITION()
D) GROUP()

Answer: A

---

3. The PARTITION BY clause in window functions:
A) Divides result set into partitions
B) Sorts rows within partitions
C) Filters rows
D) Groups rows like GROUP BY

Answer: A

---

4. The ORDER BY clause in window functions:
A) Orders rows within each partition
B) Orders final result
C) Filters rows
D) Creates index

Answer: A

---

5. ROW_NUMBER() assigns:
A) Unique sequential number to each row within partition
B) Same number for ties
C) Rank with gaps
D) Percentage rank

Answer: A

---

6. RANK() vs DENSE_RANK():
A) RANK has gaps for ties, DENSE_RANK has no gaps
B) Both have gaps

C) Both have no gaps
D) RANK has no gaps, DENSE_RANK has gaps

Answer: A

---

7. For values: 100, 100, 200, RANK() returns:
A) 1, 1, 3
B) 1, 1, 2
C) 1, 2, 3
D) 1, 1, 1

Answer: A

---

8. For values: 100, 100, 200, DENSE_RANK() returns:
A) 1, 1, 2
B) 1, 1, 3
C) 1, 2, 3
D) 1, 1, 1

Answer: A

---

9. NTILE(n) function:
A) Divides rows into n approximately equal buckets
B) Returns nth row
C) Returns top n rows
D) Creates n partitions

Answer: A

---

10. LAG() function returns:
A) Value from previous row
B) Value from next row
C) First value in partition
D) Last value in partition

Answer: A

---

11. LEAD() function returns:
A) Value from next row
B) Value from previous row
C) First value in partition
D) Last value in partition

Answer: A

---

12. FIRST_VALUE() returns:
A) First value in window frame
B) Last value in window frame
C) Nth value
D) Previous value

Answer: A

---

13. LAST_VALUE() returns:
A) Last value in window frame
B) First value in window frame
C) Nth value
D) Next value

Answer: A

---

14. The default window frame is:
A) RANGE BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
B) ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING
C) RANGE BETWEEN CURRENT ROW AND UNBOUNDED FOLLOWING
D) Entire partition

Answer: A

---

15. Which query finds running total of sales?
A) SELECT sales, SUM(sales) OVER (ORDER BY date) FROM orders
B) SELECT sales, SUM(sales) FROM orders GROUP BY date

C) SELECT sales, SUM(sales) OVER () FROM orders
D) SELECT sales, SUM(sales) FROM orders

Answer: A

---

16. Which query finds moving average of 3 days?
A) SELECT sales, AVG(sales) OVER (ORDER BY date ROWS BETWEEN 1 PRECEDING AND 1 FOLLOWING)
B) SELECT sales, AVG(sales) OVER (ORDER BY date ROWS 3 PRECEDING)
C) SELECT sales, AVG(sales) OVER (PARTITION BY date)
D) SELECT sales, AVG(sales) FROM orders GROUP BY date

Answer: A

---

17. PERCENT_RANK() returns:
A) Relative rank (0 to 1)
B) Percentage (0 to 100)
C) Rank as integer
D) Dense rank

Answer: A

---

18. CUME_DIST() returns:
A) Cumulative distribution (0 to 1)
B) Cumulative sum
C) Running total
D) Percentage rank

Answer: A

---

19. Window functions cannot be used in:
A) WHERE clause
B) SELECT clause
C) ORDER BY clause
D) HAVING clause

Answer: A
Explanation: Window functions are evaluated after WHERE clause.

---

20. Which is correct about window functions?
A) They don't collapse rows like GROUP BY
B) They collapse rows like GROUP BY
C) They cannot use ORDER BY
D) They cannot use PARTITION BY

Answer: A

SECTION 2: COMMON TABLE EXPRESSIONS (CTEs) - Questions 26-45

21. CTE stands for:
A) Common Table Expression
B) Common Table Extension
C) Complex Table Expression
D) Column Table Expression

Answer: A

---

22. CTE syntax uses which keyword?
A) WITH
B) CTE
C) TEMP
D) USING

Answer: A

---

23. A CTE is:
A) Temporary named result set
B) Permanent table
C) View
D) Index

Answer: A

---

24. Correct CTE syntax:
A) WITH cte_name AS (SELECT ...) SELECT * FROM cte_name
B) CTE cte_name AS (SELECT ...) SELECT * FROM cte_name
C) TEMP cte_name AS (SELECT ...) SELECT * FROM cte_name
D) CREATE CTE cte_name AS SELECT ...

Answer: A

---

25. CTEs are useful for:
A) Simplifying complex queries
B) Creating permanent tables
C) Creating indexes
D) Creating constraints

Answer: A

---

26. Multiple CTEs can be defined using:
A) WITH cte1 AS (...), cte2 AS (...) SELECT ...
B) WITH cte1 AS (...) AND cte2 AS (...) SELECT ...
C) WITH cte1 AS (...), WITH cte2 AS (...) SELECT ...
D) CTE1 ..., CTE2 ...

Answer: A

---

27. A CTE can be referenced:
A) Multiple times in the main query
B) Only once
C) Only in subqueries
D) Only in WHERE clause

Answer: A

---

28. The scope of a CTE is:
A) Single statement
B) Entire session
C) Entire database
D) Permanent

Answer: A

---

29. Recursive CTEs must include:
A) UNION ALL with anchor and recursive members
B) UNION only
C) JOIN
D) WHERE clause

Answer: A

---

30. Recursive CTE syntax requires which keyword?
A) RECURSIVE
B) LOOP
C) WHILE
D) ITERATE

Answer: A

---

31. Which query generates numbers 1 to 10 using recursive CTE?
A) WITH RECURSIVE numbers(n) AS (SELECT 1 UNION ALL SELECT n+1 FROM numbers WHERE n < 10) SELECT * FROM numbers
B) WITH numbers(n) AS (SELECT 1 UNION SELECT n+1 FROM numbers WHERE n < 10) SELECT * FROM numbers
C) WITH RECURSIVE numbers AS SELECT 1 TO 10
D) WITH numbers AS GENERATE_SERIES(1,10)

Answer: A

---

32. Recursive CTEs are commonly used for:
A) Hierarchical data (org charts, bill of materials)
B) Simple aggregations

C) Joins
D) Filtering

Answer: A

---

33. The MAXRECURSION option:
A) Limits recursion depth
B) Enables recursion
C) Disables recursion
D) Shows recursion count

Answer: A

---

34. Without proper termination, recursive CTE can cause:
A) Infinite loop
B) Syntax error
C) Faster execution
D) Index creation

Answer: A

---

35. CTEs vs Derived Tables:
A) CTEs can be referenced multiple times, derived tables cannot
B) Derived tables are more powerful
C) CTEs are permanent
D) Both same

Answer: A

SECTION 3: ADVANCED JOINS (Questions 46-65)

36. CROSS JOIN returns:
A) Cartesian product of two tables
B) Matching rows only
C) All rows from left table
D) All rows from right table

Answer: A

---

37. NATURAL JOIN automatically joins on:
A) Columns with same name
B) Primary keys
C) Foreign keys
D) First column

Answer: A

---

38. Which join returns all rows from both tables with NULLs for non-matches?
A) FULL OUTER JOIN
B) INNER JOIN
C) LEFT JOIN
D) RIGHT JOIN

Answer: A

---

39. Self-join is used for:
A) Comparing rows within same table
B) Joining two different tables
C) Joining three tables
D) Creating Cartesian product

Answer: A

---

40. Find employees earning more than their managers (self-join):
A) SELECT e.name FROM emp e JOIN emp m ON e.manager_id = m.id WHERE e.salary > m.salary
B) SELECT name FROM emp WHERE salary > manager_salary
C) SELECT e.name FROM emp e, emp m WHERE e.salary > m.salary
D) Both A and similar queries

Answer: A

---

41. LATERAL JOIN allows:
A) Subquery to reference columns from preceding tables
B) Regular join only
C) Cross join only
D) Outer join only

Answer: A

---

42. APPLY operator (SQL Server) is similar to:
A) LATERAL join
B) CROSS JOIN
C) INNER JOIN
D) OUTER JOIN

Answer: A

---

43. CROSS APPLY returns:
A) Only rows where subquery returns rows
B) All rows from left table
C) Cartesian product
D) Matching rows only

Answer: A

---

44. OUTER APPLY returns:
A) All rows from left table (NULLs where subquery empty)
B) Only matching rows
C) All rows from right table
D) Cartesian product

Answer: A

---

45. Anti-join returns:
A) Rows in left table with no match in right
B) Rows with matches
C) All rows
D) Cartesian product

Answer: A

---

46. Anti-join can be written using:
A) NOT EXISTS or LEFT JOIN WHERE right IS NULL
B) INNER JOIN
C) UNION
D) INTERSECT

Answer: A

---

47. Semi-join returns:
A) Rows from left table that have at least one match in right
B) Only matching columns
C) All rows from both
D) Rows with no matches

Answer: A

SECTION 4: PIVOTING AND UNPIVOTING (Questions 66-80)

48. PIVOT operation converts:
A) Rows to columns
B) Columns to rows
C) Tables to views
D) Data types

Answer: A

---

49. UNPIVOT operation converts:
A) Columns to rows
B) Rows to columns
C) Tables to views
D) Data types

Answer: A

---

50. Which clause in SQL Server performs pivoting?
A) PIVOT
B) ROTATE
C) TURN
D) FLIP

Answer: A

---

51. PIVOT requires:
A) Aggregate function and column to pivot
B) Only column to pivot
C) Only aggregate function
D) No special requirements

Answer: A

---

52. In Oracle, pivoting can be done using:
A) PIVOT clause or CASE with aggregation
B) Only PIVOT clause
C) Only CASE
D) UNPIVOT

Answer: A

---

53. Convert monthly sales columns to rows uses:
A) UNPIVOT
B) PIVOT
C) JOIN
D) UNION

Answer: A

SECTION 5: HIERARCHICAL QUERIES (Questions 81-95)

54. In Oracle, hierarchical queries use:
A) CONNECT BY PRIOR
B) RECURSIVE WITH
C) TREE JOIN
D) LEVEL UP

Answer: A

---

55. The LEVEL pseudo-column in Oracle hierarchical queries:
A) Indicates depth in hierarchy
B) Shows salary level
C) Indicates access level
D) Shows importance level

Answer: A

---

56. START WITH clause in hierarchical queries:
A) Specifies root row(s)
B) Starts transaction
C) Begins query
D) Initializes variable

Answer: A

---

57. CONNECT BY PRIOR emp_id = manager_id:
A) Navigates from child to parent
B) Navigates from parent to child
C) Joins tables
D) Creates index

Answer: A

---

58. ORDER SIBLINGS BY:
A) Orders children within each parent
B) Orders entire result
C) Orders siblings only
D) Orders by level

Answer: A

---

59. In SQL Server, hierarchical data uses:
A) Recursive CTEs
B) CONNECT BY
C) LEVEL
D) TREE

Answer: A

---

60. The hierarchyid data type in SQL Server:
A) Stores hierarchical position
B) Stores tree structure
C) Stores level number
D) Stores parent ID

Answer: A

SECTION 6: JSON AND XML IN SQL (Questions 96-110)

61. JSON support in modern SQL includes:
A) JSON_VALUE, JSON_QUERY, JSON_MODIFY
B) JSON only as string
C) No JSON support
D) XML only

Answer: A

---

62. JSON_VALUE extracts:
A) Scalar value from JSON
B) JSON object/array
C) Entire JSON
D) XML data

Answer: A

---

63. JSON_QUERY extracts:
A) JSON object/array from JSON
B) Scalar value
C) Text value
D) Number

Answer: A

---

64. ISJSON() function:
A) Checks if string contains valid JSON
B) Parses JSON
C) Creates JSON
D) Modifies JSON

Answer: A

---

65. JSON_MODIFY updates:
A) Value in JSON
B) JSON structure only
C) JSON type
D) JSON validation

Answer: A

---

66. OPENJSON (SQL Server) is a:
A) Table-valued function to parse JSON
B) JSON validator
C) JSON creator
D) JSON modifier

Answer: A

---

67. XML in SQL Server uses:
A) XML data type, XQuery, XML indexes
B) TEXT only

C) VARCHAR only
D) No XML support

Answer: A

---

68. exist() method on XML column:
A) Checks if XQuery expression returns nodes
B) Returns XML nodes
C) Modifies XML
D) Validates XML

Answer: A

---

69. nodes() method in SQL Server XML:
A) Shreds XML into relational format
B) Creates XML
C) Validates XML
D) Indexes XML

Answer: A

---

70. value() method extracts:
A) Scalar value from XML
B) XML fragment
C) Entire XML
D) Attribute

Answer: A

SECTION 7: STORED PROCEDURES (Questions 111-125)

71. A stored procedure is:
A) Precompiled SQL code stored in database
B) Regular SQL query
C) View
D) Index

Answer: A

---

72. Stored procedures can have:
A) Input and output parameters
B) Only input parameters
C) Only output parameters
D) No parameters

Answer: A

---

73. Benefits of stored procedures:
A) Performance, security, reusability
B) Only performance
C) Only security
D) Only reusability

Answer: A

---

74. CREATE PROCEDURE syntax:
A) CREATE PROCEDURE proc_name AS BEGIN ... END
B) CREATE PROC proc_name AS BEGIN ... END
C) Both A and B
D) CREATE SP proc_name

Answer: C

---

75. EXEC or EXECUTE is used to:
A) Run a stored procedure
B) Create procedure
C) Drop procedure
D) Alter procedure

Answer: A

---

76. Output parameters in stored procedures:
A) Return values to caller
B) Are not allowed
C) Are input only
D) Are constants

Answer: A

---

77. RETURN in stored procedure:
A) Exits procedure and returns integer status
B) Returns dataset
C) Continues execution
D) Commits transaction

Answer: A

---

78. Try-Catch in stored procedures handles:
A) Errors
B) Warnings
C) Messages
D) Information

Answer: A

SECTION 8: FUNCTIONS (Questions 126-140)

79. Scalar functions return:
A) Single value
B) Table
C) Multiple values
D) Cursor

Answer: A

---

80. Table-valued functions return:
A) Table

B) Single value
C) Scalar
D) Cursor

Answer: A

---

81. Inline table-valued function uses:
A) Single SELECT statement
B) Multiple statements
C) BEGIN...END block
D) No SELECT

Answer: A

---

82. Multi-statement table-valued function:
A) Uses BEGIN...END, can have multiple statements
B) Single SELECT only
C) Cannot return table
D) Returns scalar

Answer: A

---

83. Deterministic functions:
A) Always return same result for same input
B) May return different results
C) Depend on database state
D) Use random values

Answer: A

---

84. Non-deterministic functions include:
A) GETDATE(), RAND()
B) ABS(), ROUND()
C) UPPER(), LOWER()
D) CONCAT()

Answer: A

---

85. Functions cannot:
A) Modify database state (no INSERT/UPDATE/DELETE)
B) Have parameters
C) Return values
D) Use variables

Answer: A

SECTION 9: TRIGGERS (Questions 141-155)

86. A trigger is:
A) Automatic code execution on data changes
B) Manual query
C) View
D) Index

Answer: A

---

87. DML triggers fire on:
A) INSERT, UPDATE, DELETE
B) SELECT only
C) CREATE only
D) DROP only

Answer: A

---

88. DDL triggers fire on:
A) CREATE, ALTER, DROP statements
B) Data changes
C) SELECT queries
D) LOGIN events

Answer: A

---

89. INSTEAD OF triggers:
A) Replace the original operation
B) Execute after operation
C) Execute before operation
D) Don't execute

Answer: A

---

90. AFTER triggers:
A) Execute after operation completes
B) Execute before operation
C) Replace operation
D) Execute instead of operation

Answer: A

---

91. FOR triggers in SQL Server:
A) Same as AFTER triggers
B) Same as INSTEAD OF
C) Different type
D) Not used

Answer: A

---

92. Inserted and Deleted virtual tables in triggers contain:
A) Affected rows
B) All table rows
C) Schema information
D) Index information

Answer: A

---

93. In INSERT trigger, the Inserted table contains:
A) New rows
B) Old rows
C) Both old and new
D) No rows

Answer: A

---

94. In DELETE trigger, the Deleted table contains:
A) Removed rows
B) New rows
C) Both
D) No rows

Answer: A

---

95. In UPDATE trigger:
A) Inserted has new values, Deleted has old values
B) Inserted has old values, Deleted has new
C) Both have same
D) Only one table used

Answer: A

---

96. Nested triggers:
A) Trigger that causes another trigger to fire
B) Triggers inside triggers
C) Multiple triggers on same table
D) Recursive triggers

Answer: A

---

97. Recursive triggers:
A) Trigger that calls itself
B) Nested triggers
C) Multiple triggers
D) System triggers

Answer: A

SECTION 10: QUERY OPTIMIZATION (Questions 156-175)

98. An execution plan shows:
A) How SQL engine executes query
B) Query syntax
C) Table structure
D) Index definitions

Answer: A

---

99. Estimated execution plan vs Actual execution plan:
A) Estimated is predicted, Actual is executed
B) Both same
C) Actual is predicted
D) Estimated is executed

Answer: A

---

100. Table scan means:
A) Reading entire table
B) Using index
C) Fast access
D) No access

Answer: A

---

101. Index seek is:
A) Efficient direct access via index
B) Scanning entire index
C) Table scan
D) No index used

Answer: A

---

102. Index scan is:
A) Scanning entire index structure

B) Direct lookup
C) Table scan
D) No access

Answer: A

---

103. Key lookup (Bookmark lookup):
A) Retrieving data from table after index find
B) Looking up key in index
C) Finding primary key
D) Foreign key check

Answer: A

---

104. SARGable queries are:
A) Can use indexes effectively
B) Cannot use indexes
C) Always slow
D) Use table scans

Answer: A

---

105. Which is SARGable?
A) WHERE salary > 50000
B) WHERE YEAR(date) = 2023
C) WHERE UPPER(name) = 'SMITH'
D) WHERE salary + 1000 > 50000

Answer: A

---

106. Statistics in SQL Server:
A) Help query optimizer estimate cardinality
B) Store data
C) Create indexes
D) Enforce constraints

Answer: A

---

107. Missing index suggestions appear in:
A) Execution plan
B) Query results
C) Error log
D) System tables only

Answer: A

---

108. Covering index includes:
A) All columns needed by query
B) Only key columns
C) Only included columns
D) Primary key only

Answer: A

---

109. Filtered index is:
A) Index on subset of rows
B) Index with WHERE clause
C) Index on all rows
D) Full table index

Answer: A

---

110. Index fragmentation occurs when:
A) Index pages become disorganized
B) Index is too large
C) Index is missing
D) Index is unique

Answer: A

---

111. Rebuilding index:
A) Defragments and updates statistics
B) Only updates statistics

C) Only defragments
D) Drops index

Answer: A

SECTION 11: DYNAMIC SQL (Questions 176-185)

112. Dynamic SQL is:
A) SQL constructed and executed at runtime
B) Static SQL statements
C) Stored procedures
D) Views

Answer: A

---

113. sp_executesql in SQL Server:
A) Executes dynamic SQL
B) Creates stored procedure
C) Drops procedure
D) Alters table

Answer: A

---

114. EXECUTE IMMEDIATE in Oracle:
A) Executes dynamic SQL
B) Immediate execution
C) Fast query
D) Direct execution

Answer: A

---

115. SQL injection is a risk with:
A) Dynamic SQL
B) Static SQL
C) Stored procedures
D) Views

Answer: A

---

116. Parameterized queries help prevent:
A) SQL injection
B) Syntax errors
C) Performance issues
D) Deadlocks

Answer: A

SECTION 12: PERFORMANCE TUNING (Questions 186-195)

117. SET STATISTICS IO ON:
A) Shows IO statistics for query
B) Shows time statistics
C) Enables indexing
D) Disables logging

Answer: A

---

118. SET STATISTICS TIME ON:
A) Shows time statistics for query
B) Shows IO statistics
C) Times out query
D) Sets timeout

Answer: A

---

119. Logical reads indicate:
A) Pages read from cache
B) Pages read from disk
C) Both
D) Index reads

Answer: A

---

120. Physical reads indicate:
A) Pages read from disk
B) Pages read from cache
C) Index reads
D) Table scans

Answer: A

---

121. Read-ahead reads:
A) Prefetching pages into cache
B) Actual reads
C) Logical reads
D) Physical reads

Answer: A

---

122. A high number of page splits indicates:
A) Index fragmentation
B) Good performance
C) Proper indexing
D) No issues

Answer: A

---

123. Fill factor determines:
A) How full index pages are initially
B) Table size
C) Database size
D) Log size

Answer: A

---

124. Low fill factor:
A) More space, less page splits
B) Less space, more page splits

C) Faster queries
D) Slower inserts

Answer: A

---

125. Wait statistics help identify:
A) Performance bottlenecks
B) Syntax errors
C) Missing tables
D) Login issues

Answer: A

SECTION 13: ADVANCED DATA TYPES (Questions 196-210)

126. HierarchyID data type stores:
A) Hierarchical position
B) Geographic data
C) Spatial data
D) JSON data

Answer: A

---

127. Geography data type stores:
A) Spatial data (round earth)
B) Planar spatial data
C) JSON
D) XML

Answer: A

---

128. Geometry data type stores:
A) Planar spatial data
B) Round earth data
C) Geographic data
D) Hierarchical data

Answer: A

---

129. FILESTREAM in SQL Server:
A) Stores BLOB data in file system with transactional consistency
B) Stores files in database only
C) Streams video
D) File system only

Answer: A

---

130. FileTable in SQL Server:
A) Integrates files with database
B) Regular table
C) File system
D) Temp table

Answer: A

SECTION 14: ADVANCED QUERY PATTERNS (Questions 211-225)

131. Gaps and islands problems deal with:
A) Consecutive sequences and missing values
B) Database gaps
C) Index fragmentation
D) Deadlocks

Answer: A

---

132. Finding consecutive days present uses:
A) Window functions with difference of row numbers
B) Simple GROUP BY
C) JOIN only
D) UNION only

Answer: A

---

133. Median calculation in SQL can use:
A) PERCENTILE_CONT or NTILE with window functions
B) AVG only
C) SUM only
D) COUNT only

Answer: A

---

134. Mode (most frequent value) can be found using:
A) RANK() with COUNT(*) and window functions
B) AVG()
C) SUM()
D) MIN()

Answer: A

---

135. Running totals use:
A) SUM() OVER (ORDER BY ...)
B) GROUP BY
C) JOIN
D) UNION

Answer: A

---

136. Year-over-year comparison uses:
A) LAG() with same period previous year
B) JOIN with self
C) UNION
D) GROUP BY

Answer: A

---

137. Pareto analysis (80/20 rule) uses:
A) Window functions with running totals
B) Simple COUNT
C) AVG only
D) MIN only

Answer: A

SECTION 1: INTRODUCTION TO DATABASE DESIGN (Questions 1-20)

1. Database design is the process of:
A) Creating a detailed data model of a database
B) Writing SQL queries
C) Creating indexes
D) Backing up data

Answer: A
Explanation: Database design involves creating a detailed structure representing real-world requirements.

---

2. The two main approaches to database design are:
A) Top-down and bottom-up
B) Left-right and right-left
C) Inside-out and outside-in
D) Forward and reverse

Answer: A

---

3. Top-down design starts with:
A) Entity-Relationship model development
B) Identifying individual attributes
C) Creating tables
D) Writing queries

Answer: A

---

4. Bottom-up design starts with:
A) Identifying attributes and grouping into relations
B) Creating ER diagrams
C) Defining business rules
D) Identifying entities

Answer: A

---

5. The main phases of database design are:
A) Conceptual, Logical, Physical
B) Planning, Analysis, Implementation
C) Design, Development, Testing
D) Requirements, Coding, Deployment

Answer: A

---

6. Conceptual design produces:
A) ER diagram independent of DBMS
B) SQL tables
C) Index definitions
D) Storage specifications

Answer: A

---

7. Logical design produces:
A) Relational schema (tables, constraints)
B) Physical storage details
C) Index structures
D) Query optimization plans

Answer: A

---

8. Physical design produces:
A) Storage structures, indexes, partitioning
B) ER diagram
C) Business rules
D) Requirements document

Answer: A

---

9. A well-designed database should have:
A) Minimal redundancy and no anomalies
B) Maximum redundancy for performance
C) Duplicate data for safety
D) No constraints for flexibility

Answer: A

---

10. Data redundancy leads to:
A) Update anomalies and wasted storage
B) Better performance
C) Easier queries
D) Faster backups

Answer: A

SECTION 2: FUNCTIONAL DEPENDENCIES (Questions 21-45)

11. A functional dependency X → Y means:
A) X uniquely determines Y
B) Y uniquely determines X
C) X and Y are related
D) X equals Y

Answer: A

---

12. In functional dependency, X is called:
A) Determinant
B) Dependent
C) Attribute set
D) Key

Answer: A

---

13. Given R(A,B,C) and A → B, if two tuples have same A value, they must have:
A) Same B value
B) Different B values
C) Same C value
D) Different C values

Answer: A

---

14. A functional dependency is trivial if:
A) $Y \subseteq X$
B) $X \subseteq Y$
C) $X = Y$
D) $X \cap Y = \emptyset$

Answer: A

---

15. Example of trivial FD:
A) A,B → A
B) A → B
C) A,B → C
D) A → B,C

Answer: A

---

16. Armstrong's axioms are used for:
A) Inferring all functional dependencies
B) Creating indexes
C) Query optimization
D) Data encryption

Answer: A

---

17. Armstrong's axioms include:
A) Reflexivity, Augmentation, Transitivity
B) Reflexivity, Symmetry, Transitivity
C) Reflexivity, Augmentation, Union
D) Decomposition, Union, Pseudo-transitivity

Answer: A

---

18. Reflexivity rule states:
A) If $Y \subseteq X$, then $X \rightarrow Y$
B) If $X \rightarrow Y$, then $X \rightarrow Y$
C) If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$
D) If $X \rightarrow Y$, then $XZ \rightarrow YZ$

Answer: A

---

19. Augmentation rule states:
A) If $X \rightarrow Y$, then $XZ \rightarrow YZ$
B) If $X \rightarrow Y$, then $X \rightarrow YZ$
C) If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$
D) If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$

Answer: A

---

20. Transitivity rule states:
A) If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$
B) If $X \rightarrow Y$, then $XZ \rightarrow YZ$
C) If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$
D) If $X \rightarrow Y$, then $X \rightarrow Y$

Answer: A

---

21. Additional inference rules derived from Armstrong's axioms include:
A) Union, Decomposition, Pseudo-transitivity
B) Union, Intersection, Difference
C) Join, Product, Division
D) Select, Project, Rename

Answer: A

---

22. Union rule states:
A) If X → Y and X → Z, then X → YZ
B) If X → YZ, then X → Y and X → Z
C) If X → Y and YZ → W, then XZ → W
D) If X → Y, then XZ → Y

Answer: A

---

23. Decomposition rule states:
A) If X → YZ, then X → Y and X → Z
B) If X → Y and X → Z, then X → YZ
C) If X → Y and YZ → W, then XZ → W
D) If X → Y, then XZ → Y

Answer: A

---

24. Pseudo-transitivity rule states:
A) If X → Y and YZ → W, then XZ → W
B) If X → Y and Y → Z, then X → Z
C) If X → YZ, then X → Y and X → Z
D) If X → Y and X → Z, then X → YZ

Answer: A

---

25. The closure of a set of attributes X+ is:
A) Set of all attributes functionally determined by X
B) Set of all keys
C) Set of all FDs
D) Set of all relations

Answer: A

---

26. To find if X → Y holds, we can:
A) Compute X+ and check if Y ⊆ X+
B) Check all tuples manually
C) Use trial and error
D) Assume it holds

Answer: A

---

27. The closure of a set of FDs F+ is:
A) All FDs that can be derived from F
B) All attributes in the relation
C) All keys
D) All superkeys

Answer: A

---

28. A candidate key can be found by computing:
A) Attribute closure that includes all attributes
B) Any attribute
C) First attribute only
D) Last attribute only

Answer: A

---

29. A superkey is:
A) A set of attributes that determines all attributes
B) A primary key
C) A foreign key
D) Any attribute

Answer: A

---

30. A candidate key is:
A) Minimal superkey
B) Any superkey
C) Primary key
D) Foreign key

Answer: A

---

31. Given R(A,B,C) with FDs: A → B, B → C. Candidate key(s) is/are:
A) A
B) B
C) C
D) AB

Answer: A

---

32. Given R(A,B,C) with FDs: AB → C, C → A. Candidate key(s) is/are:
A) AB and BC
B) A only
C) B only
D) C only

Answer: A

---

33. The closure of FDs can be computed using:
A) Armstrong's axioms repeatedly
B) Trial and error
C) Random selection
D) Table scan

Answer: A

---

34. An FD X → Y is redundant if:
A) It can be derived from other FDs
B) It is trivial
C) It is non-trivial
D) It involves all attributes

Answer: A

---

35. A minimal cover of FDs is:
A) Smallest set of FDs equivalent to original
B) Largest set of FDs
C) Set of all FDs
D) Set of candidate keys

Answer: A

---

36. Minimal cover must satisfy:
A) Right side single attribute, no redundant FDs, no extraneous attributes
B) Left side single attribute
C) All FDs trivial
D) All FDs non-trivial

Answer: A

SECTION 3: FIRST NORMAL FORM (1NF) (Questions 46-60)

37. A relation is in 1NF if:
A) All attributes are atomic (no repeating groups)
B) It has a primary key
C) No transitive dependencies
D) No partial dependencies

Answer: A

---

38. Which violates 1NF?
A) Attribute containing multiple phone numbers in one cell
B) Attribute with NULL value
C) Attribute with integer value
D) Attribute with date value

Answer: A

---

39. Repeating groups in a relation indicate:
A) Violation of 1NF

B) Compliance with 1NF
C) Good design
D) Normalized design

Answer: A

---

40. To convert to 1NF, we can:
A) Create separate table for repeating group or flatten
B) Add more columns
C) Remove all NULLs
D) Add primary key

Answer: A

---

41. Example of non-atomic attribute:
A) Address that contains street, city, state, zip
B) Salary amount
C) Employee ID
D) Hire date

Answer: A

---

42. 1NF allows:
A) Only atomic values per cell
B) Multi-valued attributes
C) Composite attributes
D) Nested relations

Answer: A

---

43. A table with columns (EmpID, Phone1, Phone2, Phone3) is in 1NF but has:
A) Design problem (limited number of phones)
B) Good design
C) No issues
D) Atomic values problem

Answer: A

SECTION 4: SECOND NORMAL FORM (2NF) (Questions 61-75)

44. A relation is in 2NF if:
A) It is in 1NF and no partial dependencies
B) It is in 1NF only
C) No transitive dependencies
D) All determinants are candidate keys

Answer: A

---

45. Partial dependency means:
A) Non-key attribute depends on part of composite key
B) Attribute depends on whole key
C) Attribute depends on another non-key
D) Key depends on non-key

Answer: A

---

46. Given R(StudentID, CourseID, StudentName, Grade) with key (StudentID, CourseID). Which is a partial dependency?
A) StudentID → StudentName
B) StudentID, CourseID → Grade
C) CourseID → Grade
D) StudentID, CourseID → StudentName

Answer: A

---

47. 2NF violation occurs only in relations with:
A) Composite primary keys
B) Single attribute keys
C) No keys
D) Foreign keys

Answer: A

---

48. To convert to 2NF, we:
A) Remove partial dependencies by decomposing
B) Add more attributes
C) Remove all dependencies
D) Add surrogate key

Answer: A

---

49. Given R(A,B,C,D) with key (A,B) and FDs: A → C, B → D. This relation is in:
A) 1NF but not 2NF
B) 2NF
C) 3NF
D) BCNF

Answer: A

---

50. After decomposing the above into 2NF, we get:
A) R1(A,C), R2(B,D), R3(A,B)
B) R1(A,B,C), R2(B,D)
C) R1(A,C), R2(A,B,D)
D) R1(A,B), R2(C,D)

Answer: A

SECTION 5: THIRD NORMAL FORM (3NF) (Questions 76-95)

51. A relation is in 3NF if:
A) It is in 2NF and no transitive dependencies
B) It is in 2NF only
C) No partial dependencies
D) All determinants are candidate keys

Answer: A

---

52. Transitive dependency means:
A) Non-key attribute depends on another non-key attribute
B) Attribute depends on key
C) Key depends on non-key
D) Attribute depends on part of key

Answer: A

---

53. Given R(EmpID, DeptID, DeptName) with EmpID → DeptID, DeptID → DeptName. This has:
A) Transitive dependency EmpID → DeptName
B) Partial dependency
C) No dependency
D) Direct dependency only

Answer: A

---

54. This relation is in:
A) 2NF but not 3NF
B) 3NF
C) BCNF
D) 1NF only

Answer: A

---

55. To convert to 3NF, we decompose into:
A) R1(EmpID, DeptID), R2(DeptID, DeptName)
B) R1(EmpID, DeptName), R2(DeptID, DeptName)
C) R1(EmpID, DeptID, DeptName)
D) R1(EmpID), R2(DeptID, DeptName)

Answer: A

---

56. 3NF allows:
A) A non-key to determine another non-key if the determinant is a candidate key

B) No dependencies at all
C) Only key dependencies
D) Partial dependencies

Answer: A

---

57. The 3NF definition: For every FD X → A, either:
A) X is a superkey or A is part of some candidate key
B) X is a key only
C) A is non-key
D) X is non-key

Answer: A

---

58. A relation in 3NF but not BCNF has:
A) Overlapping candidate keys with dependencies
B) No candidate keys
C) Single attribute key
D) No FDs

Answer: A

---

59. Example of 3NF but not BCNF:
A) R(A,B,C) with FDs: AB → C, C → B
B) R(A,B) with A → B
C) R(A,B,C) with A → B, B → C
D) R(A,B) with no FDs

Answer: A

SECTION 6: BOYCE-CODD NORMAL FORM (BCNF) (Questions 96-115)

60. A relation is in BCNF if:
A) For every non-trivial FD X → Y, X is a superkey

B) For every FD, Y is part of key
C) No FDs exist
D) All attributes are keys

Answer: A

---

61. BCNF is stronger than 3NF because:
A) Every determinant must be a superkey
B) No transitive dependencies
C) No partial dependencies
D) All attributes atomic

Answer: A

---

62. A relation in 3NF may not be in BCNF if:
A) There are overlapping candidate keys with FDs
B) There is a single candidate key
C) All attributes are prime
D) No FDs exist

Answer: A

---

63. Given R(A,B,C) with FDs: AB → C, C → B. Candidate keys are:
A) AB and AC
B) AB only
C) AC only
D) BC

Answer: A

---

64. In this relation, C → B violates BCNF because:
A) C is not a superkey
B) C is a superkey
C) B is part of key
D) No violation

Answer: A

---

65. To convert to BCNF, we decompose into:
A) R1(C,B), R2(A,C)
B) R1(A,B,C), R2(C,B)
C) R1(A,B), R2(C,B)
D) R1(A,C), R2(A,B)

Answer: A

---

66. BCNF decomposition is:
A) Lossless but may not preserve all FDs
B) Always preserves all FDs
C) May be lossy
D) Cannot be done

Answer: A

---

67. Lossless join decomposition means:
A) Natural join of decomposed relations yields original without spurious tuples
B) No data is lost
C) All FDs preserved
D) No redundancy

Answer: A

---

68. Dependency preservation means:
A) All FDs can be checked on individual relations
B) All data preserved
C) All keys preserved
D) No constraints needed

Answer: A

---

69. The chase algorithm is used to test:
A) Lossless join property
B) Dependency preservation

C) Normal form
D) Key status

Answer: A

SECTION 7: HIGHER NORMAL FORMS (Questions 116-135)

70. Fourth Normal Form (4NF) deals with:
A) Multivalued dependencies
B) Functional dependencies
C) Join dependencies
D) Transitive dependencies

Answer: A

---

71. A multivalued dependency (MVD) X →→ Y means:
A) Y values are independent of other attributes
B) X determines Y uniquely
C) Y determines X
D) X and Y are related

Answer: A

---

72. MVD occurs when:
A) Two independent multi-valued facts exist in same relation
B) There is a functional dependency
C) There is a transitive dependency
D) There is a partial dependency

Answer: A

---

73. Example of MVD: Person has multiple degrees and multiple languages:
A) Person →→ Degree, Person →→ Language
B) Person → Degree

C) Person → Language
D) Degree → Person

Answer: A

---

74. A relation in 4NF must be in BCNF and have:
A) No non-trivial MVDs except FDs
B) No MVDs at all
C) No FDs
D) Only trivial MVDs

Answer: A

---

75. Fifth Normal Form (5NF) deals with:
A) Join dependencies
B) Multivalued dependencies
C) Functional dependencies
D) Transitive dependencies

Answer: A

---

76. A join dependency (JD) means:
A) Relation can be reconstructed from its projections
B) Relation can be joined
C) Relation has joins
D) Relation has dependencies

Answer: A

---

77. A relation in 5NF has:
A) No non-trivial join dependencies
B) No FDs
C) No MVDs
D) All attributes as key

Answer: A

---

78. Domain-Key Normal Form (DKNF) is:
A) Ultimate normal form with no modification anomalies
B) Same as 5NF
C) Same as BCNF
D) Same as 3NF

Answer: A

SECTION 8: DECOMPOSITION PROPERTIES (Questions 136-150)

79. Lossless join decomposition ensures:
A) No information is lost when joining back
B) All FDs preserved
C) No redundancy
D) All keys preserved

Answer: A

---

80. The condition for lossless join decomposition of R into R1 and R2 is:
A) $R1 \cap R2 \rightarrow R1$ or $R1 \cap R2 \rightarrow R2$
B) $R1 \cap R2 = \emptyset$
C) $R1 \cup R2 = R$
D) $R1 = R2$

Answer: A

---

81. Dependency preserving decomposition allows:
A) Each FD checked on individual relations
B) FDs checked only after join
C) No FD checking
D) Only key constraints

Answer: A

---

82. The projection of FDs on a decomposed relation includes:
A) FDs that involve only attributes of that relation
B) All original FDs
C) No FDs
D) Only key FDs

Answer: A

---

83. A decomposition may be lossless but not:
A) Dependency preserving
B) Useful
C) Possible
D) Normalized

Answer: A

---

84. Example of lossless but not dependency preserving:
A) BCNF decomposition of R(A,B,C) with AB → C, C → B
B) 3NF decomposition
C) 2NF decomposition
D) 1NF decomposition

Answer: A

SECTION 9: DENORMALIZATION (Questions 151-160)

85. Denormalization is:
A) Introducing redundancy for performance
B) Further normalization
C) Removing all redundancy
D) Creating indexes

Answer: A

---

86. Denormalization is used to:
A) Improve query performance at cost of redundancy
B) Improve data integrity
C) Reduce storage
D) Eliminate anomalies

Answer: A

---

87. Examples of denormalization include:
A) Pre-joining tables, storing derived columns
B) Splitting tables
C) Creating more tables
D) Adding foreign keys

Answer: A

---

88. Denormalization risks:
A) Update anomalies, data inconsistency
B) Slower queries
C) Better integrity
D) Less storage

Answer: A

SECTION 10: DESIGN ANOMALIES (Questions 161-170)

89. Insertion anomaly occurs when:
A) Cannot insert data due to unrelated data requirement
B) Cannot insert duplicate
C) Insert is slow
D) Insert fails

Answer: A

---

90. Example of insertion anomaly:
A) Cannot add new department without employee

B) Cannot add employee without ID
C) Cannot add duplicate employee
D) Cannot add NULL

Answer: A

---

91. Deletion anomaly occurs when:
A) Deleting data removes unintended information
B) Cannot delete data
C) Delete is slow
D) Delete fails

Answer: A

---

92. Example of deletion anomaly:
A) Deleting last employee removes department info
B) Deleting employee with ID
C) Deleting duplicate
D) Deleting NULL

Answer: A

---

93. Update anomaly occurs when:
A) Updating data requires multiple changes due to redundancy
B) Cannot update
C) Update is slow
D) Update fails

Answer: A

---

94. Example of update anomaly:
A) Changing department name in multiple employee rows
B) Changing employee salary
C) Changing employee name
D) Changing employee ID

Answer: A

SECTION 11: PRACTICAL DESIGN (Questions 171-185)

95. Surrogate keys are:
A) Artificially generated unique identifiers
B) Natural keys from data
C) Composite keys
D) Foreign keys

Answer: A

---

96. Advantages of surrogate keys:
A) Stable, simple, independent of business changes
B) Meaningful
C) Based on real data
D) Always composite

Answer: A

---

97. Natural keys are:
A) Keys that exist in real-world data
B) Generated by system
C) Always single attribute
D) Always numeric

Answer: A

---

98. When to use natural keys:
A) When natural key is stable and unique
B) Always
C) Never
D) Only for small tables

Answer: A

---

99. Recursive relationships in design:
A) Entity relates to itself (e.g., employee-manager)
B) Entity relates to another entity
C) Entity has no relationships
D) Entity relates to multiple entities

Answer: A

---

100. Associative entities are used for:
A) Many-to-many relationships
B) One-to-many relationships
C) One-to-one relationships
D) Recursive relationships

Answer: A

---

101. Generalization in database design:
A) Subtype-supertype relationships
B) Specialization only
C) Aggregation
D) Composition

Answer: A

---

102. Specialization is:
A) Creating subtypes from supertype
B) Creating supertype from subtypes
C) Joining tables
D) Normalizing

Answer: A

---

103. Aggregation represents:
A) Has-a relationship
B) Is-a relationship
C) Uses-a relationship
D) Creates-a relationship

Answer: A

---

104. Composition is:
A) Strong form of aggregation with ownership
B) Weak aggregation
C) Generalization
D) Specialization

Answer: A

SECTION 12: NORMALIZATION ALGORITHMS (Questions 186-195)

105. The 3NF synthesis algorithm:
A) Produces dependency-preserving 3NF decomposition
B) Produces BCNF decomposition
C) Always lossless
D) Always dependency preserving

Answer: A

---

106. The BCNF decomposition algorithm:
A) May not preserve all FDs
B) Always preserves FDs
C) Always produces 3NF
D) Cannot be applied

Answer: A

---

107. To find candidate keys, we:
A) Compute attribute closures
B) Guess
C) Try all combinations
D) Use first attribute only

Answer: A

---

108. Normalization up to 3NF is usually sufficient because:
A) Higher normal forms handle rare cases
B) 3NF has no anomalies
C) BCNF is impossible
D) 4NF always achieved

Answer: A

---

109. The Universal Relation assumption:
A) All attributes come from one universal table
B) Multiple tables are needed
C) No tables exist
D) Only views exist

Answer: A

---

110. View integration in database design:
A) Combining multiple user views into conceptual schema
B) Creating views
C) Dropping views
D) Modifying views

Answer: A

SECTION 13: NORMALIZATION EXAMPLES (Questions 196-210)

111. Consider R(OrderID, ProductID, ProductName, Quantity). Assuming ProductID → ProductName, this relation is in:
A) 2NF (if key is OrderID, ProductID - partial dependency ProductID → ProductName)
B) 3NF

C) BCNF
D) 1NF only

Answer: A

---

112. Consider R(StudentID, CourseID, Instructor, Grade). Assuming CourseID → Instructor, this is in:
A) 2NF (if key is StudentID, CourseID - partial dependency exists)
B) 3NF
C) BCNF
D) 1NF only

Answer: A

---

113. Consider R(SSN, Name, Address, DeptID, DeptName). With SSN → Name, Address, DeptID and DeptID → DeptName. This is in:
A) 2NF (since single attribute key, no partial, but transitive DeptID → DeptName)
B) 3NF
C) BCNF
D) 1NF only

Answer: A

---

114. After normalizing above to 3NF:
A) R1(SSN, Name, Address, DeptID), R2(DeptID, DeptName)
B) R1(SSN, Name, Address), R2(DeptID, DeptName)
C) R1(SSN, DeptID), R2(DeptID, DeptName)
D) R1(SSN, Name, Address, DeptName)

Answer: A

---

115. Consider R(A,B,C,D) with FDs: AB → C, C → D, D → A. Candidate keys are:
A) AB, DB, CB
B) A only
C) B only
D) AB only

Answer: A

---

116. The above relation is in:
A) 3NF but not BCNF (C → D, D → A violate BCNF but determinants are part of candidate keys)
B) BCNF
C) 2NF
D) 1NF

Answer: A

SECTION 14: DESIGN ALTERNATIVES (Questions 211-220)

117. Star schema in data warehousing is:
A) Central fact table with dimension tables
B) Normalized design
C) Snowflake schema
D) 3NF design

Answer: A

---

118. Snowflake schema is:
A) Normalized dimensions in star schema
B) Central fact table only
C) Denormalized design
D) No dimensions

Answer: A

---

119. Fact table contains:
A) Measures and foreign keys to dimensions
B) Only descriptive data
C) Only keys
D) Only aggregated data

Answer: A

---

120. Dimension table contains:
A) Descriptive attributes
B) Measures
C) Facts
D) Aggregates

Answer: A

SECTION 1: INTRODUCTION TO TRANSACTIONS (Questions 1-25)

1. A transaction is:
A) A unit of program execution that accesses and updates data
B) A single SQL statement
C) A database table
D) An index structure

Answer: A
Explanation: A transaction is a logical unit of work that may consist of multiple operations.

---

2. Which of the following is NOT a state of a transaction?
A) Active
B) Partially committed
C) Serialized
D) Failed

Answer: C

---

3. The correct sequence of transaction states is:
A) Active → Partially committed → Committed
B) Active → Committed → Partially committed
C) Partially committed → Active → Committed
D) Active → Failed → Committed

Answer: A

---

4. A transaction goes from active state to failed state when:
A) Normal execution cannot continue due to error
B) Transaction completes successfully
C) Transaction is rolled back
D) Transaction is aborted

Answer: A

---

5. When a transaction is rolled back, it goes to:
A) Aborted state
B) Committed state
C) Active state
D) Partially committed state

Answer: A

---

6. After rollback, the transaction can be:
A) Restarted or killed
B) Only killed
C) Only restarted
D) Automatically committed

Answer: A

---

7. The transaction state "Partially committed" means:
A) Final statement executed, but output may still be in memory
B) Transaction is still executing
C) Transaction is aborted
D) Transaction is written to disk

Answer: A

---

8. The transaction is considered committed when:
A) All changes are written permanently to database
B) Last statement is executed

C) Transaction begins
D) Transaction is rolled back

Answer: A

SECTION 2: ACID PROPERTIES (Questions 26-50)

9. ACID stands for:
A) Atomicity, Consistency, Isolation, Durability
B) Accuracy, Consistency, Isolation, Durability
C) Atomicity, Concurrency, Isolation, Durability
D) Atomicity, Consistency, Integrity, Durability

Answer: A

---

10. Atomicity ensures:
A) Transaction executes completely or not at all
B) Transaction maintains database consistency
C) Transaction is isolated from others
D) Transaction effects persist after commit

Answer: A

---

11. Which property ensures that either all operations of a transaction are reflected or none?
A) Atomicity
B) Consistency
C) Isolation
D) Durability

Answer: A

---

12. Consistency ensures:
A) Transaction transforms database from one consistent state to another
B) Data is never lost

C) Transactions run sequentially
D) Data is encrypted

Answer: A

---

13. A consistent database state means:
A) All integrity constraints are satisfied
B) All data is backed up
C) No transactions are running
D) All tables are normalized

Answer: A

---

14. Isolation ensures:
A) Concurrent transactions appear to execute in isolation
B) Transactions are serial
C) No transactions run concurrently
D) Data is encrypted

Answer: A

---

15. Which property is responsible for handling concurrency?
A) Isolation
B) Atomicity
C) Consistency
D) Durability

Answer: A

---

16. Durability ensures:
A) Once committed, changes persist even after system failure
B) Data is always available
C) Transactions never fail
D) Data is backed up

Answer: A

---

17. Which storage component helps achieve durability?
A) Non-volatile storage (disk)
B) RAM
C) Cache
D) Buffer

Answer: A

---

18. A transaction that violates consistency will be:
A) Rolled back
B) Committed
C) Paused
D) Ignored

Answer: A

---

19. The recovery manager is responsible for:
A) Atomicity and Durability
B) Consistency and Isolation
C) Only Atomicity
D) Only Durability

Answer: A

---

20. The concurrency control manager is responsible for:
A) Isolation
B) Atomicity
C) Durability
D) Consistency

Answer: A

SECTION 3: SCHEDULES AND SERIALIZABILITY (Questions 51-75)

21. A schedule is:
A) An order of execution of operations from multiple transactions
B) A transaction plan
C) A database backup
D) An index structure

Answer: A

---

22. A serial schedule is one where:
A) Transactions execute one after another without interleaving
B) Transactions execute concurrently
C) Transactions are interleaved
D) Transactions are parallel

Answer: A

---

23. A concurrent (interleaved) schedule is:
A) Operations of different transactions are interleaved
B) Transactions execute sequentially
C) Only one transaction at a time
D) No transaction interleaving

Answer: A

---

24. Serializability ensures that:
A) Concurrent schedule produces same result as some serial schedule
B) All schedules are serial
C) Transactions run in order
D) No concurrency allowed

Answer: A

---

25. A schedule is conflict serializable if:
A) It can be transformed to a serial schedule by swapping non-conflicting operations
B) It is already serial
C) It has no conflicts
D) It has only read operations

Answer: A

---

26. Two operations conflict if:
A) They belong to different transactions, access same data, and at least one is write
B) They are from same transaction
C) They are both reads
D) They access different data

Answer: A

---

27. Read(A) and Write(A) from different transactions:
A) Conflict
B) Do not conflict
C) Are compatible
D) Can be swapped

Answer: A

---

28. Write(A) and Read(A) from different transactions:
A) Conflict
B) Do not conflict
C) Are compatible
D) Can be swapped

Answer: A

---

29. Read(A) and Read(A) from different transactions:
A) Do not conflict
B) Conflict
C) Cannot occur
D) Must be serialized

Answer: A

---

30. A precedence graph (conflict graph) has:
A) Nodes = transactions, edges = conflicts
B) Nodes = operations, edges = order
C) Nodes = data items, edges = transactions
D) Nodes = schedules, edges = conflicts

Answer: A

---

31. A schedule is conflict serializable if the precedence graph has:
A) No cycles
B) At least one cycle
C) Multiple cycles
D) No edges

Answer: A

---

32. View serializability is:
A) More general than conflict serializability
B) Less general than conflict serializability
C) Same as conflict serializability
D) Unrelated to conflict serializability

Answer: A

---

33. A schedule is view serializable if:
A) It is view equivalent to some serial schedule
B) It is conflict serializable
C) It has no conflicts
D) It has only reads

Answer: A

---

34. Which is easier to test?
A) Conflict serializability
B) View serializability
C) Both equally
D) Neither

Answer: A

---

35. A schedule that is conflict serializable is always:
A) View serializable
B) Not view serializable
C) Serializable only sometimes
D) Non-serializable

Answer: A

---

36. Recoverable schedules require that:
A) A transaction commits only after all transactions it read from have committed
B) Transactions can commit anytime
C) No rollbacks allowed
D) All transactions commit together

Answer: A

---

37. Cascading rollback occurs when:
A) One transaction rollback causes others to rollback
B) Multiple transactions rollback independently
C) System rolls back all transactions
D) No rollback occurs

Answer: A

---

38. Cascadeless schedules prevent:
A) Cascading rollbacks
B) All conflicts
C) All writes
D) All reads

Answer: A

---

39. Strict schedules require:
A) Dirty reads are not allowed
B) Only clean reads
C) No writes
D) No reads

Answer: A

SECTION 4: CONCURRENCY CONTROL - LOCK-BASED PROTOCOLS (Questions 76-110)

40. Lock-based concurrency control uses:
A) Locks to control access to data items
B) Timestamps
C) Validation
D) No control

Answer: A

---

41. A shared lock (S-lock) allows:
A) Read only, multiple transactions can hold simultaneously
B) Write only
C) Both read and write
D) Exclusive access

Answer: A

---

42. An exclusive lock (X-lock) allows:
A) Both read and write, only one transaction can hold
B) Read only
C) Multiple transactions to hold
D) Only write

Answer: A

---

43. Lock compatibility matrix: S-lock and S-lock on same item:
A) Compatible
B) Incompatible
C) Not allowed
D) Depends

Answer: A

---

44. S-lock and X-lock on same item:
A) Incompatible
B) Compatible
C) Both allowed
D) Shared only

Answer: A

---

45. X-lock and X-lock on same item:
A) Incompatible
B) Compatible
C) Both allowed
D) Shared only

Answer: A

---

46. Two-phase locking (2PL) protocol requires:
A) All locks acquired before first unlock
B) Locks released anytime
C) No lock acquisition order
D) Only read locks

Answer: A

---

47. The two phases of 2PL are:
A) Growing phase (acquire locks) and Shrinking phase (release locks)
B) Lock phase and unlock phase
C) Read phase and write phase
D) Start phase and end phase

Answer: A

---

48. In growing phase, a transaction may:
A) Only acquire locks, not release
B) Only release locks
C) Both acquire and release
D) Neither acquire nor release

Answer: A

---

49. In shrinking phase, a transaction may:
A) Only release locks, not acquire
B) Only acquire locks
C) Both acquire and release
D) Neither

Answer: A

---

50. 2PL ensures:
A) Conflict serializability
B) No deadlocks
C) Always cascadeless
D) Always recoverable

Answer: A

---

51. Strict 2PL requires:
A) All exclusive locks held until commit
B) Locks released immediately after use
C) No exclusive locks
D) Only shared locks

Answer: A

---

52. Strict 2PL ensures:
A) Cascadeless schedules
B) No serializability
C) No concurrency
D) No locking

Answer: A

---

53. Rigorous 2PL requires:
A) All locks (shared and exclusive) held until commit
B) Only exclusive locks held
C) Locks released early
D) No locks

Answer: A

---

54. Conservative 2PL (preclaiming) requires:
A) All locks acquired before transaction begins
B) Locks acquired as needed
C) No pre-acquisition
D) Only shared locks pre-acquired

Answer: A

---

55. Conservative 2PL prevents:
A) Deadlocks
B) Serializability
C) Concurrency
D) All conflicts

Answer: A

---

56. Deadlock occurs when:
A) Each transaction waits for locks held by others in a cycle
B) Transaction waits too long
C) Lock is unavailable
D) System crashes

Answer: A

---

57. Deadlock prevention strategies include:
A) Wait-die and wound-wait
B) Only detection
C) Only avoidance
D) No prevention

Answer: A

---

58. In wait-die scheme:
A) Older transaction waits for younger, younger dies (rolls back)
B) Younger waits for older
C) Both wait
D) Both die

Answer: A

---

59. In wound-wait scheme:
A) Older wounds (aborts) younger, younger waits for older
B) Younger wounds older
C) Both wait
D) Both die

Answer: A

---

60. Deadlock detection uses:
A) Wait-for graph
B) Precedence graph
C) Conflict graph
D) Lock table

Answer: A

---

61. A cycle in wait-for graph indicates:
A) Deadlock
B) Serializability
C) No deadlock
D) Conflict

Answer: A

---

62. When deadlock is detected, we resolve by:
A) Aborting one or more transactions (victim selection)
B) Waiting indefinitely
C) Ignoring
D) Restarting system

Answer: A

---

63. Victim selection considers:
A) Transaction age, how many locks held, how far progressed
B) Random selection
C) First come first served
D) Last come first served

Answer: A

---

64. Starvation occurs when:
A) A transaction waits indefinitely for locks
B) Transaction commits quickly
C) No locks needed
D) System is fast

Answer: A

SECTION 5: TIMESTAMP-BASED PROTOCOLS (Questions 111-130)

65. Timestamp-based concurrency control uses:
A) Timestamps to order transactions
B) Locks
C) Validation
D) No ordering

Answer: A

---

66. Each transaction is assigned a unique:
A) Timestamp (TS)
B) Lock ID
C) Transaction number
D) Priority

Answer: A

---

67. The timestamp ordering rule for read operations:
A) Read only if transaction started after last write
B) Read anytime
C) Read only if last write is older
D) No rule

Answer: A

---

68. Thomas Write Rule allows:
A) Ignoring obsolete writes
B) All writes
C) No writes
D) Only reads

Answer: A

---

69. In basic timestamp ordering, if a transaction tries to read a data item written by a future transaction:
A) Transaction is aborted and restarted
B) Read is allowed
C) Write is ignored
D) Transaction waits

Answer: A

---

70. Timestamp protocols ensure:
A) Conflict serializability
B) No deadlocks (but may cause cascading aborts)
C) No aborts
D) Always recoverable

Answer: A and B

SECTION 6: MULTIVERSION CONCURRENCY CONTROL (Questions 131-140)

71. MVCC (Multiversion Concurrency Control) maintains:
A) Multiple versions of data items
B) Single version only
C) No versions
D) Only current version

Answer: A

---

72. In MVCC, readers:
A) Never wait for writers
B) Always wait for writers
C) Are blocked
D) Cannot read

Answer: A

---

73. Snapshot isolation allows:
A) Transaction sees snapshot of data at start
B) Always sees latest data
C) Sees only committed data
D) Sees uncommitted data

Answer: A

---

74. Snapshot isolation may allow:
A) Write skew anomaly
B) No anomalies
C) Dirty reads
D) Lost updates

Answer: A

SECTION 7: OPTIMISTIC CONCURRENCY CONTROL (Questions 141-150)

75. Optimistic concurrency control assumes:
A) Conflicts are rare, validate at commit
B) Conflicts are frequent
C) Always lock first
D) No validation needed

Answer: A

---

76. The three phases of optimistic concurrency control are:
A) Read, Validation, Write
B) Lock, Read, Write
C) Start, Execute, Commit
D) Begin, Read, End

Answer: A

---

77. Validation phase checks:
A) If transaction conflicts with others
B) If data is valid
C) If locks are held
D) If transaction is active

Answer: A

---

78. Optimistic concurrency is good for:
A) Low-conflict environments
B) High-conflict environments
C) Always
D) Never

Answer: A

SECTION 8: RECOVERY TECHNIQUES (Questions 151-175)

79. Recovery manages:
A) Restoring database to consistent state after failure
B) Only backups
C) Only transactions
D) Only indexes

Answer: A

---

80. Types of failures include:
A) Transaction failure, system crash, disk failure
B) Only transaction failure
C) Only system crash
D) Only disk failure

Answer: A

---

81. Log-based recovery uses:
A) A log of all database modifications

B) Only checkpoints
C) Only backups
D) No logs

Answer: A

---

82. The log contains:
A) <T, X, old value, new value>
B) Only transaction ID
C) Only data item
D) Only new values

Answer: A

---

83. Write-ahead logging (WAL) requires:
A) Log record written before data update
B) Data updated before log
C) Log and data written simultaneously
D) No log needed

Answer: A

---

84. Undo operation restores:
A) Old values for uncommitted transactions
B) New values
C) Transaction state
D) System state

Answer: A

---

85. Redo operation applies:
A) New values for committed transactions
B) Old values
C) Transaction start
D) System restart

Answer: A

---

86. Deferred update (NO-UNDO/REDO):
A) Updates only after commit, no undo needed
B) Updates immediately, undo needed
C) No logging
D) Only undo

Answer: A

---

87. Immediate update (UNDO/REDO):
A) Updates may happen before commit, both undo and redo needed
B) No undo needed
C) No redo needed
D) Only undo

Answer: A

---

88. Checkpoint is used to:
A) Reduce recovery time
B) Increase recovery time
C) Stop transactions
D) Backup database

Answer: A

---

89. At checkpoint:
A) All dirty buffers are written to disk
B) Transactions are aborted
C) System shuts down
D) Log is cleared

Answer: A

---

90. ARIES recovery algorithm uses:
A) Analysis, Redo, Undo phases
B) Only Redo

C) Only Undo
D) Only Analysis

Answer: A

---

91. In ARIES, the Log Sequence Number (LSN) identifies:
A) Each log record
B) Transaction ID
C) Data item
D) Page number

Answer: A

---

92. Dirty page table tracks:
A) Pages modified in buffer but not on disk
B) Clean pages
C) Index pages
D) System tables

Answer: A

SECTION 9: RECOVERY BASED ON DEFERRED UPDATE (Questions 176-185)

93. In deferred update protocol:
A) All updates are done after transaction commits
B) Updates done immediately
C) Updates done before commit
D) No updates

Answer: A

---

94. Deferred update requires:
A) REDO but no UNDO
B) UNDO but no REDO

C) Both UNDO and REDO
D) No recovery

Answer: A

---

95. If system crashes before commit:
A) No recovery needed (no updates written)
B) Must undo
C) Must redo
D) Both undo and redo

Answer: A

SECTION 10: RECOVERY BASED ON IMMEDIATE UPDATE (Questions 186-195)

96. In immediate update protocol:
A) Updates may be applied before commit
B) Updates only after commit
C) No updates
D) Updates in memory only

Answer: A

---

97. Immediate update may require:
A) Both UNDO and REDO
B) Only UNDO
C) Only REDO
D) No recovery

Answer: A

---

98. With immediate update and no logging, we cannot:
A) Undo uncommitted transactions
B) Redo committed transactions

C) Recover at all
D) Commit

Answer: A

---

99. Shadow paging technique:
A) Uses shadow pages for atomic updates
B) Uses logs
C) Uses checkpoints only
D) No recovery

Answer: A

---

100. Shadow paging advantages:
A) No logging overhead
B) Requires logging
C) Slow recovery
D) Complex implementation

Answer: A

SECTION 11: ADVANCED TOPICS (Questions 196-210)

101. Distributed transactions involve:
A) Multiple databases across network
B) Single database
C) Local only
D) No transactions

Answer: A

---

102. Two-phase commit (2PC) ensures:
A) Atomic commitment across distributed sites
B) Local commit only
C) No commit
D) Fast commit

Answer: A

---

103. 2PC phases are:
A) Prepare and Commit
B) Start and End
C) Begin and Finish
D) Lock and Unlock

Answer: A

---

104. In prepare phase, coordinator:
A) Asks all participants if ready to commit
B) Commits immediately
C) Aborts all
D) Does nothing

Answer: A

---

105. If any participant votes NO in prepare phase:
A) Global abort
B) Global commit
C) Partial commit
D) Retry

Answer: A

---

106. Long transactions require special handling because:
A) They hold locks for long time, reducing concurrency
B) They are fast
C) They never conflict
D) They don't need concurrency control

Answer: A

---

107. Nested transactions allow:
A) Transactions within transactions
B) Only flat transactions
C) No hierarchy
D) Single level only

Answer: A

---

108. Savepoints allow:
A) Partial rollback within transaction
B) Only full rollback
C) No rollback
D) Commit only

Answer: A

---

109. ACID in distributed systems is harder due to:
A) Network latency, partial failures
B) Single system
C) No failures
D) Fast networks

Answer: A

---

110. BASE (Basically Available, Soft state, Eventually consistent) is:
A) Alternative to ACID for distributed systems
B) Same as ACID
C) Stricter than ACID
D) Not used

Answer: A

SECTION 12: PRACTICAL CONSIDERATIONS (Questions 211-220)

111. Transaction isolation levels in SQL include:
A) READ UNCOMMITTED, READ COMMITTED, REPEATABLE READ, SERIALIZABLE
B) Only SERIALIZABLE
C) Only READ COMMITTED
D) No levels

Answer: A

---

112. READ UNCOMMITTED allows:
A) Dirty reads
B) No dirty reads
C) Phantom reads only
D) Repeatable reads

Answer: A

---

113. READ COMMITTED prevents:
A) Dirty reads
B) Non-repeatable reads
C) Phantom reads
D) All anomalies

Answer: A

---

114. REPEATABLE READ prevents:
A) Dirty reads and non-repeatable reads
B) Phantom reads
C) All anomalies
D) Only dirty reads

Answer: A

---

115. SERIALIZABLE prevents:
A) All anomalies (dirty read, non-repeatable read, phantom read)
B) Only dirty reads
C) Only phantom reads
D) No anomalies prevented

Answer: A

---

116. Phantom read occurs when:
A) New rows appear in subsequent reads due to inserts
B) Same row changes value
C) Dirty data is read
D) Transaction aborts

Answer: A

---

117. Lost update problem occurs when:
A) Two concurrent updates, one overwrites other without awareness
B) Update is lost due to system failure
C) Update is rolled back
D) Update never happens

Answer: A

---

118. Dirty read occurs when:
A) Reading uncommitted data that may be rolled back
B) Reading committed data
C) Reading old data
D) Reading new data

Answer: A

---

119. Non-repeatable read occurs when:
A) Same row read twice gets different values due to updates
B) Same query returns different rows
C) Different rows read
D) No reads

Answer: A

---

120. Choosing isolation level involves trade-off between:
A) Consistency and concurrency
B) Speed and security
C) Size and speed
D) Cost and performance

Answer: A

SECTION 1: INTRODUCTION TO CONCURRENCY CONTROL (Questions 1-20)

1. Concurrency control is responsible for:
A) Managing simultaneous access to database to ensure consistency
B) Controlling user access permissions
C) Managing database backups
D) Optimizing query performance

Answer: A
Explanation: Concurrency control coordinates simultaneous transactions to prevent interference.

---

2. The main problems that concurrency control addresses are:
A) Lost update, dirty read, non-repeatable read, phantom read
B) Deadlock only
C) System crashes only
D) Data redundancy only

Answer: A

---

3. Lost update problem occurs when:
A) Two transactions update same data item, one overwrites the other without awareness
B) Transaction reads uncommitted data
C) Same query returns different results
D) New rows appear in subsequent reads

Answer: A

---

4. Example of lost update:
A) T1 reads X=100, T2 reads X=100, T1 updates X=X+50, T2 updates X=X-30 → final X=70 (lost T1's update)
B) T1 reads X, T2 updates X
C) T1 updates X, T2 reads X
D) T1 and T2 both read X

Answer: A

---

5. Dirty read problem occurs when:
A) Transaction reads uncommitted data that may be rolled back
B) Transaction reads committed data
C) Transaction reads its own updates
D) Transaction reads from multiple tables

Answer: A

---

6. Example of dirty read:
A) T1 updates X=200 (uncommitted), T2 reads X=200, T1 aborts → T2 has read invalid data
B) T1 reads X, T2 reads X
C) T1 commits, T2 reads
D) T1 and T2 update same data

Answer: A

---

7. Non-repeatable read occurs when:
A) Same row read twice gets different values due to updates by other transaction
B) Same query returns different number of rows
C) Transaction reads its own updates
D) Transaction reads from different tables

Answer: A

---

8. Example of non-repeatable read:
A) T1 reads X=100, T2 updates X=200 and commits, T1 reads X again → gets 200
B) T1 reads X, T1 reads X again
C) T1 reads X, T2 reads X
D) T1 updates X, T2 reads X

Answer: A

---

9. Phantom read occurs when:
A) New rows appear in subsequent reads due to inserts by other transaction
B) Same row changes value
C) Transaction reads uncommitted data
D) Transaction updates same data

Answer: A

---

10. Example of phantom read:
A) T1 reads all employees with salary > 50000 (5 rows), T2 inserts new employee with salary 60000 and commits, T1 reads again → gets 6 rows
B) T1 reads employee record, T2 updates same record
C) T1 reads employee, T1 reads again
D) T2 inserts, T2 commits

Answer: A

---

11. Serializability ensures that:
A) Concurrent execution produces same result as some serial execution
B) All transactions run one after another
C) No concurrency is allowed
D) Transactions are isolated from each other

Answer: A

---

12. Conflict serializability is based on:
A) Swapping non-conflicting operations
B) Timestamp ordering

C) Lock acquisition
D) Validation

Answer: A

---

13. Two operations conflict if:
A) They belong to different transactions, access same data, and at least one is write
B) They are from same transaction
C) They access different data
D) They are both reads

Answer: A

SECTION 2: LOCK-BASED PROTOCOLS (Questions 21-50)

14. In lock-based concurrency control, a transaction must:
A) Acquire appropriate locks before accessing data
B) Access data without locks
C) Release locks before accessing
D) Never use locks

Answer: A

---

15. A shared lock (S-lock) allows:
A) Read only; multiple transactions can hold simultaneously
B) Write only
C) Both read and write
D) Exclusive access

Answer: A

---

16. An exclusive lock (X-lock) allows:
A) Both read and write; only one transaction can hold
B) Read only
C) Multiple transactions to hold
D) Write only

Answer: A

---

17. The lock compatibility matrix shows:
A) Which locks can be held simultaneously on same item
B) Which transactions are compatible
C) Which schedules are serializable
D) Which data items can be locked

Answer: A

---

18. S-lock and S-lock on same data item are:
A) Compatible
B) Incompatible
C) Not allowed
D) Depends on transaction

Answer: A

---

19. S-lock and X-lock on same data item are:
A) Incompatible
B) Compatible
C) Always allowed
D) Never used

Answer: A

---

20. X-lock and X-lock on same data item are:
A) Incompatible
B) Compatible
C) Always allowed
D) Shared locks only

Answer: A

---

21. Lock manager maintains:
A) Lock table with entries for each locked data item
B) Transaction table only
C) Index structures
D) Buffer cache

Answer: A

---

22. Lock upgrade occurs when:
A) Transaction converts S-lock to X-lock
B) Transaction releases lock
C) Transaction acquires new lock
D) Transaction commits

Answer: A

---

23. Lock downgrade occurs when:
A) Transaction converts X-lock to S-lock
B) Transaction acquires exclusive lock
C) Transaction releases lock
D) Transaction aborts

Answer: A

---

24. Two-Phase Locking (2PL) protocol requires:
A) All lock acquisitions before any lock releases
B) Locks can be released anytime
C) No order for locks
D) Only exclusive locks

Answer: A

---

25. The two phases of 2PL are:
A) Growing phase (acquire locks) and Shrinking phase (release locks)
B) Lock phase and unlock phase
C) Read phase and write phase
D) Start phase and commit phase

Answer: A

---

26. In growing phase, a transaction can only:
A) Acquire locks, not release
B) Release locks, not acquire
C) Both acquire and release
D) Neither

Answer: A

---

27. In shrinking phase, a transaction can only:
A) Release locks, not acquire
B) Acquire locks, not release
C) Both acquire and release
D) Neither

Answer: A

---

28. 2PL ensures:
A) Conflict serializability
B) Freedom from deadlocks
C) No cascading aborts
D) Always recoverable

Answer: A

---

29. Basic 2PL may allow:
A) Cascading rollbacks
B) No serializability
C) No concurrency
D) Phantom reads only

Answer: A

---

30. Strict 2PL requires:
A) All exclusive locks held until transaction commits
B) All locks released immediately after use
C) Only shared locks held until commit
D) No locks held until commit

Answer: A

---

31. Strict 2PL ensures:
A) Cascadeless schedules
B) No serializability
C) More concurrency
D) Fewer locks

Answer: A

---

32. Rigorous 2PL requires:
A) All locks (shared and exclusive) held until commit
B) Only exclusive locks held until commit
C) Locks released early
D) No locks held until commit

Answer: A

---

33. Conservative 2PL (preclaiming) requires:
A) All locks acquired before transaction begins
B) Locks acquired as needed
C) Locks acquired after start
D) No pre-acquisition

Answer: A

---

34. Conservative 2PL prevents:
A) Deadlocks
B) Serializability
C) All conflicts
D) All concurrency

Answer: A

---

35. Deadlock occurs when:
A) Each transaction in a set is waiting for locks held by another in a cycle
B) Transaction waits too long
C) Lock is unavailable
D) System is busy

Answer: A

---

36. Necessary conditions for deadlock (Coffman conditions) include:
A) Mutual exclusion, hold and wait, no preemption, circular wait
B) Only mutual exclusion
C) Only circular wait
D) Only hold and wait

Answer: A

---

37. Deadlock prevention strategies:
A) Wait-die and wound-wait
B) Only detection
C) Only avoidance
D) No prevention needed

Answer: A

---

38. In wait-die scheme:
A) Older transaction waits for younger; younger dies (aborts) if it needs lock held by older
B) Younger waits for older; older dies
C) Both wait
D) Both die

Answer: A

---

39. In wound-wait scheme:
A) Older wounds (aborts) younger; younger waits for older
B) Younger wounds older
C) Both wait
D) Both die

Answer: A

---

40. Wait-die is:
A) Non-preemptive (transaction never preempted, only aborts itself)
B) Preemptive
C) Both
D) Neither

Answer: A

---

41. Wound-wait is:
A) Preemptive (older can abort younger)
B) Non-preemptive
C) Both
D) Neither

Answer: A

---

42. Deadlock detection uses:
A) Wait-for graph
B) Precedence graph
C) Conflict graph
D) Lock table only

Answer: A

---

43. A cycle in wait-for graph indicates:
A) Deadlock
B) No deadlock
C) Serializability
D) Conflict

Answer: A

---

44. Deadlock detection frequency affects:
A) Overhead vs response time trade-off
B) Only overhead
C) Only response time
D) No effect

Answer: A

---

45. When deadlock detected, victim selection considers:
A) Transaction age, locks held, progress, rows updated
B) Random selection only
C) First come first served only
D) Transaction ID only

Answer: A

---

46. Starvation occurs when:
A) Transaction waits indefinitely for locks
B) Transaction commits quickly
C) No locks needed
D) System is idle

Answer: A

---

47. Starvation can be prevented by:
A) Fairness policies (e.g., first-come-first-served, aging)
B) Deadlock detection
C) Wait-die only
D) Wound-wait only

Answer: A

SECTION 3: TIMESTAMP-BASED PROTOCOLS (Questions 51-75)

48. Timestamp-based concurrency control assigns:
A) Unique timestamp to each transaction
B) Lock ID to each transaction
C) Priority number
D) Sequence number to each operation

Answer: A

---

49. Timestamp ordering ensures:
A) Conflict serializability by ordering transactions based on timestamps
B) No deadlocks always
C) Maximum concurrency
D) Minimum overhead

Answer: A

---

50. In basic timestamp ordering (BTO), each data item has:
A) Read-timestamp and write-timestamp
B) Only read-timestamp
C) Only write-timestamp
D) Lock information

Answer: A

---

51. Read-timestamp (R-TS) is:
A) Largest timestamp of transaction that successfully read the item
B) Smallest timestamp of reader
C) Average of read timestamps
D) Current time

Answer: A

---

52. Write-timestamp (W-TS) is:
A) Largest timestamp of transaction that successfully wrote the item
B) Smallest timestamp of writer
C) Average of write timestamps
D) Current time

Answer: A

---

53. When transaction T with timestamp TS tries to read item X:
A) If TS < W-TS(X), reject and abort T; else allow read and update R-TS(X)
B) If TS > W-TS, reject
C) Always allow read
D) Always reject

Answer: A

---

54. When transaction T tries to write item X:
A) If TS < R-TS(X) or TS < W-TS(X), reject and abort T; else allow write and update W-TS(X)
B) Always allow write
C) Always reject write
D) Allow if TS > R-TS only

Answer: A

---

55. Thomas Write Rule optimizes by:
A) Ignoring obsolete writes (when TS < W-TS)
B) Always performing writes
C) Never performing writes
D) Delaying writes

Answer: A

---

56. With Thomas Write Rule, if TS < W-TS:
A) Ignore the write (don't execute, don't abort)
B) Abort the transaction

C) Perform write anyway
D) Delay write

Answer: A

---

57. Timestamp protocols are:
A) Deadlock-free (no waiting for locks)
B) Prone to deadlocks
C) Lock-based
D) Validation-based

Answer: A

---

58. However, timestamp protocols may cause:
A) Cascading aborts
B) No aborts
C) No rollbacks
D) Always commit

Answer: A

---

59. Multiversion Timestamp Ordering maintains:
A) Multiple versions of data items
B) Single version only
C) No versions
D) Only current version

Answer: A

SECTION 4: MULTIVERSION CONCURRENCY CONTROL (MVCC) (Questions 76-95)

60. MVCC maintains:
A) Multiple versions of each data item
B) Single version only

C) No versions
D) Only current version

Answer: A

---

61. In MVCC, readers:
A) Never wait for writers
B) Always wait for writers
C) Are blocked by writers
D) Cannot read while write occurs

Answer: A

---

62. MVCC provides:
A) Snapshot isolation
B) Serializability always
C) No isolation
D) Only read committed

Answer: A

---

63. Snapshot isolation gives transaction:
A) A snapshot of committed data at start time
B) Latest data always
C) Uncommitted data
D) No data

Answer: A

---

64. First-committer wins rule in snapshot isolation:
A) First transaction to commit succeeds, others abort on conflict
B) Last transaction wins
C) All commit
D) All abort

Answer: A

---

65. First-updater wins rule:
A) First to update gets lock, others wait/abort
B) Last updater wins
C) All update
D) No updates

Answer: A

---

66. Snapshot isolation may allow:
A) Write skew anomaly
B) Lost update
C) Dirty read
D) Phantom read

Answer: A

---

67. Write skew example:
A) T1 reads X and Y, updates X based on Y; T2 reads X and Y, updates Y based on X; both commit → inconsistency
B) T1 updates X, T2 updates X
C) T1 reads X, T2 writes X
D) T1 writes X, T2 reads X

Answer: A

---

68. Oracle's default isolation level is:
A) Snapshot isolation (read consistency)
B) Serializable
C) Read uncommitted
D) Repeatable read

Answer: A

---

69. PostgreSQL's MVCC implementation:
A) Uses transaction IDs and visibility maps

B) Uses locks only
C) No versioning
D) Timestamp only

Answer: A

---

70. Version garbage collection removes:
A) Old versions no longer needed by any transaction
B) All versions
C) New versions
D) Current versions

Answer: A

SECTION 5: OPTIMISTIC CONCURRENCY CONTROL (Questions 96-115)

71. Optimistic concurrency control assumes:
A) Conflicts are rare, validate at commit
B) Conflicts are frequent
C) Always lock first
D) Never validate

Answer: A

---

72. The three phases of optimistic concurrency control are:
A) Read, Validation, Write
B) Lock, Read, Write
C) Start, Execute, Commit
D) Begin, Validate, End

Answer: A

---

73. In read phase:
A) Transaction reads data, makes updates to private workspace

B) Transaction writes to database
C) Transaction acquires locks
D) Transaction validates

Answer: A

---

74. In validation phase:
A) Check if transaction conflicts with others
B) Write changes to database
C) Read data
D) Release locks

Answer: A

---

75. In write phase:
A) If validation succeeds, write changes to database
B) Read data again
C) Acquire locks
D) Abort transaction

Answer: A

---

76. Backward validation checks:
A) Against already committed transactions
B) Against active transactions
C) Against future transactions
D) No validation

Answer: A

---

77. Forward validation checks:
A) Against other active transactions
B) Against committed transactions
C) Against future transactions
D) No validation

Answer: A

---

78. Optimistic concurrency is good for:
A) Low-conflict environments
B) High-conflict environments
C) Always
D) Never

Answer: A

---

79. In high-conflict environments, optimistic concurrency leads to:
A) Many aborts and wasted work
B) Few aborts
C) Better performance
D) No aborts

Answer: A

SECTION 6: GRANULARITY OF LOCKS (Questions 116-130)

80. Lock granularity refers to:
A) Size of data item that can be locked (database, table, page, row, field)
B) Number of locks
C) Lock duration
D) Lock type

Answer: A

---

81. Coarse granularity (table-level locks):
A) Less overhead, less concurrency
B) More overhead, more concurrency
C) Less overhead, more concurrency
D) More overhead, less concurrency

Answer: A

---

82. Fine granularity (row-level locks):
A) More overhead, more concurrency
B) Less overhead, less concurrency
C) Less overhead, more concurrency
D) More overhead, less concurrency

Answer: A

---

83. Multiple granularity locking uses:
A) Intention locks (IS, IX, SIX)
B) Only shared locks
C) Only exclusive locks
D) No locks

Answer: A

---

84. Intention-shared (IS) lock indicates:
A) Intention to set shared locks at finer granularity
B) Intention to set exclusive locks
C) Shared lock held
D) Exclusive lock held

Answer: A

---

85. Intention-exclusive (IX) lock indicates:
A) Intention to set exclusive locks at finer granularity
B) Intention to set shared locks
C) Exclusive lock held
D) Shared lock held

Answer: A

---

86. Shared-intention-exclusive (SIX) lock indicates:
A) Shared lock at current level with intention for exclusive at finer level
B) Only shared
C) Only exclusive
D) No intention

Answer: A

---

87. Compatibility matrix for intention locks ensures:
A) Proper lock hierarchy is maintained
B) No locks needed
C) All locks compatible
D) No hierarchy

Answer: A

SECTION 7: CONCURRENCY CONTROL IN PRACTICE (Questions 131-145)

88. SQL isolation levels are:
A) READ UNCOMMITTED, READ COMMITTED, REPEATABLE READ, SERIALIZABLE
B) Only SERIALIZABLE
C) Only READ COMMITTED
D) No levels

Answer: A

---

89. READ UNCOMMITTED allows:
A) Dirty reads, non-repeatable reads, phantom reads
B) No anomalies
C) Only phantom reads
D) Only dirty reads

Answer: A

---

90. READ COMMITTED prevents:
A) Dirty reads
B) Non-repeatable reads
C) Phantom reads
D) All anomalies

Answer: A

---

91. REPEATABLE READ prevents:
A) Dirty reads and non-repeatable reads
B) Phantom reads
C) All anomalies
D) Only dirty reads

Answer: A

---

92. SERIALIZABLE prevents:
A) All anomalies (dirty read, non-repeatable read, phantom read)
B) Only dirty reads
C) Only phantom reads
D) No anomalies prevented

Answer: A

---

93. In SQL Server, default isolation level is:
A) READ COMMITTED
B) SERIALIZABLE
C) READ UNCOMMITTED
D) REPEATABLE READ

Answer: A

---

94. In Oracle, default isolation level is:
A) READ COMMITTED (snapshot-based)
B) SERIALIZABLE
C) READ UNCOMMITTED
D) REPEATABLE READ

Answer: A

---

95. In MySQL InnoDB, default isolation level is:
A) REPEATABLE READ
B) READ COMMITTED
C) SERIALIZABLE
D) READ UNCOMMITTED

Answer: A

---

96. PostgreSQL's default isolation level is:
A) READ COMMITTED
B) SERIALIZABLE
C) REPEATABLE READ
D) READ UNCOMMITTED

Answer: A

---

97. Serializable isolation in PostgreSQL uses:
A) Serializable Snapshot Isolation (SSI)
B) Regular locking
C) No isolation
D) Read committed only

Answer: A

---

98. Serializable Snapshot Isolation detects:
A) Serialization anomalies using predicate locks and conflict tracking
B) Only dirty reads
C) Only lost updates
D) No anomalies

Answer: A

SECTION 8: DEADLOCK HANDLING (Questions 146-160)

99. Deadlock avoidance strategies include:
A) Transaction ordering, lock acquisition ordering

B) Only detection
C) Only prevention
D) No strategies

Answer: A

---

100. Lock timeout mechanism:
A) Transaction waits for lock only up to timeout, then aborts
B) Waits forever
C) Never waits
D) Immediately aborts

Answer: A

---

101. Wait-die and wound-wait are:
A) Deadlock prevention schemes
B) Deadlock detection schemes
C) Deadlock avoidance schemes
D) Recovery schemes

Answer: A

---

102. In wait-die, if younger transaction requests lock held by older:
A) Younger dies (aborts)
B) Younger waits
C) Older dies
D) Older waits

Answer: A

---

103. In wait-die, if older requests lock held by younger:
A) Older waits
B) Older dies
C) Younger dies
D) Younger waits

Answer: A

---

104. In wound-wait, if older requests lock held by younger:
A) Older wounds (aborts) younger
B) Older waits
C) Younger waits
D) Both wait

Answer: A

---

105. In wound-wait, if younger requests lock held by older:
A) Younger waits
B) Younger wounds older
C) Older waits
D) Both wait

Answer: A

---

106. No-wait algorithm:
A) Transaction aborts if lock not available
B) Waits indefinitely
C) Waits for timeout
D) Spins

Answer: A

SECTION 9: INDEX CONCURRENCY CONTROL (Questions 161-175)

107. B-tree concurrency control uses:
A) Crabbing (latch coupling) protocol
B) No locking
C) Table locks only
D) Page locks only

Answer: A

---

108. Latch vs Lock:
A) Latches are short-term, protect in-memory structures; locks are long-term, protect logical data
B) Latches are long-term
C) Both same
D) Locks are short-term

Answer: A

---

109. Latch crabbing (coupling) ensures:
A) Safe traversal of tree structures
B) No tree access
C) Slow access
D) No concurrency

Answer: A

---

110. In crabbing, when moving from parent to child:
A) Hold parent latch until child latched, then release parent
B) Release parent before latching child
C) Hold both forever
D) No latches needed

Answer: A

---

111. B-link trees use:
A) Right links for sibling traversal
B) No links
C) Left links only
D) Parent links only

Answer: A

SECTION 10: ADVANCED TOPICS (Questions 176-190)

112. Nested transactions allow:
A) Subtransactions within a transaction
B) Only flat transactions
C) No hierarchy
D) Single level only

Answer: A

---

113. In nested transactions, abort of subtransaction:
A) Does not necessarily abort parent
B) Always aborts parent
C) Commits parent
D) No effect

Answer: A

---

114. Distributed concurrency control uses:
A) Two-phase commit (2PC) with distributed locking
B) Local locks only
C) No coordination
D) Centralized only

Answer: A

---

115. Distributed deadlock detection uses:
A) Global wait-for graph
B) Local graphs only
C) No detection
D) Centralized only

Answer: A

---

116. Path pushing algorithm for distributed deadlock detection:
A) Transactions push wait information along paths
B) No information sharing
C) Local detection only
D) Centralized only

Answer: A

---

117. Edge chasing algorithm:
A) Probes circulated to detect cycles
B) No probes
C) Local only
D) Centralized only

Answer: A

---

118. Real-time concurrency control considers:
A) Transaction deadlines
B) Only serializability
C) Only consistency
D) Only isolation

Answer: A

---

119. Adaptive concurrency control:
A) Adjusts based on workload and conflict rates
B) Fixed protocol always
C) No adjustment
D) Random selection

Answer: A

SECTION 11: COMPARISON AND TRADE-OFFS (Questions 191-200)

120. Locking vs Timestamp:
A) Locking may deadlock, timestamp may abort
B) Locking never aborts
C) Timestamp never aborts
D) Both same

Answer: A

---

121. Optimistic vs Pessimistic:
A) Optimistic better for low conflict, pessimistic for high conflict
B) Optimistic always better
C) Pessimistic always better
D) No difference

Answer: A

---

122. MVCC advantages:
A) Readers never block writers, good concurrency
B) Readers always block
C) Writers always block
D) No concurrency

Answer: A

---

123. MVCC disadvantages:
A) Storage overhead for versions, garbage collection needed
B) No overhead
C) Simple implementation
D) No version management

Answer: A

---

124. Choosing concurrency control depends on:
A) Workload characteristics, conflict rate, system requirements
B) Only database size
C) Only number of users
D) Only query types

Answer: A

SECTION 1: INTRODUCTION TO RECOVERY MANAGEMENT (Questions 1-20)

1. Recovery management is responsible for:
A) Restoring database to a consistent state after failures
B) Managing user access permissions
C) Optimizing query performance
D) Creating database indexes

Answer: A
Explanation: Recovery management ensures database consistency despite various failures.

---

2. The main types of failures in database systems are:
A) Transaction failure, system crash, disk failure
B) Only transaction failure
C) Only system crash
D) Only disk failure

Answer: A

---

3. Transaction failure occurs when:
A) A transaction cannot complete normally due to logical error or system condition
B) The entire system crashes
C) Disk storage is damaged
D) Network connection fails

Answer: A

---

4. System crash (soft failure) refers to:
A) Failure of volatile memory contents, but non-volatile storage intact
B) Permanent disk damage
C) Transaction abort
D) Network failure

Answer: A

---

5. Disk failure (hard failure) refers to:
A) Loss or corruption of data in non-volatile storage
B) Memory failure
C) Transaction rollback
D) CPU failure

Answer: A

---

6. The recovery manager ensures which ACID properties?
A) Atomicity and Durability
B) Consistency and Isolation
C) Only Atomicity
D) Only Durability

Answer: A

---

7. Atomicity requires that:
A) Either all operations of transaction are reflected or none
B) Transaction runs in isolation
C) Data is consistent
D) Changes persist after commit

Answer: A

---

8. Durability requires that:
A) Once transaction commits, changes persist despite failures
B) Transaction is atomic
C) Transaction is isolated
D) Data is consistent

Answer: A

SECTION 2: STORAGE HIERARCHY (Questions 21-35)

9. Volatile storage:
A) Loses contents when power fails (main memory, cache)
B) Retains data after power failure
C) Never loses data
D) Only used for backups

Answer: A

---

10. Non-volatile storage:
A) Retains data after power failure (disk, SSD, tape)
B) Loses data on power failure
C) Faster than volatile storage
D) Not used for permanent storage

Answer: A

---

11. Stable storage is:
A) Theoretical ideal where data is never lost (achieved via redundancy)
B) Volatile memory
C) Cache memory
D) Register storage

Answer: A

---

12. Stable storage implementation uses:
A) Multiple storage devices with redundant copies
B) Single disk
C) Only memory
D) No redundancy

Answer: A

---

13. RAID (Redundant Array of Independent Disks) helps achieve:
A) Fault tolerance through redundancy
B) Faster processing only
C) Smaller storage
D) Cheaper storage

Answer: A

SECTION 3: LOG-BASED RECOVERY (Questions 36-65)

14. A log is:
A) A sequential record of all database modifications
B) A backup of the database
C) An index structure
D) A transaction list

Answer: A

---

15. Each log record contains:
A) Transaction ID, data item, old value, new value
B) Only transaction ID
C) Only new value
D) Only operation type

Answer: A

---

16. Log records are written:
A) In the order transactions execute
B) Randomly
C) Only at commit time
D) Only at abort time

Answer: A

---

17. Write-Ahead Logging (WAL) protocol requires:
A) Log record written to stable storage before corresponding data update
B) Data update before log record
C) Log and data written simultaneously
D) No log needed

Answer: A

---

18. WAL ensures:
A) Recovery can use log to undo/redo changes
B) Faster writes
C) Less storage
D) No recovery needed

Answer: A

---

19. The two main recovery techniques using logs are:
A) Deferred update and Immediate update
B) Shadow paging and Snapshot
C) Backup and Restore
D) Checkpoint and Archive

Answer: A

---

20. In deferred update (NO-UNDO/REDO):
A) Updates applied only after transaction commits; no undo needed, redo for committed transactions
B) Updates applied immediately; undo needed
C) Updates never applied
D) No logging needed

Answer: A

---

21. In deferred update, during recovery:
A) Only REDO for committed transactions (no UNDO)
B) Only UNDO for uncommitted
C) Both UNDO and REDO
D) No action needed

Answer: A

---

22. In immediate update (UNDO/REDO):
A) Updates may be applied before commit; undo for uncommitted, redo for

committed
B) Updates only after commit
C) No undo needed
D) No redo needed

Answer: A

---

23. With immediate update and no logging, we cannot:
A) Undo uncommitted transactions
B) Redo committed transactions
C) Commit transactions
D) Start transactions

Answer: A

---

24. UNDO operation restores:
A) Old values for uncommitted transactions
B) New values
C) Transaction state
D) System state

Answer: A

---

25. REDO operation applies:
A) New values for committed transactions
B) Old values
C) Transaction start
D) System restart

Answer: A

---

26. Transaction rollback uses:
A) Log records to undo changes in reverse order
B) Forward scan of log
C) Random access to log
D) No log needed

Answer: A

---

27. During rollback, we process log:
A) From end backward (latest to earliest)
B) From beginning forward
C) Randomly
D) Only commit records

Answer: A

---

28. Log sequence number (LSN) is:
A) Unique identifier for each log record
B) Transaction identifier
C) Data item identifier
D) Page number

Answer: A

---

29. Dirty page table tracks:
A) Pages modified in buffer but not yet written to disk
B) Clean pages
C) Index pages
D) System tables

Answer: A

---

30. Transaction table tracks:
A) Status of each active transaction (active, committed, aborted)
B) Only committed transactions
C) Only aborted transactions
D) Data items

Answer: A

SECTION 4: CHECKPOINTS (Questions 66-85)

31. A checkpoint is:
A) Point where all dirty buffers are written to disk to reduce recovery time
B) Database backup
C) Transaction commit point
D) System shutdown

Answer: A

---

32. Checkpoints help:
A) Reduce the amount of log to scan during recovery
B) Increase recovery time
C) Eliminate need for logs
D) Prevent failures

Answer: A

---

33. During a checkpoint:
A) All dirty pages in buffer are written to disk
B) Only committed transactions are written
C) Log is cleared
D) Database is shut down

Answer: A

---

34. After checkpoint, recovery only needs to consider:
A) Transactions active after checkpoint
B) All transactions ever
C) Transactions before checkpoint only
D) No transactions

Answer: A

---

35. Fuzzy checkpoint allows:
A) Other transactions to continue during checkpoint

B) No concurrent activity
C) Database shutdown
D) Only read-only transactions

Answer: A

---

36. Checkpoint log record contains:
A) List of active transactions and dirty pages
B) Only time stamp
C) Only transaction IDs
D) No information

Answer: A

---

37. Recovery with checkpoints involves:
A) Finding last checkpoint, then scanning forward from there
B) Scanning entire log from beginning
C) No log scan needed
D) Only checkpoint record needed

Answer: A

SECTION 5: ARIES RECOVERY ALGORITHM (Questions 86-110)

38. ARIES stands for:
A) Algorithm for Recovery and Isolation Exploiting Semantics
B) Automatic Recovery Indexing System
C) Advanced Recovery Implementation System
D) Atomic Recovery with Integrated Security

Answer: A

---

39. ARIES is based on:
A) Write-ahead logging and repeating history

B) No logging
C) Shadow paging only
D) Snapshot isolation

Answer: A

---

40. The three phases of ARIES recovery are:
A) Analysis, Redo, Undo
B) Scan, Recovery, Restart
C) Begin, Middle, End
D) Read, Write, Commit

Answer: A

---

41. Analysis phase:
A) Determines dirty pages and transactions active at crash
B) Redoes all changes
C) Undoes all changes
D) Creates checkpoint

Answer: A

---

42. Redo phase:
A) Repeats history by reapplying all changes from appropriate point
B) Undoes uncommitted transactions
C) Analyzes log
D) Creates checkpoint

Answer: A

---

43. Undo phase:
A) Rolls back uncommitted transactions
B) Redoes all changes
C) Analyzes log
D) Creates checkpoint

Answer: A

---

44. In ARIES, each page has a:
A) Page LSN (last LSN of update affecting that page)
B) Transaction ID
C) Page number only
D) No identifier

Answer: A

---

45. Page LSN helps determine:
A) Whether redo is needed for that page
B) Page size
C) Page type
D) Page owner

Answer: A

---

46. Compensation Log Records (CLR) record:
A) Undo operations to ensure idempotence
B) Committed transactions
C) Checkpoints
D) Aborted transactions only

Answer: A

---

47. CLRs ensure that:
A) Undo operations are not repeated if system crashes during undo
B) Redo is faster
C) Checkpoints are efficient
D) No logging needed

Answer: A

---

48. ARIES supports:
A) Partial rollbacks using savepoints
B) Only full rollbacks

C) No rollbacks
D) Only commits

Answer: A

---

49. In ARIES, transaction rollback writes:
A) CLRs for each undo operation
B) No log records
C) Only commit record
D) Only abort record

Answer: A

SECTION 6: SHADOW PAGING (Questions 111-125)

50. Shadow paging is an alternative to:
A) Log-based recovery
B) Locking
C) Indexing
D) Query optimization

Answer: A

---

51. Shadow paging maintains:
A) Two page tables: current and shadow
B) Single page table
C) No page tables
D) Only log

Answer: A

---

52. During transaction, updates go to:
A) Current page table (new pages)
B) Shadow page table
C) Both tables
D) Log only

Answer: A

---

53. On commit:
A) Current page table becomes active, shadow discarded
B) Shadow page table becomes active
C) Both tables kept
D) No change

Answer: A

---

54. On abort:
A) Current page table discarded, shadow restored
B) Shadow discarded
C) Both kept
D) No action

Answer: A

---

55. Shadow paging advantages:
A) No logging overhead, fast recovery
B) Requires complex logging
C) Slow recovery
D) High overhead

Answer: A

---

56. Shadow paging disadvantages:
A) Data fragmentation, garbage collection needed
B) No fragmentation
C) Simple implementation
D) Low overhead

Answer: A

SECTION 7: RECOVERY BASED ON DEFERRED UPDATE (Questions 126-140)

57. In deferred update protocol:
A) All updates buffered in memory until commit
B) Updates written to disk immediately
C) Updates written before commit
D) No buffering

Answer: A

---

58. If transaction fails before commit:
A) No recovery needed (no updates on disk)
B) Must undo changes
C) Must redo changes
D) Both undo and redo

Answer: A

---

59. If system crashes after commit but before all updates written:
A) Redo needed (committed updates not on disk)
B) Undo needed
C) No action
D) Both undo and redo

Answer: A

---

60. Deferred update requires:
A) REDO but no UNDO
B) UNDO but no REDO
C) Both UNDO and REDO
D) No recovery

Answer: A

SECTION 8: RECOVERY BASED ON IMMEDIATE UPDATE (Questions 141-155)

61. In immediate update protocol:
A) Updates may be applied to disk before commit
B) Updates only after commit
C) No disk updates until commit
D) Updates never to disk

Answer: A

---

62. If transaction aborts:
A) Must undo changes already on disk
B) No action needed
C) Must redo
D) Commit anyway

Answer: A

---

63. If system crashes with uncommitted updates on disk:
A) Must undo those updates
B) Must redo them
C) No action
D) Commit them

Answer: A

---

64. Immediate update with logging requires:
A) Both UNDO and REDO capability
B) Only UNDO
C) Only REDO
D) No logging

Answer: A

---

65. UNDO of uncommitted transactions during recovery uses:
A) Log records with old values
B) Log records with new values
C) No log needed
D) Shadow pages

Answer: A

SECTION 9: RECOVERY WITH CONCURRENT TRANSACTIONS (Questions 156-170)

66. With concurrent transactions, recovery must consider:
A) Interactions between transactions (cascading rollbacks)
B) Only individual transactions
C) No interactions
D) Only committed transactions

Answer: A

---

67. Cascading rollback occurs when:
A) One transaction abort causes others that read its data to also abort
B) Multiple transactions abort independently
C) System rolls back all transactions
D) No rollback occurs

Answer: A

---

68. Recoverable schedules require:
A) Transaction commits only after all transactions it read from have committed
B) Transactions can commit anytime
C) No dependencies
D) All transactions commit together

Answer: A

---

69. Cascadeless schedules prevent:
A) Dirty reads (transactions only read committed data)
B) All conflicts
C) All writes
D) All reads

Answer: A

---

70. Strict schedules require:
A) Writes only after previous transactions that wrote same data have committed/aborted
B) No ordering
C) Reads before writes
D) Writes before reads

Answer: A

SECTION 10: REMOTE BACKUP AND REPLICATION (Questions 171-185)

71. Remote backup systems provide:
A) Disaster recovery through geographically separated copies
B) Local recovery only
C) No recovery
D) Only performance improvement

Answer: A

---

72. Primary site sends log records to:
A) Remote backup site
B) Only local storage
C) No backup
D) Archive

Answer: A

---

73. In hot standby mode:
A) Backup site actively applies logs and is ready to take over
B) Backup site is offline
C) No backup exists
D) Backup only for reads

Answer: A

---

74. Failover is the process of:
A) Switching to backup site when primary fails
B) Switching back to primary
C) Creating backup
D) Removing backup

Answer: A

---

75. Failback is the process of:
A) Switching back to primary after recovery
B) Switching to backup
C) Initial setup
D) Testing

Answer: A

---

76. Synchronous replication ensures:
A) Primary waits for backup acknowledgment before commit
B) Faster commits
C) No waiting
D) Data loss possible

Answer: A

---

77. Asynchronous replication:
A) Primary commits without waiting for backup
B) Always consistent
C) No data loss risk
D) Slower commits

Answer: A

SECTION 11: DATABASE BACKUP TECHNIQUES (Questions 186-200)

78. Full database backup contains:
A) Entire database at a point in time
B) Only changes since last backup
C) Only schema
D) Only indexes

Answer: A

---

79. Incremental backup contains:
A) Changes since last backup (full or incremental)
B) Entire database
C) Only schema
D) Only logs

Answer: A

---

80. Differential backup contains:
A) Changes since last full backup
B) Changes since last incremental
C) Entire database
D) Only schema

Answer: A

---

81. Online (hot) backup is taken:
A) While database is running and accessible
B) During downtime
C) Only when database is shut down
D) Never

Answer: A

---

82. Offline (cold) backup is taken:
A) When database is shut down
B) During normal operation
C) While users are active
D) During peak hours

Answer: A

---

83. Point-in-time recovery (PITR) allows:
A) Restoring database to any past time using full backup + logs
B) Only latest state
C) Only specific times
D) No time selection

Answer: A

SECTION 12: ADVANCED RECOVERY TOPICS (Questions 201-220)

84. Media recovery handles:
A) Disk failures requiring restoration from backup plus logs
B) Transaction failures
C) System crashes
D) Memory failures

Answer: A

---

85. Crash recovery handles:
A) System failures with intact disk
B) Disk failures
C) Media failures
D) Network failures

Answer: A

---

86. Fast-Start recovery in modern DBMS:
A) Minimizes recovery time using techniques like partial rollback
B) Always slow
C) No optimization
D) Only full recovery

Answer: A

---

87. Recovery in distributed databases uses:
A) Two-phase commit (2PC) and distributed logs
B) Local recovery only
C) No coordination
D) Single site recovery

Answer: A

---

88. Presumed Abort (PA) protocol in 2PC:
A) If coordinator fails, assume abort unless commit record found
B) Assume commit always
C) No assumption
D) Always commit

Answer: A

---

89. Presumed Commit (PC) protocol:
A) Assume commit unless abort record found
B) Assume abort always
C) No assumption
D) Always abort

Answer: A

---

90. In-doubt transactions in 2PC:
A) Transactions where coordinator failed after prepare but before decision
B) Committed transactions
C) Aborted transactions
D) Completed transactions

Answer: A

---

91. Heuristic decisions in distributed recovery:
A) Local decision to commit/abort without coordinator (risks inconsistency)
B) Always coordinated
C) Never used
D) Always consistent

Answer: A

---

92. Recovery in main-memory databases:
A) Uses logging to disk and checkpointing for durability
B) No recovery needed
C) Memory only
D) No logging

Answer: A

SECTION 13: PRACTICAL RECOVERY IN DBMS (Questions 221-235)

93. Oracle recovery uses:
A) REDO logs, UNDO segments, control files
B) Only REDO logs
C) Only UNDO
D) No logs

Answer: A

---

94. SQL Server recovery uses:
A) Transaction log, checkpoints, ARIES-based recovery
B) Only data files
C) No logs
D) Shadow paging

Answer: A

---

95. MySQL InnoDB recovery uses:
A) REDO logs, UNDO logs, doublewrite buffer
B) Only binary log
C) No logs
D) Shadow paging

Answer: A

---

96. PostgreSQL recovery uses:
A) Write-ahead log (WAL), checkpoints, point-in-time recovery
B) Only data files
C) No logs
D) Shadow paging

Answer: A

---

97. MongoDB recovery uses:
A) Journal (write-ahead log) and checkpoints
B) No logging
C) Only data files
D) Shadow paging

Answer: A

---

98. Recovery testing is important to:
A) Verify that backup and recovery procedures work
B) Only for development
C) Never needed
D) Only for performance

Answer: A

---

99. Recovery time objective (RTO) is:
A) Maximum acceptable time to recover after failure
B) Time to backup
C) Time to commit
D) Time to abort

Answer: A

---

100. Recovery point objective (RPO) is:
A) Maximum acceptable data loss measured in time
B) Point of recovery
C) Recovery start time
D) Backup time

Answer: A

SECTION 1: INTRODUCTION TO FILE ORGANIZATION (Questions 1-20)

1. File organization refers to:
A) How records are structured and stored on disk
B) How queries are optimized
C) How transactions are managed
D) How users access the database

Answer: A
Explanation: File organization deals with physical storage and arrangement of records on disk.

---

2. The basic unit of data storage on disk is:
A) Block (or page)
B) Byte

C) Record
D) Field

Answer: A

---

3. A disk block typically contains:
A) Multiple records
B) Single record only
C) Only index information
D) Only metadata

Answer: A

---

4. Blocking factor is:
A) Number of records per block
B) Size of a block
C) Number of blocks in file
D) Time to read a block

Answer: A

---

5. The time to access a disk block consists of:
A) Seek time + Rotational latency + Transfer time
B) Only seek time
C) Only transfer time
D) Only rotational latency

Answer: A

---

6. Seek time is the time to:
A) Position the disk arm over the correct track
B) Rotate disk to correct sector
C) Transfer data
D) Process the data

Answer: A

---

7. Rotational latency is the time to:
A) Wait for desired sector to rotate under head
B) Move disk arm
C) Transfer data
D) Process data

Answer: A

---

8. Transfer time is the time to:
A) Move data between disk and memory
B) Position disk arm
C) Rotate to correct sector
D) Process data

Answer: A

---

9. Sequential file organization stores records:
A) In order based on a key field
B) Randomly
C) By insertion time
D) By record size

Answer: A

---

10. Sequential files are efficient for:
A) Processing all records in order
B) Random access by key
C) Frequent insertions and deletions
D) Concurrent access

Answer: A

SECTION 2: HEAP FILE ORGANIZATION (Questions 21-35)

11. In heap (unordered) file organization:
A) Records are placed where space is available, no particular order
B) Records are sorted by key
C) Records are stored in separate files
D) Records are indexed only

Answer: A

---

12. Heap files are efficient for:
A) Bulk loading and full table scans
B) Search by key value
C) Range queries
D) Ordered retrieval

Answer: A

---

13. Insertion in heap file is:
A) Fast (append or use free space)
B) Slow (requires sorting)
C) Requires index update
D) Not allowed

Answer: A

---

14. Search in heap file without index requires:
A) Linear scan of all records
B) Binary search
C) Hash lookup
D) Index lookup

Answer: A

---

15. Deletion in heap file typically uses:
A) Mark record as deleted, may need reorganization
B) Immediate physical removal
C) No action needed
D) Automatic compaction

Answer: A

SECTION 3: SORTED FILE ORGANIZATION (Questions 36-50)

16. In sorted (sequential) file organization:
A) Records are ordered by a key field
B) Records are in insertion order
C) Records are hashed
D) Records are indexed separately

Answer: A

---

17. Sorted files are efficient for:
A) Range queries and ordered retrieval
B) Random insertions
C) Frequent updates
D) Equality searches without index

Answer: A

---

18. Binary search on sorted file requires:
A) O(log N) block accesses
B) O(N) block accesses
C) O(1) block accesses
D) O(N log N) accesses

Answer: A

---

19. Insertion in sorted file is expensive because:
A) May require moving many records to maintain order
B) Always fast
C) Requires no changes
D) Only appends at end

Answer: A

---

20. Overflow area in sorted files handles:
A) Records that don't fit in their correct position
B) Deleted records
C) Index entries
D) Metadata

Answer: A

SECTION 4: INTRODUCTION TO INDEXING (Questions 51-70)

21. An index is:
A) A data structure that speeds up data retrieval
B) A backup of the database
C) A type of file organization
D) A transaction log

Answer: A

---

22. Indexing is analogous to:
A) Book index (keyword to page number mapping)
B) Table of contents
C) Appendix
D) Glossary

Answer: A

---

23. The two main types of indexes are:
A) Ordered indexes and Hash indexes
B) Primary and Secondary
C) Clustered and Non-clustered
D) Dense and Sparse

Answer: A

---

24. An index entry consists of:
A) Search key value and pointer to record(s)
B) Only search key
C) Only pointer
D) Entire record

Answer: A

---

25. Search key is:
A) Attribute(s) used to look up records in index
B) Primary key of table
C) Foreign key
D) Any attribute

Answer: A

---

26. Indexes improve performance of:
A) SELECT queries with conditions on indexed columns
B) INSERT operations
C) DELETE operations without conditions
D) All operations equally

Answer: A

---

27. Indexes may degrade performance of:
A) INSERT, UPDATE, DELETE (due to index maintenance)
B) SELECT queries
C) Read-only operations
D) Backups

Answer: A

---

28. Dense index has:
A) An entry for every search key value in the file
B) Entries for only some keys

C) One entry per block
D) No entries

Answer: A

---

29. Sparse index has:
A) An entry for each block of the file
B) An entry for every record
C) No entries
D) Multiple entries per record

Answer: A

---

30. Sparse index requires:
A) File to be ordered on the indexed key
B) No ordering
C) Hash organization
D) Heap organization

Answer: A

---

31. Primary index is defined on:
A) Ordered file's ordering key field
B) Any attribute
C) Foreign key
D) Non-key attribute

Answer: A

---

32. A primary index is typically:
A) Sparse (one entry per block)
B) Dense
C) Always dense
D) Always sparse

Answer: A

---

33. Clustering index is defined on:
A) Non-key field that is ordered (records with same value stored together)
B) Primary key
C) Any attribute
D) Foreign key

Answer: A

---

34. In clustering index, all records with same value for indexing field are stored:
A) Together, possibly in overflow blocks
B) Scattered randomly
C) In separate tables
D) In index only

Answer: A

---

35. Secondary index provides:
A) Alternate access path on non-ordering field
B) Primary access path
C) Only for primary key
D) Only for foreign keys

Answer: A

---

36. Secondary indexes are typically:
A) Dense (one entry per record)
B) Sparse
C) Always sparse
D) Block-level only

Answer: A

SECTION 5: MULTILEVEL INDEXING (Questions 71-85)

37. Multilevel indexing uses:
A) Index on index to reduce height
B) Single level only
C) No hierarchy
D) Flat structure

Answer: A

---

38. The outermost (top) index in multilevel indexing is called:
A) Master index
B) Leaf index
C) Inner index
D) Base index

Answer: A

---

39. Multilevel indexing reduces:
A) Number of disk accesses for search
B) Index size
C) Data size
D) Query complexity

Answer: A

---

40. With multilevel indexing, search complexity becomes:
A) O(log_n N) where n is fanout
B) O(N)
C) O(1)
D) O(N log N)

Answer: A

SECTION 6: B-TREE AND B+ TREE INDEXES (Questions 86-120)

41. B-tree is a:
A) Balanced tree data structure for indexing
B) Binary tree
C) Unbalanced tree
D) Linear structure

Answer: A

---

42. B-tree properties include:
A) All leaf nodes at same depth, nodes at least half full
B) Leaves at different depths
C) No balancing
D) Fixed node size only

Answer: A

---

43. In B-tree of order m:
A) Each node has at most m children, at least ⌈m/2⌉ children (except root)
B) Each node has exactly m children
C) Each node has at most 2 children
D) No restrictions

Answer: A

---

44. B+ tree differs from B-tree in that:
A) All data pointers in leaves only, leaves linked for sequential access
B) Data pointers in all nodes
C) No leaf links
D) Same structure

Answer: A

---

45. In B+ tree, internal nodes contain:
A) Only keys and pointers to child nodes
B) Keys and data pointers
C) Only data
D) Only pointers

Answer: A

---

46. In B+ tree, leaf nodes contain:
A) Keys and pointers to actual records
B) Only keys
C) Only pointers
D) Child pointers

Answer: A

---

47. Leaf nodes in B+ tree are linked to provide:
A) Efficient sequential access (range queries)
B) Random access only
C) No sequential access
D) Faster inserts

Answer: A

---

48. Height of B+ tree with millions of records is typically:
A) 3-4 levels
B) 10-20 levels
C) 100+ levels
D) 1 level only

Answer: A

---

49. Search in B+ tree requires:
A) Traversal from root to leaf (height number of accesses)
B) Full scan
C) Binary search on tree
D) Hash lookup

Answer: A

---

50. Insertion in B+ tree may cause:
A) Node splits when node overflows
B) Node merges always
C) No changes
D) Tree shrink

Answer: A

---

51. Deletion in B+ tree may cause:
A) Node merges or redistribution when node underflows
B) Always splits
C) No changes
D) Tree expansion

Answer: A

---

52. Node split in B+ tree distributes keys:
A) Evenly between two nodes, promotes one key to parent
B) All keys to new node
C) All keys to old node
D) Randomly

Answer: A

---

53. The fanout of a B+ tree node is:
A) Number of pointers (children) in the node
B) Height of tree
C) Number of records
D) Size of node

Answer: A

---

54. Larger fanout results in:
A) Shorter tree height, fewer disk accesses
B) Taller tree
C) More disk accesses
D) Slower search

Answer: A

---

55. B+ trees are widely used because:
A) They remain balanced, support range queries, have logarithmic search cost
B) They are simple
C) They are unordered
D) They don't support updates

Answer: A

---

56. The order of a B+ tree affects:
A) Node size, fanout, and tree height
B) Only search time
C) Only insert time
D) Only delete time

Answer: A

SECTION 7: HASH-BASED INDEXING (Questions 121-145)

57. Hash indexing uses:
A) Hash function to map search keys to bucket addresses
B) Tree structure
C) Sequential ordering
D) No mapping

Answer: A

---

58. Static hashing has:
A) Fixed number of buckets
B) Dynamic bucket count
C) No buckets
D) Variable buckets always

Answer: A

---

59. Hash function h(K) returns:
A) Bucket address for key K
B) Record data
C) Index size
D) Key value

Answer: A

---

60. Collision occurs when:
A) Two different keys hash to same bucket
B) Bucket is empty
C) Key not found
D) Hash function fails

Answer: A

---

61. Overflow handling techniques include:
A) Chaining (overflow buckets) and open addressing
B) Only rehashing
C) Only linear probing
D) No handling

Answer: A

---

62. In chaining, overflow records are stored in:
A) Linked list of overflow buckets
B) Same bucket
C) Different file
D) Index structure

Answer: A

---

63. Static hashing performance degrades due to:
A) Overflow chains growing long as file grows
B) Tree imbalance

C) Index height
D) Sorting overhead

Answer: A

---

64. Dynamic hashing allows:
A) Buckets to grow and shrink as needed
B) Fixed bucket count
C) No bucket changes
D) Only shrinking

Answer: A

---

65. Extendible hashing uses:
A) Directory of bucket addresses that doubles as needed
B) Fixed directory
C) No directory
D) Tree structure

Answer: A

---

66. In extendible hashing, global depth indicates:
A) Number of bits used for directory indexing
B) Tree height
C) Bucket size
D) Hash function complexity

Answer: A

---

67. Local depth of a bucket indicates:
A) Number of bits used for keys in that bucket
B) Bucket size
C) Directory depth
D) Overflow count

Answer: A

---

68. When bucket overflows in extendible hashing:
A) Split bucket, double directory if local depth = global depth
B) Always double directory
C) Never split
D) Use overflow chain

Answer: A

---

69. Linear hashing grows:
A) One bucket at a time without directory
B) By doubling directory
C) By rebuilding
D) Never grows

Answer: A

---

70. Linear hashing uses:
A) Split pointer to track next bucket to split
B) Directory
C) Tree structure
D) No structure

Answer: A

---

71. Hash indexes are excellent for:
A) Equality searches (exact match)
B) Range queries
C) Ordered retrieval
D) Pattern matching

Answer: A

---

72. Hash indexes are not suitable for:
A) Range queries (e.g., salary BETWEEN 50000 AND 70000)
B) Exact match queries

C) Point queries
D) Primary key lookups

Answer: A

SECTION 8: BITMAP INDEXES (Questions 146-160)

73. Bitmap indexes use:
A) Bit arrays (bitmaps) for each distinct key value
B) Tree structure
C) Hash tables
D) Sequential files

Answer: A

---

74. Bitmap indexes are efficient for:
A) Low-cardinality columns (few distinct values)
B) High-cardinality columns
C) Primary keys
D) Unique columns

Answer: A

---

75. In bitmap index, each bitmap has:
A) One bit per record indicating whether record has that value
B) One byte per record
C) One integer per record
D) Multiple bits per record

Answer: A

---

76. Bitmap indexes support efficient:
A) Boolean operations (AND, OR, NOT) for complex conditions
B) Range scans
C) Sorting
D) Joins only

Answer: A

---

77. Example columns suitable for bitmap indexes:
A) Gender, status, region (few distinct values)
B) Employee ID
C) Salary
D) Name

Answer: A

SECTION 9: CLUSTERED VS NON-CLUSTERED INDEXES (Questions 161-175)

78. Clustered index determines:
A) Physical order of data rows in table
B) Logical order only
C) No ordering
D) Separate structure

Answer: A

---

79. A table can have:
A) Only one clustered index
B) Multiple clustered indexes
C) Any number of clustered indexes
D) No clustered indexes

Answer: A

---

80. Non-clustered index:
A) Stores index separately from data, with pointers to data rows
B) Stores data in index order
C) Determines physical order
D) Only one allowed

Answer: A

---

81. A table can have:
A) Multiple non-clustered indexes
B) Only one non-clustered index
C) No non-clustered indexes
D) Same as clustered

Answer: A

---

82. Clustered index scan is faster than non-clustered for:
A) Range queries because data is physically ordered
B) Point queries
C) Inserts
D) Updates

Answer: A

---

83. Non-clustered index lookup may require:
A) Key lookup (bookmark lookup) to fetch actual data if index doesn't cover query
B) Only index access
C) No data access
D) Table scan

Answer: A

---

84. Covering index contains:
A) All columns needed by a query, avoiding data access
B) Only key columns
C) Only included columns
D) No columns

Answer: A

---

85. Included columns in non-clustered index:
A) Stored only at leaf level, not part of key
B) Part of key
C) Not stored
D) In internal nodes

Answer: A

SECTION 10: INDEX SELECTION AND TUNING (Questions 176-190)

86. Index selection considers:
A) Query workload, update frequency, selectivity
B) Only table size
C) Only number of columns
D) Only data types

Answer: A

---

87. Selective queries (returning few rows) benefit from:
A) Indexes on selective columns
B) Full table scans
C) No indexes
D) Any index

Answer: A

---

88. Composite index on (A, B) is useful for queries with:
A) Conditions on A or (A and B)
B) Conditions on B only
C) Any condition
D) No conditions

Answer: A

---

89. Index on (A, B) supports:
A) A = value, A = value AND B = value, A = value AND B > value
B) Only exact match on both
C) Only A conditions
D) Only B conditions

Answer: A

---

90. Index on (A, B) does NOT support efficiently:
A) B = value alone (without A)
B) A = value
C) A = value AND B = value
D) A > value

Answer: A

---

91. Index Intersection uses:
A) Multiple indexes to satisfy query using AND conditions
B) Single index only
C) No indexes
D) Table scan

Answer: A

---

92. Index Union uses:
A) Multiple indexes for OR conditions
B) Single index only
C) No indexes
D) Table scan

Answer: A

---

93. Index hint forces:
A) Query optimizer to use specific index
B) Ignore indexes
C) Create index
D) Drop index

Answer: A

SECTION 11: INDEX MAINTENANCE (Questions 191-205)

94. Index fragmentation occurs when:
A) Index pages become disorganized due to inserts/updates/deletes
B) Index is too large
C) Index is missing
D) Index is unique

Answer: A

---

95. Fragmentation types include:
A) Internal (unused space in pages) and External (logical order mismatch)
B) Only internal
C) Only external
D) No fragmentation

Answer: A

---

96. Index rebuild:
A) Creates new index with defragmented structure
B) Updates statistics only
C) Drops index
D) Renames index

Answer: A

---

97. Index reorganize:
A) Defragments leaf level without full rebuild
B) Rebuilds entire index
C) Drops index
D) Updates only

Answer: A

---

98. Fill factor determines:
A) How full index pages are initially (leaves space for inserts)
B) Index size
C) Number of levels
D) Key count

Answer: A

---

99. Low fill factor (e.g., 70%):
A) Reduces page splits, increases space usage
B) Increases page splits
C) Saves space
D) No effect

Answer: A

---

100. High fill factor (e.g., 100%):
A) Saves space, increases page splits on insert
B) Wastes space
C) Reduces splits
D) No effect

Answer: A

SECTION 12: SPECIALIZED INDEXING TECHNIQUES (Questions 206-220)

101. Full-text indexes support:
A) Text search on large text columns (contains, freetext)
B) Exact match only
C) Numeric comparisons
D) Date ranges

Answer: A

---

102. Spatial indexes (R-tree, Grid) support:
A) Geographic queries (nearby, within distance, contains)
B) Text search
C) Numeric equality
D) Pattern matching

Answer: A

---

103. XML indexes support:
A) Queries on XML data using XPath/XQuery
B) Text search only
C) Numeric data
D) Date data

Answer: A

---

104. JSON indexes in modern DBMS:
A) Support indexing on JSON properties
B) No JSON indexing
C) Only full-text
D) Only text

Answer: A

---

105. Expression-based indexes index:
A) Result of expression or function on columns
B) Only raw columns
C) Only computed columns
D) No expressions

Answer: A

---

106. Partial (filtered) indexes index:
A) Only subset of rows (WHERE clause)
B) All rows

C) Only NULL rows
D) Only non-NULL rows

Answer: A

---

107. Inverted indexes are used in:
A) Search engines, document retrieval
B) B-trees
C) Hash tables
D) Sequential files

Answer: A

SECTION 13: INDEXING IN VARIOUS DBMS (Questions 221-230)

108. In MySQL InnoDB, the primary key index is:
A) Clustered index (data stored with index)
B) Non-clustered
C) Heap
D) Hash

Answer: A

---

109. In MySQL, secondary indexes store:
A) Primary key values as row pointers (since clustered)
B) Physical row addresses
C) No pointers
D) Data directly

Answer: A

---

110. In SQL Server, clustered index:
A) Determines physical order, data at leaf level
B) Separate from data

C) Not available
D) Always hash

Answer: A

---

111. In Oracle, index-organized tables (IOT):
A) Store data in index itself (like clustered)
B) Separate data and index
C) Heap organized
D) No indexes

Answer: A

---

112. In PostgreSQL, indexes are:
A) Separate structures, can be B-tree, Hash, GiST, GIN, SP-GiST, BRIN
B) Only B-tree
C) Only hash
D) No indexes

Answer: A

SECTION 14: COMPARISON AND TRADE-OFFS (Questions 231-240)

113. B-tree vs Hash:
A) B-tree supports range, hash supports only equality
B) Both support range
C) Both support only equality
D) Hash supports range

Answer: A

---

114. B-tree vs Bitmap:
A) B-tree good for high cardinality, bitmap for low cardinality
B) Both same

C) Bitmap for high cardinality
D) B-tree for low cardinality

Answer: A

---

115. Clustered vs Non-clustered:
A) Clustered faster for range, one per table; non-clustered multiple, may need lookup
B) Both same
C) Non-clustered faster for range
D) Clustered multiple allowed

Answer: A

---

116. Index maintenance overhead affects:
A) INSERT, UPDATE, DELETE performance
B) SELECT performance only
C) Read-only queries
D) Backups

Answer: A

SECTION 1: INTRODUCTION TO QUERY PROCESSING (Questions 1-20)

1. Query processing refers to:
A) Series of steps to extract data from database
B) Writing SQL queries
C) Creating indexes
D) Backing up data

Answer: A
Explanation: Query processing involves parsing, optimization, and execution of queries.

---

2. The phases of query processing in order are:
A) Parsing & translation → Optimization → Evaluation

B) Optimization → Parsing → Evaluation
C) Evaluation → Parsing → Optimization
D) Parsing → Evaluation → Optimization

Answer: A

---

3. Parsing and translation converts SQL into:
A) Internal form (parse tree, relational algebra)
B) Machine code
C) Assembly language
D) Byte code

Answer: A

---

4. Query optimization chooses:
A) Most efficient execution plan
B) First available plan
C) Random plan
D) Simplest plan

Answer: A

---

5. Query evaluation executes:
A) The chosen execution plan
B) The original SQL
C) Any plan
D) No plan

Answer: A

---

6. An execution plan contains:
A) Sequence of operations (scans, joins, sorts) with algorithms
B) Only SQL text
C) Only table names
D) Only index names

Answer: A

---

7. The query optimizer uses:
A) Statistics about data (size, distribution, indexes)
B) Random guesses
C) User preferences
D) Default settings only

Answer: A

---

8. Statistics used by optimizer include:
A) Number of tuples, distinct values, index existence, data distribution
B) Only table names
C) Only column names
D) Only query text

Answer: A

SECTION 2: MEASURING QUERY COST (Questions 21-40)

9. The cost of query execution is measured in terms of:
A) Disk accesses, CPU time, memory usage, network cost
B) Only disk accesses
C) Only CPU time
D) Only memory usage

Answer: A

---

10. Disk access is often the dominant cost because:
A) I/O is orders of magnitude slower than memory/CPU
B) CPU is slower
C) Memory is slower
D) Network is faster

Answer: A

---

11. Number of block transfers (I/Os) is a common cost metric:
A) Because it correlates well with execution time
B) Because it's easy to measure
C) Because CPU is free
D) Because memory is infinite

Answer: A

---

12. Seek time vs transfer time:
A) Seek time is often larger than transfer time for random access
B) Transfer time is always larger
C) Both equal
D) Seek time is zero

Answer: A

---

13. Sequential I/O is faster than random I/O because:
A) Multiple blocks can be read with one seek
B) No data is read
C) CPU is faster
D) Disk spins faster

Answer: A

---

14. Cost estimates depend on:
A) Database statistics (relation sizes, index info)
B) Query complexity only
C) User input
D) System load

Answer: A

---

15. The number of tuples in a relation is denoted as:
A) n_r (cardinality)
B) b_r
C) f_r
D) V_r

Answer: A

---

16. The number of blocks containing relation r is:
A) $b_r = \lceil n_r / f_r \rceil$ where f_r is blocking factor
B) $n_r \times f_r$
C) $n_r + f_r$
D) $n_r - f_r$

Answer: A

---

17. The blocking factor f_r is:
A) Number of tuples per block
B) Number of blocks
C) Number of tuples
D) Block size

Answer: A

---

18. Selection cardinality $s = n_r / V(A,r)$ where $V(A,r)$ is:
A) Number of distinct values of attribute A
B) Number of tuples
C) Number of blocks
D) Index height

Answer: A

---

19. $V(A,r)$ represents:
A) Number of distinct values of attribute A in relation r
B) Number of tuples
C) Number of blocks
D) Index size

Answer: A

---

20. Histograms store:
A) Frequency distribution of attribute values

B) Only minimum and maximum
C) Only average
D) No statistics

Answer: A

SECTION 3: SELECTION OPERATIONS (Questions 41-60)

21. File scan (linear search) cost:
A) b_r block transfers (entire relation)
B) log(b_r) transfers
C) 1 transfer
D) 0 transfers

Answer: A

---

22. Linear search cost for equality on key (assuming one match):
A) b_r/2 average (scan until found)
B) b_r always
C) log b_r
D) 1

Answer: A

---

23. Index scan using primary index (B+ tree) equality on key:
A) Height of tree + 1 block access
B) b_r accesses
C) log b_r
D) 1

Answer: A

---

24. Primary index equality on non-key retrieves:
A) Multiple records (selection cardinality s)
B) Single record always

C) No records
D) All records

Answer: A

---

25. Cost with primary index and equality on non-key:
A) Height + s block accesses (if clustered, records in ≈ s/f_r blocks)
B) Height + b_r
C) b_r only
D) s only

Answer: A

---

26. Secondary index equality on key (unique):
A) Height + 1 (index lookup + data access)
B) Height only
C) b_r
D) 1

Answer: A

---

27. Secondary index equality on non-key (non-unique):
A) Height + s (each record may be in different block - worst case)
B) Height only
C) s only
D) b_r

Answer: A

---

28. Index scan for range condition (e.g., A > value):
A) Use index to find first match, then scan leaves/data
B) Full table scan only
C) No index use
D) Hash lookup

Answer: A

---

29. Complex selections (AND/OR) can use:
A) Index intersection, index union, or bitmaps
B) Only sequential scan
C) Only single index
D) No optimization

Answer: A

---

30. Conjunctive selection ($\sigma\theta_1 \wedge \theta_2 \wedge ... \wedge \theta_n$) can use:
A) Most selective index first, then filter
B) Random index
C) No index
D) All indexes

Answer: A

SECTION 4: SORTING (Questions 61-75)

31. External sorting is used when:
A) Data doesn't fit in main memory
B) Data fits in memory
C) No sorting needed
D) Data is small

Answer: A

---

32. External sort-merge algorithm uses:
A) Run generation phase and merge phase
B) Only one phase
C) Quick sort in memory
D) Bubble sort

Answer: A

---

33. In run generation phase, we create:
A) Sorted runs (chunks) of data
B) Unsorted chunks
C) Indexes
D) Hash tables

Answer: A

---

34. Number of passes in merge phase =:
A) ⌈log_{M-1}(N/M)⌉ where M = buffer pages, N = runs
B) N × M
C) N / M
D) M^N

Answer: A

---

35. Total I/O cost for external sort:
A) 2 × N × (number of passes + 1) where N = number of blocks
B) N × passes
C) N^2
D) log N

Answer: A

SECTION 5: JOIN OPERATIONS (Questions 76-110)

36. Nested-loop join (r ⋈ s) cost:
A) n_r × b_s + b_r (for block nested loop)
B) n_r × n_s
C) b_r + b_s
D) b_r × b_s

Answer: A

---

37. Block nested-loop join uses:
A) One block of outer, compare with all blocks of inner

B) One tuple of outer, compare with all of inner
C) No blocks
D) Only indexes

Answer: A

---

38. Block nested-loop cost with buffer of M blocks:
A) $b_r + \lceil b_r/(M-1) \rceil \times b_s$
B) $b_r \times b_s$
C) $b_r + b_s$
D) $n_r \times n_s$

Answer: A

---

39. Indexed nested-loop join uses:
A) Index on inner relation for each outer tuple
B) No index
C) Full scan always
D) Hash join

Answer: A

---

40. Cost of indexed nested-loop join:
A) $b_r + n_r \times c$ (where c is index lookup cost)
B) $b_r \times b_s$
C) $n_r \times n_s$
D) $b_r + b_s$

Answer: A

---

41. Sort-merge join first:
A) Sorts both relations on join attribute
B) Scans both relations
C) Creates indexes
D) Hashes relations

Answer: A

---

42. Sort-merge join then:
A) Merges sorted relations by scanning simultaneously
B) Performs nested loops
C) Builds hash table
D) Uses index

Answer: A

---

43. Cost of sort-merge join includes:
A) Sorting cost + scanning cost (b_r + b_s)
B) Only scanning
C) Only sorting
D) Only indexing

Answer: A

---

44. Hash join uses:
A) Hash function to partition relations into buckets
B) Sorting
C) Indexes
D) Nested loops

Answer: A

---

45. Hash join phases are:
A) Build (hash smaller relation) and Probe (hash larger relation)
B) Sort and Merge
C) Scan and Index
D) Only Build

Answer: A

---

46. For hash join to be efficient:
A) Smaller relation should fit in memory hash table
B) Both relations must be large

C) No memory needed
D) Both relations must be sorted

Answer: A

---

47. Grace hash join handles:
A) Large relations by partitioning to disk
B) Small relations only
C) Sorted data only
D) Indexed data only

Answer: A

---

48. Hybrid hash join combines:
A) In-memory and partitioned approaches
B) Sort and hash
C) Nested loop and hash
D) Index and hash

Answer: A

---

49. Join selectivity factor is:
A) Estimated size of result / ($n_r \times n_s$)
B) $n_r \times n_s$
C) $n_r + n_s$
D) $n_r - n_s$

Answer: A

---

50. Choosing the smaller relation as outer in nested-loop:
A) Reduces cost
B) Increases cost
C) No effect
D) Makes join impossible

Answer: A

---

51. For joining three relations, join order matters because:
A) Intermediate result sizes vary greatly
B) All orders same cost
C) Only order of attributes matters
D) No difference

Answer: A

---

52. Dynamic programming in join ordering:
A) Finds optimal join order using cost estimates
B) Tries all orders exhaustively
C) Uses random order
D) No optimization

Answer: A

SECTION 6: OTHER OPERATIONS (Questions 111-125)

53. Projection (π) without duplicate elimination:
A) Just retrieve columns, cost of scanning
B) Requires sorting
C) Requires hashing
D) Requires index

Answer: A

---

54. Projection with duplicate elimination requires:
A) Sorting or hashing to remove duplicates
B) No extra work
C) Index only
D) Full scan only

Answer: A

---

55. Aggregation (GROUP BY) without grouping:
A) Scan relation, compute aggregate
B) Requires sorting
C) Requires index
D) No operation

Answer: A

---

56. Aggregation with GROUP BY requires:
A) Sorting or hashing to group tuples
B) Only scanning
C) No grouping
D) Index only

Answer: A

---

57. Set operations (UNION, INTERSECT, EXCEPT) require:
A) Sorting or hashing to eliminate duplicates
B) Only scanning
C) No processing
D) Index only

Answer: A

SECTION 7: QUERY OPTIMIZATION (Questions 126-150)

58. Query optimization aims to find:
A) Lowest-cost execution plan among alternatives
B) Any valid plan
C) Simplest plan
D) Most complex plan

Answer: A

---

59. Optimization techniques include:
A) Heuristic (rule-based) and Cost-based

B) Only heuristic
C) Only cost-based
D) Random selection

Answer: A

---

60. Heuristic optimization applies:
A) Transformation rules (e.g., push selections down)
B) Cost calculations
C) Random choices
D) No rules

Answer: A

---

61. Pushing selections down the query tree:
A) Reduces size of intermediate results early
B) Increases size
C) Has no effect
D) Slows query

Answer: A

---

62. Pushing projections down:
A) Reduces number of columns early
B) Increases columns
C) No effect
D) Slows query

Answer: A

---

63. Cost-based optimization:
A) Estimates cost of different plans using statistics
B) Uses fixed rules only
C) No estimation
D) Random plans

Answer: A

---

64. System catalogs store:
A) Statistics (relation sizes, index info, value distributions)
B) Query text
C) User data
D) Logs only

Answer: A

---

65. Query optimization search space includes:
A) Different join orders, join algorithms, access paths
B) Only one plan
C) Only index selection
D) Only join order

Answer: A

---

66. Dynamic programming in optimizers:
A) Stores optimal plans for subsets of relations
B) Recomputes everything
C) No storage
D) Random selection

Answer: A

---

67. Left-deep join trees:
A) Right relation always base table, easier for pipelining
B) Both sides may be intermediate
C) No structure
D) Bushy only

Answer: A

---

68. Bushy join trees:
A) Both sides may be intermediate results
B) Only left-deep allowed

C) Only right-deep allowed
D) No joins

Answer: A

---

69. Pipelined execution:
A) Results of one operation passed directly to next without materialization
B) All intermediate results written to disk
C) No execution
D) Only materialization

Answer: A

---

70. Materialization:
A) Intermediate results written to disk/temporary storage
B) No storage of intermediates
C) Faster always
D) No processing

Answer: A

SECTION 8: ACCESS PATH SELECTION (Questions 151-165)

71. Access path is:
A) Method to retrieve tuples (file scan, index scan)
B) Query plan
C) Join algorithm
D) Sort method

Answer: A

---

72. Index selection considers:
A) Whether index exists and its type (clustered, non-clustered)
B) Only table scan
C) No indexes
D) Only primary key

Answer: A

---

73. Clustered index is preferred for:
A) Range queries (data physically ordered)
B) Point queries only
C) Inserts only
D) Deletes only

Answer: A

---

74. Non-clustered index good for:
A) Highly selective queries (few rows)
B) Full table scans
C) All queries
D) Updates only

Answer: A

---

75. Index-only plan (covering index):
A) All needed columns in index, no data access
B) Requires data access always
C) No index used
D) Only table scan

Answer: A

SECTION 9: PARALLEL QUERY PROCESSING (Questions 166-180)

76. Parallel query processing uses:
A) Multiple processors/disks to speed up queries
B) Single processor only
C) No parallelism
D) Only sequential

Answer: A

---

77. Inter-query parallelism:
A) Multiple queries execute in parallel
B) Single query parallelized
C) No parallelism
D) Serial execution

Answer: A

---

78. Intra-query parallelism:
A) Single query divided across multiple processors
B) Multiple queries
C) No parallelism
D) Serial execution

Answer: A

---

79. Horizontal partitioning (parallel scan):
A) Relation divided across disks, each processor scans its partition
B) All data on one disk
C) No partitioning
D) Vertical only

Answer: A

---

80. Parallel join algorithms include:
A) Partitioned join, asymmetric join
B) Only sequential join
C) No joins
D) Only nested loop

Answer: A

SECTION 10: ADVANCED OPTIMIZATION TOPICS (Questions 181-200)

81. Adaptive query processing:
A) Adjusts plan during execution based on observed statistics
B) Fixed plan always
C) No adjustment
D) Random changes

Answer: A

---

82. Parametric query optimization:
A) Generates multiple plans for different parameter values
B) Single plan only
C) No parameters
D) Only for constants

Answer: A

---

83. Query optimization in distributed databases considers:
A) Data transfer cost + local processing cost
B) Only local cost
C) Only network cost
D) No cost

Answer: A

---

84. Semi-join reduces:
A) Amount of data transferred by sending only joining attribute values
B) Full data always sent
C) No reduction
D) Local processing

Answer: A

---

85. Star query optimization (data warehouses) uses:
A) Star join, bitmap indexes, materialized views
B) Standard joins only
C) No optimization
D) Only nested loop

Answer: A

---

86. Materialized views can speed up queries by:
A) Precomputing and storing results for reuse
B) Always recomputing
C) No storage
D) Only base tables

Answer: A

---

87. Query rewrite uses:
A) Materialized views, predicate pushdown, transformation rules
B) Only original query
C) No changes
D) Random modifications

Answer: A

---

88. Explain plan shows:
A) Execution plan chosen by optimizer
B) Query syntax
C) Table structure
D) Index definitions

Answer: A

---

89. Hint can influence:
A) Optimizer's choices (join order, index use)
B) Query result
C) Data content
D) Table structure

Answer: A

---

90. Statistics update is important because:
A) Stale statistics lead to poor plan choices
B) Always same plan
C) No effect
D) Only for schema changes

Answer: A

SECTION 11: PRACTICAL OPTIMIZATION IN DBMS (Questions 201-220)

91. In PostgreSQL, EXPLAIN ANALYZE:
A) Shows plan and actual execution statistics
B) Only shows plan
C) Only executes
D) Shows no output

Answer: A

---

92. In MySQL, EXPLAIN shows:
A) Query execution plan with access types, key used
B) Only syntax check
C) Only errors
D) No information

Answer: A

---

93. In Oracle, autotrace:
A) Shows execution plan and statistics
B) Only syntax
C) Only errors
D) No output

Answer: A

---

94. SQL Server Management Studio shows:
A) Graphical execution plans
B) Only text
C) No plans
D) Only syntax

Answer: A

---

95. Plan caching stores:
A) Execution plans for reuse
B) Data only
C) Logs only
D) No storage

Answer: A

---

96. Parameter sniffing occurs when:
A) Plan optimized for first parameter value reused for others (may be suboptimal)
B) No parameters used
C) Always optimal
D) No plan caching

Answer: A

---

97. Optimizer hints force:
A) Specific join order, index, or algorithm
B) Random choices
C) No effect
D) Syntax errors

Answer: A

SECTION 12: COST ESTIMATION EXAMPLES (Questions 221-235)

98. Selection on key using primary index cost ≈:
A) Height of index + 1
B) b_r
C) n_r
D) log n_r

Answer: A

---

99. Selection on non-key with clustering index ≈:
A) Height + (s / f_r) where $s = n_r / V$
B) Height + s
C) b_r
D) s only

Answer: A

---

100. Block nested-loop join with R outer, S inner cost:
A) $b_R + \lceil b_R / (M-1) \rceil \times b_S$
B) $b_R \times b_S$
C) $b_R + b_S$
D) $n_R \times n_S$

Answer: A

---

101. Sort-merge join cost estimate (worst case):
A) Sort(R) + Sort(S) + ($b_R + b_S$)
B) $b_R + b_S$ only
C) $b_R \times b_S$
D) $n_R \times n_S$

Answer: A

---

102. Hash join cost (if smaller fits in memory):
A) b_R + b_S (read both, build probe)
B) b_R × b_S
C) b_R + b_S + hash
D) b_R only

Answer: A

---

103. Indexed nested-loop join cost:
A) b_R + n_R × c (c = index lookup cost)
B) b_R × n_S
C) n_R × n_S
D) b_R + b_S

Answer: A

SECTION 13: OPTIMIZATION CHALLENGES (Questions 236-250)

104. Skew in data distribution affects:
A) Accuracy of cost estimates
B) Query syntax
C) Index creation
D) No effect

Answer: A

---

105. Correlated subqueries are challenging because:
A) They may need to be executed repeatedly (can be decorrelated)
B) Always fast
C) No optimization possible
D) Always slow

Answer: A

---

106. Decorrelation transforms:
A) Correlated subquery into join for better optimization
B) Join into subquery
C) Simple query into complex
D) No transformation

Answer: A

---

107. Complex predicates (OR, LIKE) may:
A) Limit index usage, require careful optimization
B) Always use indexes
C) Never use indexes
D) No effect

Answer: A

---

108. User-defined functions (UDFs) in queries:
A) Can be optimization barriers (black box)
B) Always optimized
C) No effect
D) Faster always

Answer: A

SECTION 1: INTRODUCTION TO DATABASE SECURITY (Questions 1-20)

1. Database security primarily aims to protect:
A) Confidentiality, Integrity, and Availability (CIA triad)
B) Only data confidentiality
C) Only system performance
D) Only user access

Answer: A
Explanation: Database security encompasses confidentiality (no unauthorized

disclosure), integrity (no unauthorized modification), and availability (data accessible when needed) .

---

2. Confidentiality in database security means:
A) Preventing unauthorized disclosure of data
B) Preventing unauthorized modification
C) Ensuring data is always available
D) Backing up data regularly

Answer: A

---

3. Integrity in database security means:
A) Preventing unauthorized or improper data modification
B) Keeping data secret
C) Making data available to all users
D) Encrypting all data

Answer: A

---

4. Availability in database security means:
A) Authorized users can access data when needed
B) Data is always public
C) Data is never modified
D) Data is always encrypted

Answer: A

---

5. Data security is defined as:
A) Protection of data from destruction, corruption, or unauthorized access
B) The process of backing up data
C) Creating user accounts
D) Writing SQL queries

Answer: A

---

6. A data breach involves:
A) Unauthorized access, disclosure, or theft of data
B) Regular database maintenance
C) Creating backups
D) User authentication

Answer: A

---

7. The three states of data that require protection are:
A) Data at rest, data in transit, data in use
B) Data stored, data backed up, data archived
C) Data entered, data processed, data deleted
D) Data encrypted, data decrypted, data hashed

Answer: A

---

8. Data at rest refers to:
A) Data stored on disk, tape, or other persistent media
B) Data moving across networks
C) Data being processed in memory
D) Data being displayed to users

Answer: A

---

9. Data in transit (motion) refers to:
A) Data being transmitted across networks between systems
B) Data stored on disk
C) Data in CPU registers
D) Data in database buffers

Answer: A

---

10. Data in use (in process) refers to:
A) Data currently being accessed or modified in memory
B) Data stored on backup tapes
C) Data waiting to be processed
D) Data that is encrypted

Answer: A

SECTION 2: AUTHENTICATION (Questions 21-40)

11. Authentication is the process of:
A) Verifying the identity of a user, device, or system
B) Determining what operations a user can perform
C) Tracking user activities
D) Encrypting user data

Answer: A

---

12. Something you KNOW authentication factor includes:
A) Password or PIN
B) Fingerprint
C) Smart card
D) Retina scan

Answer: A

---

13. Something you HAVE authentication factor includes:
A) Smart card, token, or mobile device
B) Password
C) Fingerprint
D) Date of birth

Answer: A

---

14. Something you ARE authentication factor includes:
A) Biometrics (fingerprint, iris, voice)
B) Password
C) Security question answer
D) PIN

Answer: A

---

15. Multi-factor authentication (MFA) requires:
A) Two or more different authentication factors
B) Two passwords
C) Two usernames
D) Two attempts to login

Answer: A

---

16. Database authentication by the operating system means:
A) OS credentials are used to access database without separate password
B) Database stores OS passwords
C) OS and database share no credentials
D) Users must login twice

Answer: A

---

17. A database user account typically includes:
A) Username and authentication credential (password, certificate, etc.)
B) Only username
C) Only password
D) Only role name

Answer: A

---

18. Failed login attempts can be configured to:
A) Lock the account after a specified number of failures
B) Always allow access
C) Reset the password automatically
D) Delete the account

Answer: A

---

19. Password profiles can enforce:
A) Password complexity, expiration, and reuse policies
B) Only minimum length

C) Only maximum age
D) No restrictions

Answer: A

---

20. The CREATE USER statement in SQL is used to:
A) Create a new database user account
B) Create a new table
C) Grant privileges
D) Create a role

Answer: A

---

21. The CREATE SESSION privilege allows a user to:
A) Connect to the database
B) Create tables
C) Drop tables
D) Select data

Answer: A

SECTION 3: AUTHORIZATION AND PRIVILEGES (Questions 41-65)

22. Authorization is:
A) Permission given to a user to access an object or perform an action
B) Verifying user identity
C) Tracking user activities
D) Encrypting data

Answer: A

---

23. A privilege is:
A) The right to execute specific SQL statements or access objects
B) A user account

C) A password policy
D) An encryption key

Answer: A

---

24. The two main categories of privileges are:
A) System privileges and object privileges
B) Read privileges and write privileges
C) User privileges and role privileges
D) Local privileges and remote privileges

Answer: A

---

25. System privileges allow:
A) Performing actions at the system level (e.g., CREATE USER, CREATE TABLE)
B) Accessing specific tables only
C) Only connecting to database
D) Only viewing data

Answer: A

---

26. Object privileges allow:
A) Performing actions on specific schema objects (e.g., SELECT on employees)
B) Creating new databases
C) Shutting down database
D) Creating users

Answer: A

---

27. Common object privileges for tables include:
A) SELECT, INSERT, UPDATE, DELETE, ALTER, REFERENCES
B) CONNECT, RESOURCE, DBA
C) CREATE SESSION, CREATE TABLE
D) EXECUTE, DEBUG

Answer: A

---

28. The REFERENCES privilege allows:
A) Creating foreign key constraints referencing a table
B) Selecting from a table
C) Updating a table
D) Deleting from a table

Answer: A

---

29. The GRANT statement is used to:
A) Give privileges to users or roles
B) Remove privileges
C) Create users
D) Create tables

Answer: A

---

30. The REVOKE statement is used to:
A) Remove previously granted privileges
B) Grant new privileges
C) Create roles
D) Drop users

Answer: A

---

31. WITH GRANT OPTION allows:
A) The grantee to grant the same privilege to others
B) Unlimited privileges
C) Granting all privileges
D) Only viewing grants

Answer: A

---

32. When privileges are revoked, they should be treated as if:
A) They had never been granted (proper propagation)
B) They still exist for backward compatibility

C) Only the first user loses access
D) They remain until session ends

Answer: A

SECTION 4: ROLES (Questions 66-85)

33. A role is:
A) A named group of related privileges
B) A user account
C) A password policy
D) An encryption key

Answer: A

---

34. Roles simplify privilege management by:
A) Allowing privileges to be granted to groups rather than individual users
B) Eliminating need for privileges
C) Automatically granting all privileges
D) Removing all access control

Answer: A

---

35. A user can be granted:
A) Multiple roles
B) Only one role
C) No roles
D) Only the PUBLIC role

Answer: A

---

36. Roles can be granted to:
A) Users and other roles
B) Only users
C) Only tables
D) Only schemas

Answer: A

---

37. The PUBLIC role represents:
A) All users of the database
B) Only administrators
C) Only application users
D) No users

Answer: A

---

38. Roles can be password-protected to:
A) Prevent unauthorized enabling of the role
B) Encrypt role data
C) Hide role name
D) Delete the role

Answer: A

---

39. Secure application roles are:
A) Roles implemented by a package that validates conditions before enabling
B) Roles with no privileges
C) Public roles
D) Default roles

Answer: A

---

40. Enterprise roles in Oracle:
A) Contain global roles on multiple databases and are granted to enterprise users
B) Are local to one database only
C) Cannot be granted to users
D) Have no privileges

Answer: A

---

41. Default roles are:
A) Roles automatically enabled when user connects
B) Roles that cannot be disabled
C) System-defined roles only
D) Roles with no privileges

Answer: A

SECTION 5: ACCESS CONTROL MODELS (Questions 86-105)

42. Discretionary Access Control (DAC) allows:
A) Object owners to grant access to others at their discretion
B) System-wide mandatory rules
C) Role-based assignments only
D) No user control

Answer: A

---

43. Mandatory Access Control (MAC) uses:
A) Labels and clearances to enforce system-wide security policy
B) User discretion for access
C) Only roles for access
D) No labels

Answer: A

---

44. Role-Based Access Control (RBAC) assigns:
A) Permissions to roles, and roles to users based on job functions
B) Permissions directly to users
C) No roles
D) Only system privileges

Answer: A

---

45. The principle of least privilege means:
A) Users get only the minimum permissions needed to perform their job

B) Users get all possible permissions
C) Users get no permissions
D) Permissions are unlimited

Answer: A

---

46. Separation of duties ensures:
A) No single user has all privileges to complete a critical task
B) All users have same privileges
C) Users share accounts
D) No security controls

Answer: A

---

47. Access control is defined as:
A) Mechanism to allow users different levels of access to database resources
B) Process of user identification
C) Method of encryption
D) Backup procedure

Answer: A

---

48. Row-level security restricts access to:
A) Specific rows based on user attributes or policies
B) Entire tables only
C) Only columns
D) No data

Answer: A

---

49. Column-level security restricts access to:
A) Specific columns within tables
B) Entire tables only
C) Only rows
D) No data

Answer: A

---

50. Virtual Private Database (VPD) provides:
A) Fine-grained access control by dynamically adding predicates to queries
B) No access control
C) Only table-level security
D) Only network security

Answer: A

SECTION 6: ENCRYPTION (Questions 106-130)

51. Encryption is:
A) Transforming data into unreadable format using algorithm and key
B) Making data readable
C) Deleting data
D) Backing up data

Answer: A

---

52. Decryption is:
A) Converting encrypted data back to original form using key
B) Encrypting data
C) Hashing data
D) Compressing data

Answer: A

---

53. Transparent Data Encryption (TDE) automatically:
A) Encrypts data written to disk and decrypts when read, without application changes
B) Requires application modification
C) Only works for backups
D) Only encrypts network traffic

Answer: A

---

54. TDE protects data:
A) At rest (on disk)
B) In transit
C) In memory
D) Only during backups

Answer: A

---

55. Network encryption (SSL/TLS) protects data:
A) In transit between client and server
B) At rest on disk
C) In CPU registers
D) Only during authentication

Answer: A

---

56. Encryption algorithms include:
A) AES (Advanced Encryption Standard), DES, 3DES
B) MD5, SHA (these are hash functions)
C) SQL, PL/SQL
D) HTML, XML

Answer: A

---

57. Symmetric encryption uses:
A) Same key for encryption and decryption
B) Different keys for encryption and decryption
C) No keys
D) Public keys only

Answer: A

---

58. Asymmetric encryption uses:
A) Public key for encryption, private key for decryption
B) Same key for both
C) No keys
D) Only passwords

Answer: A

---

59. Hashing is different from encryption because:
A) Hashing is one-way (cannot be reversed)
B) Hashing can be decrypted
C) Hashing uses keys
D) Hashing is slower

Answer: A

---

60. PCI-DSS compliance often requires:
A) Encryption of cardholder data
B) No encryption
C) Public data only
D) Backup only

Answer: A

---

61. HIPAA security rule requires protection of:
A) Electronic protected health information (ePHI)
B) Only financial data
C) Only public data
D) No specific data

Answer: A

SECTION 7: DATA MASKING AND REDACTION (Questions 131-145)

62. Data masking (redaction) is:
A) Hiding sensitive data by displaying modified values
B) Encrypting all data
C) Deleting sensitive data
D) Backing up data

Answer: A

---

63. Dynamic data masking:
A) Masks data in query results in real-time without changing underlying data
B) Permanently changes data
C) Encrypts data on disk
D) Only works for backups

Answer: A

---

64. Full data redaction displays:
A) Completely redacted values (e.g., all columns show spaces or zeros)
B) Original data
C) Encrypted data
D) No data

Answer: A

---

65. Partial data redaction displays:
A) Only portions of data (e.g., credit card shows last 4 digits)
B) Complete data
C) No data
D) Encrypted data

Answer: A

---

66. Random data redaction displays:
A) Randomly generated values for each query
B) Consistent masked values
C) Original values
D) No values

Answer: A

SECTION 8: AUDITING AND MONITORING (Questions 146-165)

67. Database auditing is:
A) Tracking and logging database activities for security analysis
B) Encrypting data
C) Creating users
D) Backing up data

Answer: A

---

68. Audit records typically capture:
A) User, operation, timestamp, object accessed, success/failure
B) Only username
C) Only timestamp
D) No information

Answer: A

---

69. An intrusion detection system (IDS):
A) Monitors for suspicious activity and alerts administrators
B) Prevents all intrusions
C) Encrypts all data
D) Manages users

Answer: A

---

70. SIEM (Security Information and Event Management) systems:
A) Collect and analyze logs from multiple sources for real-time monitoring
B) Only store logs
C) Only encrypt data
D) Only manage users

Answer: A

---

71. Failed login attempts should be audited to detect:
A) Brute force attacks
B) Successful logins
C) Normal activity
D) Scheduled maintenance

Answer: A

---

72. Privilege usage auditing helps identify:
A) Unnecessary privileges that can be revoked
B) All users
C) All tables
D) All indexes

Answer: A

---

73. Fine-grained auditing (FGA) can track:
A) Access to specific rows or columns based on conditions
B) All activities on all objects
C) Only logins
D) Only backups

Answer: A

SECTION 9: SQL INJECTION AND VULNERABILITIES (Questions 166-185)

74. SQL injection is:
A) Inserting malicious SQL code into application input to manipulate database
B) A type of encryption
C) A backup method
D) A user authentication technique

Answer: A

---

75. SQL injection can lead to:
A) Data theft, data corruption, authentication bypass
B) Faster queries
C) Better performance
D) Automatic backups

Answer: A

---

76. Parameterized queries (prepared statements) prevent SQL injection by:
A) Separating SQL code from user input
B) Concatenating user input with SQL
C) Disabling all input
D) Using dynamic SQL only

Answer: A

---

77. Stored procedures can help prevent SQL injection by:
A) Encapsulating SQL logic and validating input
B) Allowing direct table access
C) Using dynamic SQL only
D) Disabling all security

Answer: A

---

78. Input validation involves:
A) Checking user input for malicious content before processing
B) Accepting all input without checking
C) Disabling all input
D) Encrypting input only

Answer: A

---

79. Least privilege for application accounts means:
A) Application connects with minimal needed privileges, not DBA
B) Application uses sa or root account
C) Application has no privileges
D) Application uses public role only

Answer: A

SECTION 10: INFERENCE AND PRIVACY (Questions 186-200)

80. Inference attacks occur when:
A) Legitimate data is used to deduce unknown sensitive information
B) Data is directly accessed without authorization
C) Passwords are stolen
D) Encryption is broken

Answer: A

---

81. Cell suppression protects against inference by:
A) Removing or leaving blank some cells in published data
B) Encrypting all cells
C) Duplicating cells
D) Adding more data

Answer: A

---

82. Generalization protects against inference by:
A) Replacing specific values with broader categories (e.g., age 25 → 20-30)
B) Showing exact values
C) Encrypting values
D) Deleting values

Answer: A

---

83. Noise addition protects against inference by:
A) Adding random values to data to obscure exact values
B) Removing all data
C) Encrypting data
D) Duplicating data

Answer: A

---

84. k-anonymity requires that:
A) Each record is indistinguishable from at least k-1 others on quasi-identifiers
B) All records are unique
C) All data is encrypted
D) No data is released

Answer: A

SECTION 11: BACKUP AND RECOVERY SECURITY (Questions 201-215)

85. Backup security requires:
A) Encrypting backups and securing backup storage
B) Leaving backups unsecured for quick recovery
C) Only storing backups locally
D) Never testing backups

Answer: A

---

86. Backup encryption protects against:
A) Theft of backup media
B) Only online attacks
C) Only network attacks
D) Only user errors

Answer: A

---

87. Offsite backup storage protects against:
A) Physical disasters at primary site
B) Only logical corruption
C) Only user errors
D) Only performance issues

Answer: A

---

88. Recovery testing verifies that:
A) Backups can be successfully restored
B) Backups exist
C) Backups are encrypted
D) Backups are large enough

Answer: A

SECTION 12: PATCH MANAGEMENT AND HARDENING (Questions 216-230)

89. Database hardening involves:
A) Removing unnecessary services, securing configuration, applying patches
B) Adding more services
C) Opening all ports
D) Disabling all security

Answer: A

---

90. Patch management is important because:
A) Unpatched vulnerabilities are common attack vectors
B) Patches slow down database
C) Patches are optional
D) Patches never fix security issues

Answer: A

---

91. The Common Vulnerability Scoring System (CVSS) provides:
A) Standardized severity scores for vulnerabilities
B) Patch installation instructions
C) Encryption algorithms
D) User management tools

Answer: A

---

92. Testing patches before production deployment prevents:
A) Unforeseen issues affecting production systems
B) Faster deployment
C) Security fixes
D) Compliance

Answer: A

---

93. Removing default accounts and passwords is:
A) A critical security hardening step
B) Optional
C) Not recommended
D) Only for test systems

Answer: A

SECTION 13: NETWORK SECURITY (Questions 231-240)

94. Firewalls protect databases by:
A) Restricting network access to authorized hosts and ports
B) Encrypting all data
C) Managing users
D) Creating backups

Answer: A

---

95. Database should be placed in:
A) Private network segment, not directly exposed to internet
B) Public internet for easy access
C) DMZ with web servers
D) Same network as all users

Answer: A

---

96. VPNs provide:
A) Encrypted tunnels for remote database access
B) Public access
C) No encryption
D) User management

Answer: A

SECTION 14: COMPLIANCE AND STANDARDS (Questions 241-250)

97. GDPR requires:
A) Protection of personal data of EU citizens
B) Only US citizen data
C) No data protection
D) Only financial data

Answer: A

---

98. SOX (Sarbanes-Oxley) requires:
A) Controls over financial reporting data
B) Only IT general controls
C) No data protection
D) Only physical security

Answer: A

---

99. PCI-DSS applies to:
A) Organizations handling payment card data
B) All businesses
C) Only banks
D) Only government agencies

Answer: A

---

100. HIPAA applies to:
A) Healthcare providers handling protected health information
B) All organizations
C) Only insurance companies
D) Only patients

Answer: A

SECTION 1: INTRODUCTION TO DISTRIBUTED DATABASES (Questions 1-20)

1. A distributed database is:
A) A collection of logically interrelated data distributed across multiple physical sites connected by a network
B) Multiple copies of the same database at one site
C) A single database on one server
D) A backup database at remote location

Answer: A
Explanation: Distributed database has logically integrated data physically distributed across network nodes.

---

2. A distributed database management system (DDBMS) is:
A) Software that manages distributed database making distribution transparent to users
B) Multiple DBMSs running independently
C) A single centralized DBMS
D) A file sharing system

Answer: A

---

3. The main reasons for adopting distributed databases include:
A) Reliability, scalability, local autonomy, cost-effectiveness
B) Only better security

C) Only faster queries
D) Only easier management

Answer: A

---

4. Local autonomy in distributed databases means:
A) Each site has control over its own local data and processing
B) All sites controlled by central site
C) No local control
D) Only remote access allowed

Answer: A

---

5. Transparency in distributed databases refers to:
A) Hiding details of distribution from users and applications
B) Making distribution visible to users
C) No network connectivity
D) Only local access

Answer: A

SECTION 2: DISTRIBUTED DATABASE ARCHITECTURE (Questions 21-40)

6. The two main types of distributed database systems are:
A) Homogeneous and Heterogeneous
B) Centralized and Decentralized
C) Client-Server and Peer-to-Peer
D) Local and Remote

Answer: A

---

7. Homogeneous distributed databases have:
A) Same DBMS software at all sites
B) Different DBMS software at different sites

C) No DBMS software
D) Only one site

Answer: A

---

8. Heterogeneous distributed databases have:
A) Different DBMS software, possibly different data models
B) Same DBMS software everywhere
C) No network connectivity
D) Only one DBMS type

Answer: A

---

9. Federated database system is a type of:
A) Heterogeneous distributed database with local autonomy
B) Homogeneous database
C) Centralized database
D) Single-user database

Answer: A

---

10. Client-server architecture in distributed databases has:
A) Clients request services, servers provide them
B) All nodes are equal
C) No distinction between client and server
D) Only servers exist

Answer: A

---

11. Peer-to-peer distributed database has:
A) All nodes have equal capabilities and cooperate
B) Central coordinator node
C) Only client nodes
D) Only server nodes

Answer: A

---

12. Multi-database system (MDBS) is:
A) System that integrates multiple autonomous, possibly heterogeneous databases
B) Single database with multiple tables
C) Multiple copies of same database
D) Backup system

Answer: A

SECTION 3: DATA FRAGMENTATION (Questions 41-65)

13. Data fragmentation divides:
A) A relation into smaller pieces (fragments) stored at different sites
B) The database into tables
C) The index into parts
D) The transaction into subtransactions

Answer: A

---

14. The main types of fragmentation are:
A) Horizontal, Vertical, and Mixed (Hybrid)
B) Primary and Secondary
C) Dense and Sparse
D) Clustered and Non-clustered

Answer: A

---

15. Horizontal fragmentation divides:
A) A relation by rows (tuples) into subsets of rows
B) A relation by columns
C) A relation by both rows and columns
D) The index structure

Answer: A

---

16. Horizontal fragmentation is typically based on:
A) Predicates (conditions) on attributes
B) Random assignment
C) Row size
D) Row creation time

Answer: A

---

17. Example of horizontal fragmentation:
A) Employees in different regions stored at respective regional sites
B) Employees table split by columns (name at one site, salary at another)
C) Employees table with all rows at all sites
D) Employees table with no fragmentation

Answer: A

---

18. Vertical fragmentation divides:
A) A relation by columns (attributes) into subsets of columns
B) A relation by rows
C) The entire database
D) The index only

Answer: A

---

19. Vertical fragmentation requires:
A) A common attribute (usually primary key) in each fragment for reconstruction
B) No common attributes
C) All columns in every fragment
D) Random column assignment

Answer: A

---

20. Mixed (hybrid) fragmentation applies:
A) Both horizontal and vertical fragmentation
B) Only one type
C) No fragmentation
D) Only replication

Answer: A

---

21. Correctness rules for fragmentation include:
A) Completeness, Reconstruction, Disjointness
B) Atomicity, Consistency, Isolation
C) Selectivity, Cardinality, Degree
D) Redundancy, Replication, Distribution

Answer: A

---

22. Completeness rule ensures:
A) Every data item appears in at least one fragment
B) Data appears in multiple fragments
C) Data appears in no fragments
D) Only some data appears

Answer: A

---

23. Reconstruction rule ensures:
A) Original relation can be reconstructed from fragments
B) Fragments cannot be combined
C) Data is lost after fragmentation
D) No reconstruction possible

Answer: A

---

24. Disjointness rule ensures (for horizontal fragmentation):
A) Fragments are disjoint (no overlapping rows)
B) Fragments overlap
C) All rows in all fragments
D) No rows in fragments

Answer: A

---

25. For vertical fragmentation, disjointness applies to:
A) Non-primary key attributes (overlap only on primary key)
B) All attributes including primary key
C) No attributes
D) Only foreign keys

Answer: A

SECTION 4: DATA REPLICATION (Questions 66-90)

26. Data replication means:
A) Storing copies of data fragments at multiple sites
B) Storing data only once
C) Fragmenting data
D) Indexing data

Answer: A

---

27. The main types of replication are:
A) Full replication and Partial replication
B) Horizontal and Vertical replication
C) Synchronous and Asynchronous (main distinction)
D) Primary and Secondary replication

Answer: A and C are both valid

---

28. Full replication means:
A) Entire database is copied to all sites
B) Only parts of database are copied
C) No copies exist
D) Only indexes are copied

Answer: A

---

29. Partial replication means:
A) Some fragments are replicated, others not

B) All fragments replicated everywhere
C) No replication
D) Only one copy exists

Answer: A

---

30. Advantages of replication include:
A) Increased availability, improved read performance, fault tolerance
B) Slower reads, lower availability
C) Increased complexity only
D) No advantages

Answer: A

---

31. Disadvantages of replication include:
A) Update overhead, consistency maintenance, storage cost
B) Faster updates
C) Lower storage cost
D) Simpler management

Answer: A

---

32. Synchronous replication ensures:
A) All copies are updated within the same transaction (immediate consistency)
B) Updates propagate later
C) No updates
D) Only reads allowed

Answer: A

---

33. Asynchronous replication:
A) Updates propagate after transaction commits (eventual consistency)
B) Updates happen simultaneously
C) No updates allowed
D) Only synchronous allowed

Answer: A

---

34. Master-slave replication has:
A) One primary (master) for writes, multiple slaves for reads
B) All nodes can write
C) No writes allowed
D) Only reads allowed

Answer: A

---

35. Multi-master replication allows:
A) Writes at any node, with conflict resolution
B) Only one node for writes
C) No writes
D) Only reads

Answer: A

SECTION 5: DISTRIBUTED DATA INDEPENDENCE (Questions 91-105)

36. Distribution transparency types include:
A) Fragmentation transparency, Location transparency, Replication transparency
B) Only location transparency
C) Only fragmentation transparency
D) No transparency

Answer: A

---

37. Fragmentation transparency means:
A) User unaware that data is fragmented, queries as if whole relation exists
B) User must specify fragments
C) Fragments are visible to users
D) No fragmentation exists

Answer: A

---

38. Location transparency means:
A) User unaware of where data is physically stored
B) User must specify site addresses
C) Locations are visible
D) No location information

Answer: A

---

39. Replication transparency means:
A) User unaware that data is replicated
B) User must manage copies
C) Replication visible to users
D) No replication exists

Answer: A

---

40. Local mapping transparency means:
A) User must know both fragment and location names (lowest level)
B) No transparency
C) Complete transparency
D) Only fragment names needed

Answer: A

SECTION 6: DISTRIBUTED QUERY PROCESSING (Questions 106-130)

41. Distributed query processing involves:
A) Query decomposition, data localization, global optimization, local optimization
B) Only local processing
C) Only network transfer
D) No optimization

Answer: A

---

42. Query decomposition (first phase) is similar to centralized:
A) Parse, validate, rewrite into relational algebra
B) Data localization
C) Site selection
D) Execution

Answer: A

---

43. Data localization determines:
A) Which fragments contain required data and how to access them
B) Which sites to use
C) Join order
D) Index selection

Answer: A

---

44. Global optimization chooses:
A) Best strategy considering fragment locations, communication costs, local processing costs
B) Only local optimization
C) Random strategy
D) No optimization

Answer: A

---

45. Semi-join reduces:
A) Data transfer by sending only joining attribute values, then fetching matching tuples
B) All data always transferred
C) Local processing only
D) No reduction

Answer: A

---

46. Semi-join operation:
A) $R \ltimes S = \pi_R(R \bowtie S)$ - sends join attributes of S to R site
B) $R \bowtie S$ (full join)
C) $R \times S$
D) $R \cup S$

Answer: A

---

47. Join strategies in distributed databases include:
A) Ship whole relation, fetch-as-needed, semi-join based, partitioned join
B) Only nested loop
C) Only sort-merge
D) Only hash join

Answer: A

---

48. Ship whole relation strategy:
A) Transfer entire smaller relation to larger relation's site and join locally
B) Transfer both relations to third site
C) Transfer no data
D) Join at both sites

Answer: A

---

49. Fetch-as-needed strategy:
A) For each tuple in outer relation, fetch matching tuples from remote inner using index
B) Transfer all data first
C) No data transfer
D) Join at both sites

Answer: A

---

50. Cost of distributed query includes:
A) I/O cost + CPU cost + Communication cost
B) Only communication cost

C) Only I/O cost
D) Only CPU cost

Answer: A

SECTION 7: DISTRIBUTED TRANSACTION MANAGEMENT (Questions 131-160)

51. A distributed transaction is:
A) Transaction that accesses data at multiple sites
B) Transaction at single site only
C) Read-only transaction
D) Update transaction only

Answer: A

---

52. Distributed transaction has:
A) One coordinator and multiple participants (subtransactions)
B) Only one participant
C) No coordinator
D) Only local processing

Answer: A

---

53. Two-Phase Commit (2PC) protocol ensures:
A) Atomic commitment of distributed transactions across sites
B) No commit coordination
C) Local commits only
D) Partial commits

Answer: A

---

54. The two phases of 2PC are:
A) Prepare phase and Commit/Abort phase
B) Lock phase and Unlock phase

C) Read phase and Write phase
D) Start phase and End phase

Answer: A

---

55. In prepare phase, coordinator:
A) Sends "prepare" message to all participants, waits for votes (YES/NO)
B) Commits immediately
C) Aborts all
D) Does nothing

Answer: A

---

56. In commit phase, if all participants vote YES:
A) Coordinator sends "commit" message, participants commit
B) Coordinator aborts
C) Participants decide independently
D) Transaction continues

Answer: A

---

57. If any participant votes NO in prepare phase:
A) Coordinator sends "abort" message, all participants abort
B) Commit anyway
C) Only that participant aborts
D) Retry

Answer: A

---

58. Presumed Abort (PA) protocol:
A) If coordinator fails, assume abort unless commit record found
B) Assume commit always
C) No assumption
D) Always commit

Answer: A

---

59. Presumed Commit (PC) protocol:
A) Assume commit unless abort record found
B) Assume abort always
C) No assumption
D) Always abort

Answer: A

---

60. In-doubt transactions are those that:
A) Received prepare but no decision before coordinator failure
B) Committed
C) Aborted
D) Not started

Answer: A

---

61. Three-Phase Commit (3PC) adds:
A) Pre-commit phase to handle coordinator failure without blocking
B) Extra commit phase
C) No additional phases
D) Only two phases

Answer: A

---

62. 3PC is designed to be:
A) Non-blocking even if coordinator fails
B) Blocking always
C) Same as 2PC
D) Faster than 2PC

Answer: A

---

63. Distributed concurrency control protocols include:
A) Distributed locking, distributed timestamp ordering, optimistic methods
B) Only local locking

C) No concurrency control
D) Only centralized

Answer: A

---

64. Distributed 2PL requires:
A) All sites follow 2PL, global deadlock detection
B) No coordination
C) Only local locks
D) No deadlock handling

Answer: A

---

65. Global deadlock detection uses:
A) Distributed wait-for graph (centralized or hierarchical)
B) Local graphs only
C) No detection
D) Timeout only

Answer: A

---

66. Distributed timestamp ordering:
A) Uses globally unique timestamps (e.g., using site ID + local counter)
B) Local timestamps only
C) No timestamps
D) Random ordering

Answer: A

---

67. Commit protocols must handle:
A) Site failures, network partitioning, message loss
B) Only successful commits
C) Only aborts
D) No failures

Answer: A

SECTION 8: DISTRIBUTED CONCURRENCY CONTROL (Questions 161-175)

68. Primary site locking uses:
A) One site as lock manager for all data
B) Distributed lock managers
C) No locks
D) Local locks only

Answer: A

---

69. Distributed lock manager can be:
A) Centralized, primary copy, or fully distributed
B) Only centralized
C) Only distributed
D) Not used

Answer: A

---

70. Majority locking requires:
A) Lock majority of copies before update
B) Lock all copies
C) Lock one copy
D) No locks

Answer: A

---

71. Quorum-based consensus uses:
A) Read quorum and write quorum to ensure consistency
B) No quorums
C) Only read quorum
D) Only write quorum

Answer: A

SECTION 9: DISTRIBUTED DATABASE DESIGN (Questions 176-190)

72. Design issues in distributed databases include:
A) Fragmentation, allocation, replication
B) Only indexing
C) Only normalization
D) Only query optimization

Answer: A

---

73. Data allocation strategies include:
A) Centralized, partitioned, replicated
B) Only centralized
C) Only partitioned
D) Only replicated

Answer: A

---

74. Allocation should consider:
A) Access patterns, storage costs, communication costs, availability needs
B) Only storage cost
C) Only access patterns
D) Random assignment

Answer: A

---

75. Best allocation is often:
A) A combination of fragmentation and replication based on workload
B) Full replication everywhere
C) No replication
D) Centralized only

Answer: A

SECTION 10: DISTRIBUTED DATABASE RELIABILITY (Questions 191-205)

76. Reliability in distributed databases includes:
A) Fault tolerance, recovery, availability
B) Only performance
C) Only security
D) Only consistency

Answer: A

---

77. Network partitioning occurs when:
A) Network fails, dividing sites into groups that cannot communicate
B) Data is partitioned
C) Index is partitioned
D) Transaction is partitioned

Answer: A

---

78. Partition tolerance is handled by:
A) Protocols that ensure consistency or availability during partitions (CAP theorem)
B) Ignoring partitions
C) Always available
D) Always consistent

Answer: A

---

79. CAP theorem states:
A) Cannot have all three: Consistency, Availability, Partition tolerance simultaneously
B) Can have all three
C) Only two matter
D) Only one matters

Answer: A

---

80. PACELC theorem extends CAP:
A) Trade-offs during normal operation and during partitions
B) Same as CAP
C) No trade-offs
D) Only during partitions

Answer: A

---

81. Remote backup for disaster recovery:
A) Maintains copy at geographically separate site
B) Same site only
C) No backup
D) Only local backup

Answer: A

SECTION 11: DISTRIBUTED DATABASE EXAMPLES (Questions 206-215)

82. Google Spanner is:
A) Globally distributed database with strong consistency and TrueTime
B) Centralized database
C) File system
D) NoSQL only

Answer: A

---

83. Amazon DynamoDB uses:
A) Eventually consistent model with high availability
B) Strong consistency only
C) No replication
D) Single master

Answer: A

---

84. Cassandra is:
A) Distributed NoSQL database with eventual consistency, masterless
B) Relational database
C) Centralized system
D) File system

Answer: A

---

85. Oracle RAC (Real Application Clusters) provides:
A) Shared-disk cluster with cache fusion
B) Shared-nothing architecture
C) No clustering
D) Single instance only

Answer: A

SECTION 12: ADVANCED TOPICS (Questions 216-230)

86. Blockchain as distributed ledger:
A) Replicated, immutable, consensus-based
B) Centralized database
C) No replication
D) Mutable data

Answer: A

---

87. Federated databases vs Multi-databases:
A) Federated has more integration, multi-database has more autonomy
B) Same thing
C) No difference
D) Opposite meanings

Answer: A

---

88. Cloud databases as a service (DBaaS):
A) Distributed databases offered on cloud platforms
B) On-premise only
C) No distribution
D) Single user only

Answer: A

---

89. Geo-distributed databases optimize for:
A) Low latency for users worldwide, data sovereignty
B) Local access only
C) No distribution
D) Single region

Answer: A

---

90. Active-active configuration has:
A) Multiple sites accepting writes concurrently
B) Only one active site
C) No writes
D) Only reads

Answer: A

SECTION 13: TRANSACTIONAL PROPERTIES (Questions 231-240)

91. Global serializability in distributed databases:
A) Interleaved execution of distributed transactions equivalent to serial
B) Local serializability only
C) No serializability
D) Random execution

Answer: A

---

92. Distributed deadlock detection algorithms:
A) Centralized, hierarchical, distributed (e.g., edge chasing)
B) Only centralized
C) Only local
D) No detection

Answer: A

---

93. Edge chasing (path pushing) algorithm:
A) Probes circulate in wait-for graph to detect cycles
B) Centralized detection only
C) Timeout only
D) No detection

Answer: A

---

94. Timeout-based deadlock detection:
A) Transaction aborted if waits too long (may have false positives)
B) Always accurate
C) No aborts
D) Always waits

Answer: A

SECTION 14: COMPARISON AND TRADE-OFFS (Questions 241-250)

95. Distributed vs Centralized advantages:
A) Better reliability, scalability, local autonomy
B) Simpler management
C) Lower cost
D) Easier security

Answer: A

---

96. Distributed vs Centralized disadvantages:
A) Complexity, coordination overhead, security challenges
B) Simpler design
C) Faster queries
D) Easier recovery

Answer: A

---

97. Synchronous vs Asynchronous replication trade-off:
A) Sync: stronger consistency, higher latency; Async: lower latency, eventual consistency
B) Sync: lower latency; Async: stronger consistency
C) Both same
D) No difference

Answer: A

---

98. Strong consistency vs Availability trade-off:
A) CAP theorem: cannot have both during partitions
B) Can have both always
C) No trade-off
D) Both always available

Answer: A

SECTION 1: INTRODUCTION TO NOSQL (Questions 1-20)

1. NoSQL databases are:
A) Non-relational databases designed for scale, flexibility, and specific data models
B) Databases that do not use SQL at all
C) Relational databases without joins
D) Only key-value stores

Answer: A
Explanation: NoSQL encompasses various non-relational data models for modern applications.

---

2. The main drivers for NoSQL adoption include:
A) Scalability, flexibility, big data, cloud computing, agile development
B) Only faster queries
C) Only cheaper storage
D) Only easier SQL

Answer: A

---

3. NoSQL databases are often characterized by:
A) Schema flexibility, horizontal scaling, BASE transactions
B) Strict schema, vertical scaling, ACID transactions
C) Only relational model
D) Only SQL support

Answer: A

---

4. The four main types of NoSQL databases are:
A) Document, Key-Value, Column-Family, Graph
B) Relational, Object-Oriented, Hierarchical, Network
C) SQL, NoSQL, NewSQL, OldSQL
D) Centralized, Distributed, Federated, Parallel

Answer: A

---

5. NoSQL databases are designed for:
A) Horizontal scaling across commodity servers
B) Vertical scaling on single powerful server
C) Only single-server deployment
D) Only mainframe systems

Answer: A

---

6. Schema flexibility in NoSQL means:
A) Different records can have different attributes
B) All records must have same attributes
C) No schema allowed
D) Schema is fixed forever

Answer: A

---

7. NoSQL databases typically support:
A) Eventual consistency and partition tolerance
B) Strong consistency only
C) No consistency guarantees
D) Only ACID transactions

Answer: A

SECTION 2: CAP THEOREM AND BASE (Questions 21-40)

8. CAP theorem states that a distributed system can provide at most:
A) Two of three: Consistency, Availability, Partition tolerance
B) All three simultaneously
C) Only one at a time
D) None of them

Answer: A

---

9. In CAP theorem, Consistency (C) means:
A) All nodes see the same data at the same time
B) Data never changes
C) System is always available
D) Network never fails

Answer: A

---

10. In CAP theorem, Availability (A) means:
A) Every request receives a response (success or failure)
B) System is always consistent
C) Network never partitions
D) Data is always backed up

Answer: A

---

11. In CAP theorem, Partition tolerance (P) means:
A) System continues to operate despite network partitions
B) Data is partitioned across nodes
C) System can be partitioned
D) Partitions are not allowed

Answer: A

---

12. During a network partition, a system must choose between:
A) Consistency and Availability
B) Partition tolerance and Consistency
C) Availability and Partition tolerance
D) All three

Answer: A

---

13. CP systems (Consistency + Partition tolerance) during partition:
A) May become unavailable to maintain consistency
B) Remain fully available
C) Sacrifice consistency
D) Ignore partitions

Answer: A

---

14. AP systems (Availability + Partition tolerance) during partition:
A) Remain available but may return inconsistent data
B) Become unavailable
C) Maintain strong consistency
D) Sacrifice availability

Answer: A

---

15. CA systems (Consistency + Availability) assume:
A) No network partitions (theoretical, not practical)
B) Partitions always occur

C) Partitions are handled
D) Consistency is sacrificed

Answer: A

---

16. BASE acronym stands for:
A) Basically Available, Soft state, Eventual consistency
B) Basic Availability, Strong consistency, Eventual
C) Before Anything Starts, Everything
D) Balanced, Available, Scalable, Efficient

Answer: A

---

17. Basically Available means:
A) System guarantees availability (most of the time)
B) Always available 100%
C) Never available
D) Only sometimes available

Answer: A

---

18. Soft state means:
A) State may change over time even without input (due to eventual consistency)
B) State never changes
C) State is always consistent
D) State is hard to change

Answer: A

---

19. Eventual consistency means:
A) In absence of updates, all replicas will converge to same value
B) All replicas always consistent
C) Never consistent
D) Consistency after each operation

Answer: A

---

20. BASE is a trade-off compared to ACID because:
A) Prioritizes availability and performance over strong consistency
B) Prioritizes consistency over availability
C) Both are same
D) BASE is stronger

Answer: A

SECTION 3: KEY-VALUE STORES (Questions 41-60)

21. Key-value stores model data as:
A) A collection of key-value pairs
B) Tables with rows and columns
C) Documents
D) Graphs

Answer: A

---

22. In key-value stores, the key is:
A) A unique identifier to retrieve the associated value
B) The actual data
C) The index
D) The value itself

Answer: A

---

23. The value in key-value stores is:
A) Opaque blob (can be anything: string, JSON, binary)
B) Must be relational tuple
C) Must be normalized
D) Always a number

Answer: A

---

24. Examples of key-value stores include:
A) Redis, Amazon DynamoDB, Riak, Memcached
B) MongoDB, CouchDB
C) Cassandra, HBase
D) Neo4j, JanusGraph

Answer: A

---

25. Key-value stores are optimized for:
A) Simple lookups and writes by key
B) Complex queries with joins
C) Full-text search
D) Graph traversals

Answer: A

---

26. Redis is an in-memory key-value store known for:
A) High performance, data structures, persistence options
B) Disk-based storage only
C) Relational model
D) Graph queries

Answer: A

---

27. Key-value stores typically provide:
A) Simple API (GET, PUT, DELETE)
B) Complex SQL interface
C) Join operations
D) ACID transactions always

Answer: A

---

28. Use cases for key-value stores include:
A) Session storage, caching, user profiles, shopping carts
B) Complex reporting
C) Data warehousing
D) Graph analytics

Answer: A

---

29. Partitioning in key-value stores uses:
A) Consistent hashing to distribute keys across nodes
B) Range partitioning only
C) Round-robin
D) No partitioning

Answer: A

---

30. Consistent hashing minimizes:
A) Key redistribution when nodes added/removed
B) Data storage
C) Query complexity
D) Index size

Answer: A

SECTION 4: DOCUMENT STORES (Questions 61-85)

31. Document stores manage data as:
A) Semi-structured documents (JSON, BSON, XML)
B) Key-value pairs only
C) Tables with rows
D) Graph nodes

Answer: A

---

32. A document in document stores is:
A) Self-describing, hierarchical data structure with fields and values
B) A text file
C) A spreadsheet row
D) A key-value pair

Answer: A

---

33. Documents are grouped into:
A) Collections (analogous to tables)
B) Tables
C) Keyspaces
D) Graphs

Answer: A

---

34. Examples of document stores include:
A) MongoDB, CouchDB, Couchbase, Firebase Firestore
B) Redis, Memcached
C) Cassandra, HBase
D) Neo4j, OrientDB

Answer: A

---

35. MongoDB stores data in:
A) BSON (Binary JSON) format
B) Plain text
C) CSV
D) XML

Answer: A

---

36. Schema flexibility in document stores allows:
A) Documents in same collection to have different fields
B) All documents must have identical structure
C) No fields allowed
D) Fixed schema only

Answer: A

---

37. MongoDB queries use:
A) JSON-like syntax for CRUD operations
B) SQL

C) XQuery
D) SPARQL

Answer: A

---

38. Indexing in MongoDB supports:
A) Single field, compound, multikey, text, geospatial
B) Only primary key
C) No indexes
D) Only B-tree

Answer: A

---

39. Document stores are good for:
A) Content management, catalogs, user profiles, real-time analytics
B) Highly normalized financial data
C) Graph traversals
D) Simple key lookups only

Answer: A

---

40. Embedded documents in MongoDB:
A) Store related data within a single document (denormalization)
B) Require joins
C) Are not allowed
D) Must be in separate collections

Answer: A

---

41. References in MongoDB:
A) Link documents across collections (like foreign keys)
B) Embed data
C) Are not supported
D) Are always automatic

Answer: A

---

42. Aggregation pipeline in MongoDB:
A) Processes documents through stages (filter, group, sort, project)
B) Executes SQL joins
C) Performs graph traversals
D) Does nothing

Answer: A

SECTION 5: COLUMN-FAMILY STORES (Questions 86-110)

43. Column-family stores (wide-column stores) organize data by:
A) Rows and columns, but columns are grouped into families
B) Key-value pairs only
C) Documents
D) Graph edges

Answer: A

---

44. Examples of column-family stores include:
A) Apache Cassandra, HBase, Google Bigtable
B) MongoDB, CouchDB
C) Redis, Memcached
D) Neo4j, JanusGraph

Answer: A

---

45. In Cassandra, data is stored in:
A) Tables with rows and columns, but optimized for wide rows
B) Only key-value format
C) JSON documents
D) Graph nodes

Answer: A

---

46. Column families group:
A) Related columns together that are often accessed together
B) Rows together
C) Keys together
D) Values together

Answer: A

---

47. In Cassandra, each row can have:
A) Different number of columns (sparse storage)
B) Same fixed columns
C) No columns
D) Only one column

Answer: A

---

48. Partition key in Cassandra determines:
A) Which node stores the row (data distribution)
B) The primary key only
C) The column family
D) The index type

Answer: A

---

49. Clustering columns in Cassandra determine:
A) Sort order within a partition
B) Node placement
C) Replication factor
D) Consistency level

Answer: A

---

50. Cassandra's query language is:
A) CQL (Cassandra Query Language) - similar to SQL but limited
B) SQL full
C) JSON queries
D) No query language

Answer: A

---

51. Cassandra is designed for:
A) High write throughput, scalability, no single point of failure
B) Complex joins
C) ACID transactions
D) Graph traversals

Answer: A

---

52. Tunable consistency in Cassandra allows:
A) Choosing consistency level per query (ONE, QUORUM, ALL)
B) Fixed consistency always
C) No consistency
D) Only strong consistency

Answer: A

---

53. Hinted handoff in Cassandra handles:
A) Temporary node failures by storing writes for later replay
B) Permanent failures only
C) Read failures
D) Network issues only

Answer: A

---

54. Compaction in Cassandra:
A) Merges SSTables to remove tombstones and optimize storage
B) Compresses data only
C) Deletes data
D) Creates indexes

Answer: A

---

55. HBase is modeled after:
A) Google Bigtable
B) Amazon Dynamo
C) MongoDB
D) Neo4j

Answer: A

---

56. HBase uses:
A) HDFS for storage, ZooKeeper for coordination
B) Local file system only
C) No coordination
D) Only memory

Answer: A

SECTION 6: GRAPH DATABASES (Questions 111-135)

57. Graph databases represent data as:
A) Nodes (entities), Edges (relationships), and Properties
B) Tables with rows and columns
C) Key-value pairs
D) Documents

Answer: A

---

58. Examples of graph databases include:
A) Neo4j, Amazon Neptune, JanusGraph, OrientDB
B) MongoDB, CouchDB
C) Cassandra, HBase
D) Redis, Memcached

Answer: A

---

59. Graph databases excel at:
A) Relationship-heavy queries (connections, paths, patterns)

B) Simple key lookups
C) Aggregations
D) Full-text search

Answer: A

---

60. Property graph model includes:
A) Nodes with properties, relationships with properties and direction
B) Only nodes
C) Only edges
D) Triples only

Answer: A

---

61. RDF (Resource Description Framework) uses:
A) Triples (subject-predicate-object)
B) Nodes and edges only
C) Key-value pairs
D) Documents

Answer: A

---

62. SPARQL is query language for:
A) RDF graph databases
B) Property graphs
C) Document stores
D) Key-value stores

Answer: A

---

63. Neo4j uses:
A) Cypher query language for graph pattern matching
B) SQL
C) SPARQL
D) JSON queries

Answer: A

---

64. Graph traversal algorithms include:
A) Breadth-first, depth-first, shortest path, PageRank
B) Only B-tree search
C) Only hash lookup
D) Only scan

Answer: A

---

65. Index-free adjacency in graph databases means:
A) Nodes have direct pointers to neighbors (no index lookups)
B) No indexes allowed
C) Indexes not needed at all
D) Only indexes used

Answer: A

---

66. Use cases for graph databases include:
A) Social networks, recommendation engines, fraud detection, knowledge graphs
B) Session storage
C) Caching
D) Logging

Answer: A

SECTION 7: NEWSQL DATABASES (Questions 136-150)

67. NewSQL databases aim to:
A) Provide ACID transactions and SQL with NoSQL scalability
B) Replace SQL completely
C) Only be in-memory
D) Only be distributed

Answer: A

---

68. Examples of NewSQL databases include:
A) Google Spanner, CockroachDB, VoltDB, TiDB
B) MongoDB, Cassandra
C) Neo4j, JanusGraph
D) Redis, Memcached

Answer: A

---

69. Google Spanner provides:
A) Global distribution, strong consistency, SQL using TrueTime
B) Eventual consistency only
C) No SQL
D) Only key-value

Answer: A

---

70. CockroachDB is inspired by:
A) Google Spanner (distributed SQL with strong consistency)
B) MongoDB
C) Cassandra
D) Neo4j

Answer: A

---

71. VoltDB is an:
A) In-memory, ACID-compliant, relational database for high throughput
B) Disk-based database
C) Graph database
D) Document store

Answer: A

SECTION 8: TIME-SERIES DATABASES (Questions 151-165)

72. Time-series databases optimize for:
A) Data with timestamps (metrics, events, sensor data)

B) Relational data
C) Graph data
D) Key-value pairs

Answer: A

---

73. Examples of time-series databases include:
A) InfluxDB, Prometheus, TimescaleDB, OpenTSDB
B) MongoDB, CouchDB
C) Neo4j, JanusGraph
D) Redis, Memcached

Answer: A

---

74. Time-series features include:
A) Downsampling, retention policies, continuous queries
B) Joins
C) ACID transactions
D) Graph traversals

Answer: A

---

75. InfluxDB uses:
A) InfluxQL or Flux for querying time-series data
B) SQL only
C) No query language
D) Cypher

Answer: A

SECTION 9: SEARCH ENGINES (Questions 166-175)

76. Search engines like Elasticsearch are built on:
A) Apache Lucene for full-text search and analytics
B) Relational model

C) Key-value storage
D) Graph model

Answer: A

---

77. Elasticsearch stores data as:
A) JSON documents with inverted indexes for fast search
B) Tables
C) Key-value pairs
D) Graph nodes

Answer: A

---

78. The ELK Stack consists of:
A) Elasticsearch, Logstash, Kibana
B) Elastic, Log, Kibana
C) ETL, Log, Kibana
D) Elasticsearch, Logstash, Kafka

Answer: A

---

79. Inverted index maps:
A) Terms to documents containing them
B) Documents to terms
C) Keys to values
D) Nodes to edges

Answer: A

SECTION 10: POLYGLOT PERSISTENCE (Questions 176-185)

80. Polyglot persistence means:
A) Using multiple database types for different parts of an application
B) Using only one database type
C) Speaking multiple languages
D) Multiple versions of same database

Answer: A

---

81. Example of polyglot persistence:
A) Relational for orders, document for catalogs, graph for recommendations
B) Only relational for everything
C) Only NoSQL for everything
D) Only one database

Answer: A

---

82. Advantages of polyglot persistence:
A) Best tool for each job, optimized performance
B) Simpler management
C) Lower cost
D) Easier learning

Answer: A

---

83. Challenges of polyglot persistence:
A) Integration complexity, multiple skill sets, consistency across databases
B) No challenges
C) Simpler than single database
D) Always cheaper

Answer: A

SECTION 11: COMPARISON AND SELECTION (Questions 186-200)

84. When to choose Document stores:
A) Semi-structured data, evolving schema, hierarchical data
B) Highly relational data
C) Simple key lookups only
D) Graph traversals

Answer: A

---

85. When to choose Key-Value stores:
A) Simple lookups by key, high throughput, caching
B) Complex queries
C) Joins
D) Relationship queries

Answer: A

---

86. When to choose Column-family stores:
A) Write-heavy workloads, time-series, wide rows, high scalability
B) Complex joins
C) Graph traversals
D) Document nesting

Answer: A

---

87. When to choose Graph databases:
A) Highly connected data, relationship queries, path finding
B) Simple key lookups
C) Aggregations
D) Full-text search

Answer: A

---

88. When to choose Relational databases:
A) Strong consistency, ACID, complex joins, structured data
B) Unstructured data
C) Massive scale with weak consistency
D) Simple key lookups only

Answer: A

---

89. Database selection factors include:
A) Data model, scalability needs, consistency requirements, query patterns
B) Only cost
C) Only popularity
D) Only vendor

Answer: A

SECTION 12: ADVANCED TOPICS (Questions 201-220)

90. Sharding distributes data across nodes by:
A) Partitioning key range or hash
B) Replication only
C) Indexing only
D) No distribution

Answer: A

---

91. Replication provides:
A) High availability, fault tolerance, read scaling
B) Data partitioning only
C) Consistency only
D) No benefits

Answer: A

---

92. Vector databases are optimized for:
A) Embeddings, similarity search, machine learning applications
B) Relational data
C) Graph data
D) Key-value pairs

Answer: A

---

93. Ledger databases provide:
A) Immutable, cryptographically verifiable transaction log

B) Mutable data
C) No verification
D) Only relational

Answer: A

---

94. Blockchain as database:
A) Decentralized, immutable, consensus-based replicated ledger
B) Centralized database
C) Key-value store
D) Document store

Answer: A

---

95. Cloud-native databases are:
A) Designed for cloud environments (elastic, managed, serverless)
B) On-premise only
C) Not scalable
D) Only relational

Answer: A

---

96. Serverless databases:
A) Automatically scale, pay-per-use, no server management
B) Require server provisioning
C) Fixed capacity
D) Only on-premise

Answer: A

---

97. Multi-model databases support:
A) Multiple data models (document, graph, relational) in one system
B) Only one model
C) No models
D) Only key-value

Answer: A

---

98. Examples of multi-model databases:
A) ArangoDB, OrientDB, Cosmos DB
B) MongoDB only
C) Cassandra only
D) Neo4j only

Answer: A

---

99. Database as a Service (DBaaS) provides:
A) Managed database in cloud (setup, backups, scaling handled)
B) On-premise only
C) User manages everything
D) No automation

Answer: A

---

100. Hybrid transaction/analytical processing (HTAP) databases:
A) Handle both OLTP and OLAP workloads
B) Only OLTP
C) Only OLAP
D) Neither

Answer: A

SECTION 1: INTRODUCTION TO DATA WAREHOUSING (Questions 1-25)

1. A data warehouse is best defined as:
A) A subject-oriented, integrated, time-variant, non-volatile collection of data to support decision making
B) A backup copy of operational databases
C) A large relational database for OLTP
D) A collection of files on a file server

Answer: A
Explanation: Data warehouse is specifically designed for decision support with these four key characteristics .

---

2. The four key characteristics of a data warehouse are:
A) Subject-oriented, Integrated, Time-variant, Non-volatile
B) Distributed, Parallel, Scalable, Available
C) Atomic, Consistent, Isolated, Durable
D) Fast, Reliable, Secure, Accessible

Answer: A
Explanation: These are the defining features of a data warehouse according to Inmon's definition .

---

3. "Subject-oriented" in data warehouse means:
A) Data organized around major subjects (customer, product, sales) rather than applications
B) Data organized by database subjects
C) Data organized by user preferences
D) Data organized by transaction types

Answer: A

---

4. "Integrated" in data warehouse means:
A) Data from multiple sources is transformed into consistent formats and naming conventions
B) Data is stored in one integrated database
C) Data is integrated with applications
D) Data is integrated with the operating system

Answer: A

---

5. "Time-variant" in data warehouse means:
A) Data contains historical information with time stamps
B) Data changes frequently over time

C) Data is only current
D) Data expires after time

Answer: A
Explanation: Data warehouse maintains historical data (e.g., 5-10 years) for trend analysis .

---

6. "Non-volatile" in data warehouse means:
A) Data is read-only and not updated by users
B) Data changes frequently
C) Data is volatile in memory
D) Data disappears after use

Answer: A

---

7. The primary purpose of a data warehouse is to support:
A) Decision support and business intelligence
B) Online transaction processing
C) Day-to-day operations
D) Data entry

Answer: A

---

8. Data warehouse vs Database (OLTP): Data warehouse is optimized for:
A) Complex queries and analysis
B) Frequent updates and inserts
C) Small transactions
D) Real-time processing

Answer: A

---

9. OLTP stands for:
A) Online Transaction Processing
B) Online Table Processing
C) Offline Transaction Processing
D) Online Transfer Protocol

Answer: A
Explanation: OLTP systems handle day-to-day operational transactions .

---

10. OLAP stands for:
A) Online Analytical Processing
B) Online Application Processing
C) Offline Analytical Processing
D) Online Analysis Protocol

Answer: A

---

11. OLAP systems are characterized by:
A) Read-mostly operations, complex queries, historical data
B) Frequent writes, simple queries, current data
C) No queries, only updates
D) Only insert operations

Answer: A
Explanation: OLAP involves mostly read operations for analysis .

---

12. OLTP systems are characterized by:
A) Short, atomic transactions with frequent updates
B) Long-running complex queries
C) Read-only operations
D) Batch processing only

Answer: A
Explanation: OLTP consists of short, atomic transactions ensuring consistency .

---

13. Which of the following is NOT an OLAP operation?
A) INSERT
B) Roll-up
C) Drill-down
D) Slice and dice

Answer: A
Explanation: INSERT is an OLTP operation; the others are OLAP analytical operations .

---

14. Roll-up operation in OLAP:
A) Aggregates data by moving up in concept hierarchy (e.g., city → country)
B) Drills down to detail data
C) Selects a subset of dimensions
D) Rotates the cube view

Answer: A

---

15. Drill-down operation in OLAP:
A) Navigates from summary to more detailed data
B) Aggregates to higher level
C) Removes dimensions
D) Adds dimensions

Answer: A

---

16. Slice operation in OLAP:
A) Selects a single dimension value, creating a sub-cube
B) Rotates the cube view
C) Aggregates data
D) Drills down

Answer: A

---

17. Dice operation in OLAP:
A) Selects a sub-cube by applying conditions on multiple dimensions
B) Cuts the cube into pieces
C) Aggregates data
D) Drills down

Answer: A

---

18. Pivot (rotate) operation in OLAP:
A) Rotates the cube to view it from different dimensional perspectives
B) Changes the cube structure

C) Deletes dimensions
D) Adds dimensions

Answer: A

---

19. A data mart is:
A) A subset of data warehouse focused on a specific subject or department
B) The entire data warehouse
C) A backup of data warehouse
D) An operational database

Answer: A
Explanation: Data marts serve specific business areas like sales or marketing .

---

20. Data mart is appropriate when:
A) A department needs focused, summarized data without building full warehouse
B) Enterprise-wide integration is required
C) All data must be centralized
D) No analysis is needed

Answer: A

---

21. Operational Data Store (ODS) is:
A) Integrated, current-valued, subject-oriented database for operational decision making
B) Historical data repository
C) Backup storage
D) Data warehouse subset

Answer: A

---

22. Metadata in data warehouse is:
A) Data about data (definitions, structure, lineage, rules)
B) Main data stored
C) Backup data
D) Temporary data

Answer: A
Explanation: Metadata describes the warehouse contents, sources, transformations .

---

23. Metadata functionality includes:
A) Data discovery, lineage tracking, consistency checking
B) Data storage only
C) Query execution only
D) Index creation only

Answer: A

---

24. ETL stands for:
A) Extract, Transform, Load
B) Extract, Transfer, Load
C) Enter, Transform, Load
D) Extract, Transform, Link

Answer: A

---

25. Data extraction in ETL involves:
A) Reading data from source systems
B) Writing data to warehouse
C) Transforming data formats
D) Cleaning data

Answer: A

SECTION 2: DATA WAREHOUSE ARCHITECTURE (Questions 26-45)

26. The three-tier data warehouse architecture consists of:
A) Bottom tier (warehouse server), Middle tier (OLAP server), Top tier (client tools)
B) Database tier, Application tier, Web tier
C) Storage tier, Processing tier, Presentation tier
D) Source tier, Staging tier, Reporting tier

Answer: A

---

27. Bottom tier of data warehouse architecture is:
A) Data warehouse database server (relational database)
B) OLAP server
C) Client tools
D) ETL tools

Answer: A

---

28. Middle tier of data warehouse architecture is:
A) OLAP server (relational OLAP, multidimensional OLAP)
B) Data sources
C) End-user tools
D) Metadata repository

Answer: A

---

29. Top tier of data warehouse architecture includes:
A) Query and reporting tools, analysis tools, data mining tools
B) Data sources
C) Database server
D) ETL processes

Answer: A

---

30. Staging area in data warehouse architecture:
A) Temporary storage for data cleaning and transformation before loading
B) Final storage area
C) User query area
D) Backup area

Answer: A

---

31. The three data warehouse models from architecture perspective are:
A) Virtual warehouse, Data mart, Enterprise warehouse
B) Star, Snowflake, Fact constellation
C) Relational, Multidimensional, Hybrid
D) Centralized, Distributed, Federated

Answer: A
Explanation: These are the common architectural approaches .

---

32. Enterprise warehouse:
A) Contains all enterprise information, centralized, detailed and summarized data
B) Departmental subset
C) Virtual view only
D) Only summarized data

Answer: A

---

33. Dependent data mart:
A) Derived from enterprise data warehouse
B) Independent, built from source systems
C) Virtual only
D) Not related to warehouse

Answer: A

---

34. Independent data mart:
A) Built directly from operational sources without enterprise warehouse
B) Derived from enterprise warehouse
C) Same as dependent
D) Virtual only

Answer: A

---

35. Virtual warehouse:
A) Views and queries over operational databases without physical warehouse
B) Physical storage

C) Backup warehouse
D) Test environment

Answer: A

---

36. Data warehouse bus architecture defines:
A) Conformed dimensions and facts across data marts
B) Network connections
C) ETL pipelines
D) Query paths

Answer: A

---

37. Conformed dimensions ensure:
A) Same dimension attributes and meanings across multiple fact tables
B) Different dimensions for each fact
C) No dimensions
D) Random dimensions

Answer: A

---

38. The back-end tools for data warehouse include:
A) Extraction, cleaning, transformation, loading tools
B) Query tools
C) Reporting tools
D) Analysis tools

Answer: A

---

39. Data warehouse administration includes:
A) Security, backup, recovery, monitoring, performance tuning
B) Only user management
C) Only query optimization
D) Only data loading

Answer: A

---

40. Monitoring the data warehouse environment helps:
A) Track usage, performance, data quality, and system health
B) Only track users
C) Only track queries
D) Only track storage

Answer: A

SECTION 3: MULTIDIMENSIONAL DATA MODEL (Questions 46-70)

41. A data cube represents:
A) Multidimensional data with dimensions and measures
B) 3-dimensional geometric shape
C) Relational table
D) Flat file

Answer: A

---

42. Dimensions in a data cube represent:
A) Perspectives or entities (time, product, customer, location)
B) Numerical measures
C) Facts
D) Aggregations

Answer: A

---

43. Facts (measures) in a data cube represent:
A) Numerical data to be analyzed (sales amount, quantity, profit)
B) Descriptive attributes
C) Dimension members
D) Metadata

Answer: A

---

44. Fact table contains:
A) Measures and foreign keys to dimension tables
B) Only measures
C) Only dimension keys
D) Descriptive attributes

Answer: A
Explanation: Fact table is the central table with quantitative data and links to dimensions .

---

45. Dimension table contains:
A) Descriptive attributes (hierarchies, details) about dimensions
B) Measures
C) Facts
D) Aggregated data

Answer: A

---

46. Star schema has:
A) Central fact table surrounded by denormalized dimension tables
B) Normalized dimensions
C) Multiple fact tables
D) No dimensions

Answer: A

---

47. Snowflake schema has:
A) Dimension tables normalized into multiple related tables
B) Denormalized dimensions
C) No fact table
D) Only dimensions

Answer: A

---

48. Fact constellation (Galaxy schema) has:
A) Multiple fact tables sharing dimension tables
B) Single fact table
C) No shared dimensions
D) Only one fact

Answer: A

---

49. Grain of a fact table defines:
A) Level of detail stored (e.g., daily sales by product by store)
B) Table size
C) Number of dimensions
D) Measure types

Answer: A

---

50. Additive measures can be:
A) Summed across all dimensions
B) Summed only across some dimensions
C) Not summed at all
D) Averaged only

Answer: A

---

51. Semi-additive measures can be:
A) Summed across some dimensions (e.g., across products) but not across time (e.g., inventory levels)
B) Summed across all dimensions
C) Not summed at all
D) Averaged only

Answer: A

---

52. Non-additive measures:
A) Cannot be summed (e.g., ratios, percentages)
B) Can be summed across all dimensions

C) Are always additive
D) Are only counts

Answer: A

---

53. Concept hierarchy organizes:
A) Dimension attributes into levels of detail (day → month → quarter → year)
B) Facts into levels
C) Measures into levels
D) Cubes into levels

Answer: A

---

54. A cuboid represents:
A) A specific level of aggregation in a data cube
B) The entire cube
C) A dimension
D) A fact

Answer: A
Explanation: Different cuboids represent different group-by sets .

---

55. The apex cuboid (0-D cuboid) holds:
A) Highest level of summarization (single value)
B) Most detailed data
C) Base cuboid
D) Intermediate aggregation

Answer: A
Explanation: Apex cuboid aggregates across all dimensions .

---

56. The base cuboid holds:
A) Most detailed (lowest) level of summarization
B) Highest aggregation
C) No data
D) Only metadata

Answer: A

---

57. Lattice of cuboids represents:
A) All possible group-by combinations (cuboids) for a cube
B) Cube storage structure
C) Dimension hierarchy
D) Fact relationships

Answer: A

SECTION 4: OLAP SERVER TYPES (Questions 71-85)

58. ROLAP stands for:
A) Relational OLAP
B) Real-time OLAP
C) Remote OLAP
D) Rapid OLAP

Answer: A

---

59. ROLAP uses:
A) Relational or extended-relational DBMS with metadata to map to multidimensional view
B) Multidimensional arrays
C) Hybrid storage
D) No database

Answer: A
Explanation: ROLAP stores data in relational tables with star/snowflake schemas .

---

60. MOLAP stands for:
A) Multidimensional OLAP
B) Multi-user OLAP
C) Memory OLAP
D) Managed OLAP

Answer: A

---

61. MOLAP uses:
A) Specialized multidimensional storage engines with data cubes
B) Relational tables
C) Flat files
D) No storage

Answer: A

---

62. MOLAP advantages include:
A) Fast query performance, efficient aggregation, compact storage
B) Handles large volumes better
C) Better for detailed data
D) More scalable

Answer: A

---

63. ROLAP advantages include:
A) Scalability for large datasets, handles detailed data well
B) Faster aggregations
C) Compact storage
D) Pre-computed cubes

Answer: A

---

64. HOLAP stands for:
A) Hybrid OLAP (combines ROLAP and MOLAP)
B) High-performance OLAP
C) Hierarchical OLAP
D) Historical OLAP

Answer: A

---

65. HOLAP stores:
A) Detailed data in relational tables, aggregated data in multidimensional cubes
B) All data in relational
C) All data in cubes
D) No data

Answer: A

SECTION 5: INTRODUCTION TO DATA MINING (Questions 86-110)

66. Data mining is defined as:
A) Extraction of hidden predictive information from large databases
B) Simple querying of databases
C) Data storage
D) Data entry

Answer: A
Explanation: Data mining discovers non-obvious patterns and relationships .

---

67. Data mining is also known as:
A) Knowledge Discovery in Databases (KDD)
B) Data warehousing
C) Data processing
D) Data entry

Answer: A

---

68. The KDD process involves steps:
A) Selection, Preprocessing, Transformation, Data Mining, Interpretation/Evaluation
B) Only data mining
C) Only data collection
D) Only reporting

Answer: A

---

69. Data mining functionalities include:
A) Classification, Clustering, Association, Regression, Anomaly detection
B) Only querying
C) Only reporting
D) Only storage

Answer: A

---

70. Classification is:
A) Predicting categorical class labels (supervised learning)
B) Grouping without predefined labels
C) Finding frequent patterns
D) Predicting continuous values

Answer: A

---

71. Examples of classification techniques include:
A) Decision tree, Naïve Bayes, Support Vector Machine, KNN
B) K-means, DBSCAN
C) Apriori, FP-growth
D) PCA, SVD

Answer: A

---

72. Decision tree induction:
A) Builds tree-like model with nodes representing attributes, branches representing conditions, leaves representing class labels
B) Uses probability
C) Uses distance measures
D) Uses neural networks

Answer: A

---

73. Naïve Bayes classifier assumes:
A) Conditional independence of attributes given class
B) Attributes are dependent

C) No probabilities
D) Linear relationships

Answer: A

---

74. Support Vector Machine (SVM) finds:
A) Optimal hyperplane that maximizes margin between classes
B) Tree structure
C) Probability distribution
D) Distance clusters

Answer: A

---

75. K-Nearest Neighbors (KNN) classifies based on:
A) Majority class among k closest training examples
B) Probability model
C) Decision tree
D) Neural network

Answer: A

---

76. Clustering is:
A) Grouping data objects into classes (clusters) such that objects in same cluster are similar (unsupervised)
B) Predicting class labels
C) Finding frequent patterns
D) Detecting anomalies

Answer: A

---

77. Examples of clustering algorithms include:
A) K-means, Hierarchical, DBSCAN, BIRCH
B) Decision tree, Naïve Bayes
C) Apriori, FP-growth
D) Linear regression

Answer: A

---

78. K-means clustering partitions data into:
A) k clusters, each represented by centroid (mean)
B) Hierarchical tree
C) Density-based clusters
D) Arbitrary shapes

Answer: A

---

79. Hierarchical clustering builds:
A) Tree of clusters (dendrogram) by either agglomerative (bottom-up) or divisive (top-down)
B) Flat clusters
C) Centroid-based clusters
D) Density clusters

Answer: A

---

80. Agglomerative clustering starts with:
A) Each data point as its own cluster, then merges
B) All data in one cluster, then splits
C) Random clusters
D) No clusters

Answer: A

---

81. Divisive clustering starts with:
A) All data in one cluster, then recursively splits
B) Each point as own cluster
C) Random clusters
D) No clusters

Answer: A

---

82. Density-based clustering (DBSCAN) finds:
A) Clusters based on density, can discover arbitrary shapes, handles noise

B) Spherical clusters
C) Hierarchical clusters
D) Centroid-based clusters

Answer: A

---

83. Association rule mining finds:
A) Patterns like "if X then Y" (market basket analysis)
B) Class labels
C) Clusters
D) Outliers

Answer: A

---

84. Support of an association rule X → Y is:
A) Percentage of transactions containing both X and Y
B) Percentage of transactions containing X that also contain Y
C) Number of transactions
D) Strength of rule

Answer: A

---

85. Confidence of an association rule X → Y is:
A) Percentage of transactions containing X that also contain Y
B) Percentage containing both
C) Number of transactions
D) Lift

Answer: A

---

86. Apriori algorithm uses:
A) Anti-monotone property: if itemset is infrequent, all supersets are infrequent
B) Random generation
C) No pruning
D) Hierarchical search

Answer: A

---

87. Frequent itemset property:
A) All subsets of a frequent itemset are frequent
B) All supersets are frequent
C) No relationship
D) Random

Answer: A

---

88. Regression analysis predicts:
A) Continuous numeric values
B) Categorical labels
C) Clusters
D) Associations

Answer: A

---

89. Linear regression models relationship between:
A) Dependent variable and one or more independent variables using linear equation
B) Categories
C) Clusters
D) Itemsets

Answer: A

---

90. Anomaly detection (outlier analysis) identifies:
A) Data points that deviate significantly from normal behavior
B) Normal patterns
C) Clusters
D) Associations

Answer: A

SECTION 6: DATA PREPROCESSING (Questions 111-135)

91. Data preprocessing is necessary because:
A) Real-world data is incomplete, noisy, inconsistent
B) Data is always clean
C) No preparation needed
D) Only formatting needed

Answer: A

---

92. Data preprocessing tasks include:
A) Data cleaning, integration, transformation, reduction, discretization
B) Only data cleaning
C) Only data integration
D) Only data storage

Answer: A

---

93. Data cleaning addresses:
A) Missing values, noise, outliers, inconsistencies
B) Data formatting only
C) Data storage only
D) Data querying

Answer: A

---

94. Handling missing values can be done by:
A) Ignore tuple, fill with global constant, mean, median, or predict missing value
B) Delete all data
C) Ignore all missing
D) No action

Answer: A

---

95. Noise in data refers to:
A) Random errors or variance in measured variables
B) Missing values
C) Inconsistent data
D) Duplicate data

Answer: A

---

96. Binning is a technique for:
A) Smoothing noisy data by sorting and partitioning into bins, then smoothing by bin means, medians, or boundaries
B) Data integration
C) Data reduction
D) Data transformation

Answer: A

---

97. Smoothing by bin means:
A) Replace each value in bin with mean of bin
B) Replace with median
C) Replace with boundary values
D) Replace with zero

Answer: A

---

98. Smoothing by bin boundaries:
A) Replace each value with closest bin boundary (min or max)
B) Replace with mean
C) Replace with median
D) Replace with mode

Answer: A

---

99. Data integration combines:
A) Data from multiple sources into coherent data store
B) Data from single source
C) Only metadata
D) Only queries

Answer: A

---

100. Challenges in data integration include:
A) Semantic heterogeneity, schema integration, redundancy, conflict resolution
B) No challenges
C) Only performance
D) Only storage

Answer: A

---

101. Data transformation converts data into:
A) Forms suitable for mining (normalization, aggregation, generalization)
B) Raw form only
C) No conversion
D) Only sorting

Answer: A

---

102. Normalization scales data to:
A) Specific range (e.g., min-max, z-score, decimal scaling)
B) Original values
C) No scaling
D) Random values

Answer: A

---

103. Min-max normalization:
A) v' = (v - min)/(max - min) * (new_max - new_min) + new_min
B) v' = (v - mean)/stddev
C) v' = v/10^j
D) v' = log(v)

Answer: A

---

104. Z-score normalization (standardization):
A) v' = (v - mean)/standard deviation
B) v' = (v - min)/(max - min)
C) v' = v/10^j
D) v' = log(v)

Answer: A

---

105. Data reduction techniques include:
A) Dimensionality reduction, numerosity reduction, data compression
B) Only data cleaning
C) Only integration
D) Only transformation

Answer: A

---

106. Dimensionality reduction reduces:
A) Number of attributes (feature selection, PCA)
B) Number of tuples
C) Data size
D) Query complexity

Answer: A

---

107. Principal Component Analysis (PCA) is:
A) Technique for reducing data dimensionality by transforming to new orthogonal axes capturing maximum variance
B) Classification technique
C) Clustering technique
D) Association mining

Answer: A
Explanation: PCA is for data reduction, not classification .

---

108. Numerosity reduction replaces:
A) Original data with smaller representation (sampling, clustering, histograms)
B) Attributes with fewer
C) Values with constants
D) No reduction

Answer: A

---

109. Discretization converts:
A) Continuous data into intervals (bins)
B) Categorical to numeric
C) Numeric to categorical
D) No conversion

Answer: A

---

110. Concept hierarchy generation organizes:
A) Attributes into levels of abstraction (e.g., age: youth, adult, senior)
B) Data into clusters
C) Rules into hierarchy
D) Classes into hierarchy

Answer: A

SECTION 7: DATA MINING TASKS AND ISSUES (Questions 136-150)

111. Descriptive data mining tasks:
A) Characterize general properties of data (clustering, summarization, association)
B) Predict future trends
C) Classify new instances
D) Estimate values

Answer: A

---

112. Predictive data mining tasks:
A) Perform inference on current data to make predictions (classification, regression, prediction)
B) Describe patterns
C) Find associations
D) Cluster data

Answer: A

---

113. Interestingness measures for patterns include:
A) Support, confidence, lift, novelty, utility
B) Only support
C) Only confidence
D) Only size

Answer: A

---

114. Major issues in data mining include:
A) Mining methodology, user interaction, performance, data types, privacy
B) Only performance
C) Only algorithms
D) Only data storage

Answer: A

---

115. Scalability in data mining refers to:
A) Ability to handle large datasets efficiently
B) Ability to run on small data
C) Algorithm complexity
D) Storage capacity

Answer: A

---

116. Overfitting occurs when:
A) Model fits training data too well, capturing noise, and fails on new data
B) Model underfits training data
C) Model is perfect
D) Model has high bias

Answer: A

---

117. Underfitting occurs when:
A) Model is too simple, fails to capture underlying patterns
B) Model is too complex
C) Model fits perfectly
D) Model generalizes well

Answer: A

SECTION 8: APPLICATIONS AND TOOLS (Questions 151-165)

118. Data mining applications include:
A) Market basket analysis, fraud detection, customer segmentation, recommendation systems
B) Only retail
C) Only banking
D) Only healthcare

Answer: A

---

119. Market basket analysis uses:
A) Association rule mining to find products frequently bought together
B) Classification
C) Clustering
D) Regression

Answer: A

---

120. Customer segmentation uses:
A) Clustering to group customers with similar behavior
B) Classification
C) Association rules
D) Regression

Answer: A

---

121. Fraud detection uses:
A) Anomaly detection, classification to identify suspicious patterns
B) Clustering only
C) Association only
D) Regression only

Answer: A

---

122. Recommendation systems use:
A) Collaborative filtering, association rules, content-based methods
B) Only classification
C) Only clustering
D) Only regression

Answer: A

---

123. Web mining includes:
A) Web content mining, web structure mining, web usage mining
B) Only web content
C) Only web usage
D) Only web design

Answer: A

---

124. Text mining analyzes:
A) Unstructured text documents to extract patterns and insights
B) Structured data only
C) Numeric data only
D) Images only

Answer: A

---

125. Spatial mining deals with:
A) Geographic and spatial data (location, distance, topology)
B) Text data
C) Time-series data
D) Graph data

Answer: A

---

126. Time-series mining analyzes:
A) Sequential data points over time (trends, seasonality, patterns)
B) Static data
C) Categorical data
D) Spatial data

Answer: A

---

127. Data mining tools include:
A) Weka, R, Python (scikit-learn), RapidMiner, KNIME
B) Only Excel
C) Only SQL
D) Only database tools

Answer: A

---

128. Weka is:
A) Open-source data mining software with collection of machine learning algorithms
B) Commercial database
C) Programming language
D) Operating system

Answer: A

---

129. CRISP-DM stands for:
A) Cross-Industry Standard Process for Data Mining
B) Common Research in Statistical Practice
C) Comprehensive Regression in Statistical Data Mining
D) Cross-Validation Standard Process

Answer: A

---

130. CRISP-DM phases include:
A) Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, Deployment
B) Only Modeling

C) Only Data Preparation
D) Only Evaluation

Answer: A